

Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc.

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Abstract

The rapid advancement of digital technologies transformed modern manufacturing by enabling organizations to improve production monitoring, operational efficiency, and data-driven decision-making. However, many food manufacturing companies continued to rely on manual and paper-based production recording systems, which often resulted in delayed reporting, inaccurate computations, data redundancy, and limited visibility of production performance. At ALDi Foods Inc., production personnel manually recorded raw material usage, extraction outputs, and packed yield information, making it difficult for management to monitor production activities in real time and evaluate production efficiency promptly. These challenges highlighted the need for a centralized digital production monitoring system capable of providing accurate, timely, and reliable production information.

This study aimed to develop a web-based system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc."* The proposed system was designed to digitize production recording by replacing manual documentation with a centralized platform that supported real-time monitoring of production activities. It was structured to automate the computation of yield percentage and extraction efficiency, classify production stability based on predefined organizational thresholds, generate dashboard visualizations, maintain historical production records, and produce automated weekly, monthly, quarterly, and yearly raw material utilization reports. Through these features, the system was expected to improve production visibility, reduce manual computation errors, enhance reporting accuracy, and support timely managerial decision-making.

The study employed a Developmental Research Design and utilized the Agile Development Model to guide the systematic analysis, design, development, testing, and evaluation of the proposed system. System requirements were gathered through interviews with production personnel, direct observation of existing production workflows, and a review of company production documents and records. The system was developed using HTML and CSS for the user interface, PHP for server-side application development, MySQL for centralized database management, and Chart.js for graphical visualization of production analytics. A three-tier architecture consisting of the Client Layer, Application Layer, and Database Layer was implemented to ensure efficient processing, secure data management, and reliable communication among system components.

The findings of the study were expected to demonstrate that the proposed system provided a practical, accessible, and cost-effective digital solution for improving production monitoring, real-time yield analytics, extraction efficiency monitoring, and evidence-based decision-making within ALDi Foods Inc., while contributing to the continuing digital transformation of food manufacturing operations.

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

In today's highly competitive and technology-driven environment, industries continuously sought innovative ways to improve operational efficiency, reduce waste, and enhance productivity. The manufacturing sector, particularly in food production, played a critical role in meeting consumer demand while maintaining quality and cost-effectiveness. With the emergence of digital transformation, organizations began transitioning from traditional production methods toward more advanced, automated, and data-driven systems.

Digital Transformation introduced the integration of digital technologies, such as real-time analytics, cloud computing, and intelligent systems, into manufacturing processes. These technologies enabled organizations to collect, process, and analyze data instantly, which allowed for improved decision-making and enhanced operational control. A key advantage of this transformation was the ability to monitor production processes in real time, thereby minimizing inefficiencies and preventing costly errors.

Despite these advancements, many small and medium-sized enterprises (SMEs), including ALDi Foods Inc., continued to rely on manual, paper-based methods of recording production data. These traditional systems proved prone to human error, data inconsistencies, and delays in information processing. Consequently, businesses experienced data latency and lacked real-time visibility into production performance, which made it difficult to detect yield losses, machine issues, and material wastage at an early stage.

In food production, particularly within juice manufacturing, yield had served as a critical performance indicator that had represented the ratio of output (extracted juice) to input (raw materials). A high yield had consistently indicated efficient resource utilization, whereas a low yield had pointed to underlying inefficiencies in the extraction process.

Factors such as equipment condition, raw material quality, and operator performance had significantly affected those yield outcomes. However, without a reliable monitoring system in place, identifying the impact of those variables had remained difficult to analyze and optimize.

ALDi Foods Inc. relied on manual logging systems to track production data, which limited the company's ability to monitor extraction efficiency in real time. Production issues were often identified only after operations had been completed, resulting in delayed decision-making and missed opportunities for immediate corrective actions. To address these challenges, digitizing the production process through automated data collection and centralized software systems became essential for transforming raw data into actionable insights.

To solve these operational bottlenecks, this study proposed the development of *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for ALDi Foods Inc.* The system aimed to replace manual recording methods with a centralized digital platform that provided real-time monitoring, analytics, and visualization of production data. Ultimately, the project sought to improve efficiency, reduce waste, and enhance decision-making across the company's food production operations.

1.2 Statement of the Problem

The current production system of ALDi Foods Inc. relied heavily on manual, paper-based recording of production data. This approach presented several challenges that affected the efficiency, accuracy, and reliability of operations. Manual data recording consumed time, increased the possibility of human error, and limited the company's ability to monitor production performance in real time. As a result, production inefficiencies and extraction losses were often detected only after operations had been completed, making it difficult to implement immediate corrective actions.

Additionally, the absence of a centralized digital monitoring system severely limited management's access to timely and accurate production information. Without this visibility, leadership faced significant hurdles when making critical, time-sensitive decisions, evaluating overall employee productivity, and maintaining high operational efficiency.

Furthermore, this lack of real-time analytics hindered the company's capability to identify recurring production patterns and monitor extraction performance on the fly. As a result, the facility was left vulnerable to unaddressed inefficiencies and preventable material waste during its core manufacturing processes.

Specifically, this study seeks to answer the following questions:

1. How does manual recording of production data affect accuracy, consistency, and reliability at ALDi Foods Inc.?
2. What are the limitations of the current system in terms of real-time monitoring and data accessibility?

3. How can real-time monitoring improve the tracking of raw material inputs and production outputs?
4. How can yield losses during the extraction process be detected at an earlier stage?
5. How can inefficiencies in the production process be measured and analyzed effectively?
6. How can a digital system enhance managerial decision-making through timely and accurate data?
7. What features should be included in a digital production monitoring system to address existing challenges?

1.3 Objectives of the Study

1.3.1 General Objective

The primary objective of this study was to design and develop a digital production monitoring system for AIDi Foods Inc., specifically aimed at providing real-time yield analytics and extraction efficiency tracking. By completely replacing legacy, paper-based recording methods with a centralized digital solution, the system established an automated framework to streamline data collection, eliminate manual errors, and enhance the overall accuracy, efficiency, and accessibility of production data.

Focusing on juice extraction operations, the platform centralized disparate metrics—such as raw material usage, total juice output, yield percentages, and extraction efficiency—into a cohesive, real-time data stream. This automated approach minimized recording delays and provided

high-fidelity operational visibility, establishing a standardized performance baseline that ensured every input and output was accurately accounted for across the production floor.

Ultimately, the system featured an intuitive, centralized dashboard that transformed raw operational data into real-time visualizations and actionable insights. By allowing management to immediately spot yield variations, detect processing losses, and evaluate machine performance, the digital solution empowered stakeholders at AIDi Foods Inc. to make proactive, data-driven decisions that optimized daily workflows and reduced waste.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. Analyze the existing production process and identify inefficiencies in data recording and monitoring.
2. Design a user-friendly digital system that replaces manual production logs.
3. Develop a real-time monitoring feature for tracking raw material inputs and production outputs.
4. Implement automated yield calculation and analytics functionalities.
5. Monitor extraction efficiency and identify performance trends during production.
6. Provide an interactive dashboard for real-time visualization of production data.
7. Store and manage production data securely using a centralized database.
8. Evaluate the system's performance in terms of functionality, usability, accuracy, efficiency, and reliability.
9. Reduce delays in production monitoring by providing instant access to operational data and analytics.

10. Improve the company's ability to identify extraction losses and production inefficiencies through automated monitoring and reporting features.
11. Support data-driven decision-making by generating accurate and timely production reports for management.
12. Enhance productivity and operational control through the integration of digital technologies and real-time monitoring tools.
13. Minimize manual recording errors and improve data consistency within production operations.
14. Provide a scalable foundation for future integration of IoT-based sensors and automated manufacturing technologies.

1.4 Scope and Delimitation

This study focuses on the design and development of a digital production monitoring system for ALDi Foods Inc. The system is intended to improve the tracking, monitoring, and analysis of production data, particularly in terms of materials input, yield computation and extraction efficiency. The proposed system aims to replace manual recording methods with a centralized digital platform capable of providing real-time monitoring and analytical functionalities.

The scope of the study includes the development of features such as real-time production data input, automated yield calculation, extraction efficiency monitoring, dashboard visualization, and centralized database management. The system will allow authorized users to monitor production performance, track raw material usage, and evaluate

production outputs through a digital interface. Additionally, the study includes basic analytics and visualization tools that support production analysis and operational decision-making. The system will be developed as a prototype intended for use within the operational context of AIDi Foods Inc. Data may be entered manually to demonstrate real-time monitoring capabilities. The study also covers system testing and evaluation in terms of functionality, usability, efficiency, accuracy, and reliability.

However, the study has several delimitations. The proposed system will not include full-scale industrial automation or advanced machine-learning capabilities. Integration with highly sophisticated Industrial Internet of Things (IIoT) infrastructures, robotics, or enterprise-level manufacturing systems is beyond the scope of this research. The study is limited only to monitoring production yield and extraction efficiency and does not cover other business operations such as inventory management, sales, accounting, payroll, or supply chain management.

Furthermore, the system will be tested in a controlled environment and may not fully represent all real-world production conditions. The study focuses only on the operational processes of AIDi Foods Inc., and the findings may not be fully generalized to other manufacturing companies with different production environments or operational requirements. Financial analysis, large-scale deployment, and long-term implementation assessment are also excluded from the scope of this study.

1.5 Significance of the Study

This study is beneficial to the following:

ALDi Foods Inc.- The system will improve production efficiency, reduce waste, and enhance overall productivity and profitability through real-time monitoring and analytics.

Production Managers- Managers will have access to timely and accurate production data, enabling faster decision-making, improved monitoring, and immediate response to operational issues.

Employees- The system will simplify production data recording processes, reduce manual workload, minimize human error, and improve operational efficiency.

BSIT Students- The study serves as a practical application of knowledge in system development, database management, software engineering, and real-time analytics.

Researchers- This study can serve as a reference for future research related to digital production systems, real-time monitoring, Industry 4.0 technologies, and manufacturing analytics.

1.6 Definition of Terms

Analytics – The process of analyzing data to extract meaningful insights and support decision-making.

Centralized Database – A digital storage system where production data is organized and managed in a single location for easier access and monitoring.

Dashboard – A graphical user interface that displays real-time production information, analytics, and system reports.

Digitizing Production – The conversion of manual production records and monitoring processes into digital format using information technology.

Efficiency – A measure of how effectively resources are utilized to produce desired outputs.

Extraction – The process of obtaining juice or liquid product from raw materials during production.

Manufacturing Industry- The study may contribute to the advancement of digital transformation practices in manufacturing by demonstrating the benefits of real-time monitoring systems in improving efficiency and reducing production waste.

Monitoring– The continuous observation, tracking, and evaluation of production processes and operational performance.

Real-Time – Immediate processing, updating, and availability of data without significant delay.

Yield – The ratio of output produced compared to input materials, usually expressed as a percentage.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

A study by Verhoef et al. (2021) analyzed the impact of transitioning from legacy paper-based logs to centralized digital dashboards in manufacturing. The researchers highlighted that manual data silos create "blind spots" or information gaps, where production anomalies—such as waste hotspots or mechanical inefficiencies—are often only discovered long after a batch is completed. By implementing a digitized analytics system, the facility achieved a more responsive production environment, allowing managers to make data-driven decisions based on live efficiency metrics rather than historical, error-prone logs. This study serves as a foundational reference for your claim that digitizing the production floor is a prerequisite for Industry 4.0 and is essential for reducing financial losses caused by avoidable process inefficiencies.

A study by Makhmudah et al. (2025) explored the implementation of a digital production reporting system in a manufacturing environment. Using a qualitative case study approach, the researchers replaced manual reporting processes with a system integrated with sensors, automated data transmission, and dashboard visualization. The results showed improvements in data accuracy, operational efficiency, and decision-making. This study is relevant to the present research as it demonstrates the effectiveness of digital systems in improving production monitoring and data management.

A study by Mansour and Al-Hamdani (2024) examined the application of internal Key Performance Indicators (KPIs) in evaluating production efficiency within the food industry, with

a specific focus on non-conforming products identified during manufacturing operations. The researchers proposed a structured performance measurement system centered on the internal parts per million (INTppm) indicator, which quantifies defective units detected within production orders relative to the total number of items produced. This approach allows organizations to assess production quality at the operational level rather than relying solely on end-product inspection, thereby enabling immediate, data-driven decision-making. By continuously tracking these metrics, the authors emphasized that food production performance depends heavily on the continuous monitoring of quality-related deviations to maintain process stability. In their case analysis involving food-related processing lines, Mansour and Al-Hamdani (2024) monitored various parameters across manual and automated operations, including quality control checks, labeling procedures, measurement systems, and line adjustments. The findings revealed that internal defect rates varied significantly depending on the production stage, with technical problems—such as mechanical errors, clamping inconsistencies, tool wear, and alignment issues—serving as primary drivers of elevated INTppm values.

Additionally, inadequate maintenance and delayed equipment replacements further exacerbated production instability. Consequently, the internal defect rate proved to be a critical indicator of both product quality and machine reliability. To mitigate these operational inefficiencies, Mansour and Al-Hamdani (2024) highlighted the necessity of corrective and preventive measures, such as enhancing automated process controls, executing random quality checks during shifts, and updating employee training programs. The researchers also recommended integrating skill evaluation checkpoints and fostering workforce communication to minimize human errors alongside technical updates. Cultivating this long-term data collection approach allows manufacturing organizations to transition from reactive problem-solving to

proactive process optimization, shifting focus toward controllable internal variables like equipment conditions and operator behavior. Ultimately, Mansour and Al-Hamdani (2024) concluded that utilizing internal KPIs like INTppm offers a reliable methodology for optimizing food industry operations by linking real-time defect rates directly to strategic business outcomes like waste reduction and cost efficiency. This literature is highly relevant to the present study as it provides a robust structural foundation for utilizing internal performance indicators to target production inefficiencies. Furthermore, it directly supports the proposed framework of developing a data-driven, real-time tracking system for AIDi Foods Inc. to monitor extraction efficiency and establish long-term process optimization.

A comprehensive review conducted by Aftab Siddique, Ashish Gupta, Jason T. Sawyer, Tung-Shi Huang, and Amit Morey (2025), entitled “ Big Data Analytics in the Food Industry: A State-of-the-Art Literature Review”, examined the increasing influence of artificial intelligence (AI), machine learning (ML), and big data analytics on the modernization of food industry operations. Published in “npj Science of Food”, the study explored numerous scholarly works related to digital transformation and data-driven technologies across food production, processing, quality assurance, supply chain management, and food safety. The researchers emphasized that the food industry is undergoing a significant technological shift as organizations increasingly adopt digital systems capable of collecting, storing, processing, and analyzing large volumes of operational data. As production processes grow more complex and consumer demands evolve, the deployment of advanced analytics has become a pivotal strategy for improving efficiency, productivity, and competitiveness. According to Siddique et al. (2025), modern food manufacturing facilities generate substantial amounts of data from various stages of production,

including raw material handling, processing, packaging, storage, distribution, and quality control. Traditionally, much of this data remained underutilized due to legacy constraints in collection and analytical methods. However, advancements in digital technologies enable organizations to transform raw production data into actionable insights to support real-time decision-making and operational improvements.

The study explained that big data analytics provides organizations with the capacity to identify patterns, detect hidden inefficiencies, forecast outcomes, and monitor performance far more accurately than conventional, manual approaches. The review highlighted that AI and ML technologies have become essential tools for enhancing food production systems by processing large datasets to generate insights that are difficult to obtain through manual analysis. AI-based models learn from historical and real-time data, making it possible to predict trends, isolate operational bottlenecks, and optimize processing parameters. Through continuous data analysis, organizations can achieve tighter process control, minimize material waste, and enhance overall manufacturing efficiency. A major focus of the study involved the application of predictive analytics and artificial neural networks (ANNs) in food processing operations, particularly in activities such as drying, baking, fermentation, extrusion, freezing, and quality assessment. By utilizing historical operational data, these predictive systems can anticipate final product characteristics, estimate exact processing times, model complex multi-variable relationships, and dynamically identify optimal operating conditions. Furthermore, Siddique et al. (2025) underscored the importance of real-time monitoring systems over traditional monitoring methods, which heavily rely on manual recording procedures that are time-consuming and prone to human error. In contrast, digital monitoring systems continuously collect and analyze operational data, allowing managers and operators to observe production performance

dynamically. This real-time monitoring provides immediate visibility into production loops and enables organizations to respond swiftly to operational anomalies. The authors further discussed the integration of computer vision systems and image-processing technologies to automatically inspect products during production. This automated visual intelligence reduces the need for manual inspection while increasing consistency and accuracy in food grading, defect detection, and quality evaluation. In addition to production monitoring and quality control, the review explored the role of big data analytics in food safety management, supply chain optimization, and resource sustainability. Advanced analytical systems can identify contamination risks, track disease outbreaks, and improve overall traceability to mitigate economic losses associated with product recalls. From an operational standpoint, AI-powered forecasting models analyze historical sales data and consumer behavior to predict future demand, enabling manufacturers to optimize inventory levels and schedule production efficiently. Siddique et al. (2025) also noted that data-driven systems support corporate sustainability goals by optimizing the use of raw materials, energy, and water, thereby reducing environmental impacts. Despite challenges such as data quality, system integration, and workforce readiness, the authors concluded that the future of the food industry depends on intelligent systems capable of transforming raw metrics into actionable knowledge. This extensive review is highly relevant to the present capstone project, “Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc.”, because it provides strong theoretical and empirical validation for the implementation of digital monitoring and analytics systems within food processing environments. The findings directly support the technical architecture of a platform capable of monitoring production yield and extraction efficiency in real time. By establishing a system for continuous data collection, management can move away from latency-heavy, manual logs to

instantly identify process variations, evaluate extraction performance, and detect production losses. Consequently, the work of Siddique et al. (2025) serves as a foundational benchmark for the proposed system, demonstrating that real-time data integration is critical to modernizing waste management and achieving operational excellence.

A study by Rivadeneira et al. (2023), published in *Food Research*, examined the production of pectin from “saba” banana peel waste using ultrasound-assisted extraction and evaluated its performance as an emulsifying agent compared with commercial low-methoxyl pectin (LMP). The study emphasized the potential of agricultural waste valorization by converting banana peels into functional food ingredients suitable for emulsion-based systems. Findings revealed that the extracted pectin had lower equivalent weight and a reduced degree of esterification than commercial LMP, indicating differences in molecular structure that influence its functional behavior in food applications. Although UEP showed lower purity, it contained relatively higher amounts of protein and ash, which contributed positively to its interfacial and emulsifying properties.

Structural and physical analyses further confirmed these differences, where FTIR results indicated variations in esterified carboxyl groups between UEP and LMP. Microscopic observation showed that emulsions stabilized by UEP formed smaller and more evenly distributed oil droplets compared to those stabilized by commercial pectin. This droplet size reduction contributed to improved emulsion stability, as it enhanced interfacial coverage and minimized phase separation. The study also reported that changes in formulation factors such as

pH level, oil concentration, and ionic strength significantly affected emulsion behavior, with UEP consistently showing stronger adaptability under varying conditions.

In terms of functional performance, emulsions containing UEP exhibited higher viscosity and better resistance to breakdown compared to those stabilized by LMP. The study also found that UEP influenced lipid digestion, as it reduced free fatty acid release during *in vitro* digestion, suggesting a potential role in controlling lipid bioaccessibility. Overall, the researchers concluded that ultrasound-extracted pectin from banana peel waste is a promising sustainable alternative to commercial pectin, offering improved emulsifying stability and functional benefits for food formulation and processing systems.

A study by Sabih, Farid, Ejaz, Husam, Khan, and Farooq (2023) investigated the use of computer vision and machine learning technologies to automate the measurement of raw material flow rates on a conveyor belt system used in soda ash manufacturing. The researchers recognized that many industrial processes still depend on manual monitoring and measurements, which can lead to errors, inefficiencies, and delays in decision-making. To address this challenge, they proposed an automated framework capable of estimating the flow rate of raw materials transported on a conveyor belt without requiring continuous human intervention.

The study was conducted in a soda ash manufacturing plant where limestone and coke are transported through a conveyor belt system. These materials are critical inputs in the production process because maintaining the correct ratio between limestone and coke is essential for generating carbon dioxide gas with the required purity level. Traditionally, operators manually conducted experiments and observations to determine the flow rate of materials and adjust

conveyor settings accordingly. However, this approach was time-consuming and susceptible to inaccuracies. The researchers sought to improve this process through the implementation of an intelligent monitoring system.

To collect data, a high-resolution camera was installed above the conveyor belt to continuously capture video footage of the moving materials. The data collection process spanned twenty-four weeks and included videos recorded under various environmental conditions such as changes in lighting, dust levels, and weather conditions. Since the conveyor belt was located outdoors, these environmental factors introduced additional challenges that had to be addressed by the proposed system. The researchers designed their framework to be robust enough to operate effectively despite these changing conditions.

The methodology involved several stages of image processing and machine learning. First, a region of interest was extracted from each video frame so that only the conveyor belt area containing the raw materials would be analyzed. This step reduced computational requirements and eliminated unnecessary background information. The images were then preprocessed using sharpening and filtering techniques to improve image quality and reduce the effects of dust, noise, and poor lighting. Bilateral filtering was specifically selected because it reduced image noise while preserving important edge details needed for accurate material detection.

After preprocessing, the researchers focused on segmenting the raw materials from the conveyor belt background. Several segmentation approaches were evaluated, including deep learning-based methods, watershed segmentation, frame differencing, and background subtraction. However, because of limited labeled data and highly variable environmental conditions, some methods produced inconsistent results. To overcome these limitations, the

researchers developed a voting-based segmentation approach that combined frame differencing and the BackgroundSubtractorCNT algorithm. By retaining only the pixels identified by both techniques, the system was able to isolate the most reliable regions representing the raw materials. This method improved segmentation accuracy and reduced the influence of environmental noise and lighting variations.

Once the materials were successfully segmented, the system extracted various image features to support flow-rate estimation. The researchers used moment-based features to describe the shape, size, position, and distribution of the segmented materials. Additional contour-based features were extracted to measure the area occupied by limestone and coke on the conveyor belt. K-means clustering was also applied to separate different image regions based on color information. These features were aggregated over specific time windows to accommodate the variable lengths of the video recordings and ensure consistent input dimensions for machine learning models.

To improve model performance and reduce computational complexity, the study employed feature selection techniques, particularly Recursive Feature Elimination with Cross-Validation (RFECV). This method identified the most significant features while eliminating less relevant variables. The selected features were then used to train and evaluate several machine learning algorithms, including Decision Tree, Random Forest, XGBoost, Gradient Boosting, Bagging Regressor, Bayesian Ridge, Gamma Regressor, RANSAC, and Theil-Sen Regressor. Both linear and non-linear models were assessed to determine the most effective approach for predicting material flow rates.

The researchers evaluated the performance of the models using several regression metrics,

including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). They also utilized leave-one-out validation and k-fold cross-validation to ensure the reliability of the results. The findings revealed that model performance was significantly influenced by the choice of window size, aggregation method, and feature selection strategy. While feature selection improved some non-linear models, certain linear models achieved better results when all extracted features were retained.

Among the evaluated algorithms, the Bagging Regressor demonstrated the best overall performance in terms of prediction accuracy. The final model was able to estimate material flow rates with a percentage error as low as seven percent, indicating a high level of reliability. The study showed that combining computer vision techniques with machine learning algorithms can effectively automate flow-rate monitoring while minimizing the need for manual measurements. Furthermore, the proposed system successfully operated in challenging outdoor environments where variations in lighting, dust, and weather conditions could otherwise compromise accuracy.

The researchers concluded that the integration of computer vision and machine learning provides a practical and cost-effective solution for industrial process monitoring. By using only a standard RGB camera and advanced data analytics techniques, the system was able to deliver accurate real-time estimates of material flow rates. This capability supports more informed operational decisions, enhances process efficiency, and reduces the likelihood of human error in industrial settings.

A study by **Sharma et al. (2021)** examined the growing role of artificial intelligence (AI) and big data analytics (BDA) in transforming the food industry toward more sustainable, efficient, and intelligent operations. The researchers conducted a systematic literature review, gathering and analyzing previous studies from reputable academic databases, including Scopus, ScienceDirect, Wiley Online Library, Taylor & Francis, and Web of Science. Their objective was to investigate how AI and BDA technologies are being utilized across different stages of the food industry, from agricultural production and food processing to logistics, quality assurance, supply chain management, and distribution. The review emphasized that the increasing demand for food, combined with the need for higher product quality, food safety, and sustainable resource management, has encouraged food manufacturers to adopt digital technologies that can optimize production processes and improve operational performance.

The authors explained that artificial intelligence serves as a powerful computational technology capable of analyzing complex production data, recognizing patterns, making predictions, and supporting intelligent decision-making. AI encompasses various technologies such as machine learning, artificial neural networks (ANNs), computer vision, robotics, fuzzy logic, genetic algorithms, and expert systems. These technologies enable food manufacturers to automate repetitive tasks, monitor production processes, detect defects, forecast production outcomes, and improve quality control. Meanwhile, big data analytics complements AI by collecting, processing, and analyzing large volumes of structured and unstructured data generated throughout production operations. Through advanced data analysis, manufacturers can identify operational trends, predict equipment failures, optimize production schedules, reduce waste, and improve resource utilization. Together, AI and BDA create an intelligent production environment that supports real-time monitoring and continuous process improvement.

The review highlighted numerous applications of AI within food production systems. Machine learning algorithms are widely used for demand forecasting, production planning, yield prediction, and quality classification. Artificial neural networks assist manufacturers in modeling complex production processes and predicting product quality based on multiple process variables. Computer vision systems perform automated inspection by identifying defects, grading food products according to size, shape, color, and texture, and ensuring compliance with quality standards. Robotics contributes to automating labor-intensive operations such as food handling, sorting, packaging, and processing while maintaining product consistency and reducing human error. The authors also discussed intelligent systems such as electronic noses, hyperspectral imaging, infrared sensing, and machine vision technologies that improve food safety, contamination detection, and product evaluation with greater accuracy than traditional manual inspection methods. These technologies reduce processing time while maintaining high levels of precision and reliability.

Sharma et al. (2021) further explained that big data analytics strengthens decision-making by converting massive amounts of operational data into meaningful insights. The researchers described three primary categories of analytics used in food manufacturing. **Descriptive analytics** helps identify production patterns, operational trends, and performance indicators by analyzing historical production data. **Predictive analytics** estimates future outcomes such as production demand, equipment performance, product quality, weather impacts, customer behavior, and inventory requirements using statistical models and machine learning algorithms. **Prescriptive analytics** goes one step further by recommending the most effective actions based on optimization models and simulation techniques, allowing managers to make informed decisions regarding production scheduling, inventory management, logistics, and resource

allocation. These capabilities enable manufacturers to respond more effectively to production challenges while minimizing operational risks and improving efficiency.

Another important contribution of the study is its discussion of real-time monitoring technologies integrated with AI and BDA. Modern food manufacturing facilities generate enormous amounts of production data through sensors, Internet of Things (IoT) devices, radio-frequency identification (RFID), automated equipment, and intelligent monitoring systems. These technologies continuously collect information regarding machine performance, production output, environmental conditions, inventory levels, and product quality. AI algorithms analyze these data streams in real time, enabling immediate detection of production abnormalities, equipment malfunctions, or quality deviations. Instead of relying on delayed manual reporting, manufacturers receive instant information that supports faster corrective actions and more effective production management. Real-time visibility also improves traceability throughout the production process, helping organizations maintain compliance with food safety regulations while reducing waste and operational costs.

The researchers also emphasized the growing role of AI and big data analytics across various sectors of the food industry. In dairy processing, AI models predict milk yield, monitor animal health, and estimate product shelf life. In meat processing, machine learning, computer vision, and electronic nose technologies assess meat freshness, identify spoilage, and automate quality grading. For fruits and vegetables, intelligent refrigeration systems monitor product aging and shelf life, while machine vision systems automate sorting and grading based on physical characteristics. AI applications have also been successfully implemented in bakery operations through intelligent oven control systems that optimize baking conditions and improve product consistency. Similarly, beverage manufacturers utilize AI technologies to evaluate flavor, aroma,

appearance, and consumer preferences, while robotics assists in packaging and production automation. These examples demonstrate the versatility of AI technologies in addressing operational challenges across different food manufacturing environments.

Despite these advantages, Sharma et al. (2021) acknowledged several challenges associated with implementing AI and BDA in the food industry. The adoption of intelligent technologies requires substantial financial investment in digital infrastructure, sensors, automated equipment, and information systems. Organizations also need skilled personnel capable of operating AI-based technologies, interpreting analytical results, and maintaining digital production systems. Additionally, concerns regarding cybersecurity, data privacy, workforce adaptation, and organizational readiness remain significant barriers to successful implementation. However, the researchers argued that the long-term benefits—including improved operational efficiency, enhanced product quality, reduced costs, faster decision-making, better traceability, and increased production sustainability—far outweigh these implementation challenges. As digital technologies continue to evolve, their adoption is expected to become increasingly essential for maintaining competitiveness within the global food industry.

2.2 Local Studies

A study by [Bagaforo and Albason \(2024\)](#), presented at the 19th European Conference on Innovation and Entrepreneurship (ECIE), investigated the digital transformation journeys of Micro, Small, and Medium Enterprises (MSMEs) in Iloilo Province, identifying the specific enablers and barriers local businesses face when adopting new technologies. The research utilized a descriptive-correlational approach to evaluate how local infrastructure and workforce

"readiness" influence the transition from traditional to digitized operations. Their findings highlighted that while digital tools significantly enhance local business competitiveness and operational efficiency, success is often hindered by technical literacy gaps and varying levels of infrastructural support. This study provides critical contextual evidence for AlDi Foods Inc. regarding the feasibility of implementing automated systems within the local economic landscape, directly supporting the Significance of the Study by illustrating how digital integration optimizes resource management and regional market positioning.

[4]A study by [De Leon et al. \(2017\)](#), archived in the Central Philippine University (CPU) Institutional Repository, focused on the design and development of an online ordering system for Iloilo Supermart using the V-Model Software Development Life Cycle (SDLC). The researchers implemented a secure, centralized database and a user-friendly management dashboard to streamline operations for the Iloilo-based food retailer, effectively mitigating risks associated with physical data damage and manual record-keeping.

-A study conducted by [Palero et al. \(2026\)](#), titled “*Green Extraction and Sustainability Assessment of Total Carotenoids from Carrot (*Daucus carota*) Peel Waste using Soybean Oil,*” explored a sustainable and environmentally responsible approach for recovering valuable bioactive compounds from agricultural waste materials. Published in the Davao Research Journal, the study primarily addressed the growing concern over the environmental and health hazards associated with conventional solvent-based extraction methods, particularly those involving organic solvents such as hexane. These traditional solvents, while efficient in

extracting carotenoids, are widely criticized due to their toxicity, flammability, and environmental persistence. In response to these limitations, the researchers investigated soybean oil as a safer, biodegradable, and food-grade alternative solvent capable of efficiently extracting carotenoids from carrot peel waste while aligning with green chemistry and circular economy principles.

Carotenoids are naturally occurring pigments responsible for the orange, yellow, and red coloration in many fruits and vegetables, and they are widely recognized for their antioxidant, anti-inflammatory, and health-promoting properties. Because of their high commercial value and wide applicability in food, pharmaceutical, and cosmetic industries, there is increasing global interest in improving their extraction from natural sources, especially from agricultural by-products that are often discarded. Carrot peels, which are generated in large quantities during food processing, represent a significant yet underutilized source of carotenoids. The study emphasized that valorizing such waste materials not only contributes to resource efficiency but also reduces environmental pollution and supports sustainable production systems.

To achieve its objectives, the researchers employed a two-factor, three-level full factorial experimental design to systematically evaluate and optimize the extraction process. The independent variables considered were the liquid-to-solid (L/S) ratio and extraction temperature, both of which are known to significantly influence mass transfer and solute diffusion during extraction processes. The dependent variable was the total carotenoid content (TCC), expressed in micrograms of β -carotene per gram of dried carrot peel. The experimental procedure involved freeze-drying carrot peels to preserve bioactive compounds, followed by pulverization to increase surface area and enhance solvent penetration. The prepared samples were then mixed with soybean oil under controlled conditions and subjected to agitation using a shaking water

bath. This setup ensured consistent contact between the solvent and plant material, facilitating efficient extraction of carotenoids into the oil phase.

Quantitative analysis of carotenoid content was performed using UV-Visible spectrophotometry at 450 nm, applying the Beer-Lambert law for concentration determination. The study also implemented rigorous statistical tools to evaluate the reliability and validity of the experimental model. These included analysis of variance (ANOVA), regression modeling, and diagnostic tests such as the Shapiro-Wilk test for normality, Durbin-Watson test for autocorrelation, Cook's distance for identifying influential outliers, and Variance Inflation Factor (VIF) for detecting multicollinearity. Additionally, model performance was assessed using R^2 , adjusted R^2 , and predicted R^2 values to ensure both accuracy and predictive capability.

The experimental results demonstrated that both the liquid-to-solid ratio and temperature significantly influenced carotenoid extraction efficiency, although their effects were not equal in magnitude. The highest observed total carotenoid content was $84.22 \pm 5.56 \mu\text{g } \beta\text{-carotene/g}$ of dried peel, achieved at a liquid-to-solid ratio of 40 mL/g and a temperature of 45 °C. This result indicated that increasing the solvent proportion relative to the biomass significantly enhances extraction efficiency by improving concentration gradients and promoting better diffusion of carotenoids from the plant matrix into the oil solvent. In contrast, temperature exhibited a positive but comparatively weaker effect on extraction performance, suggesting that thermal enhancement plays a supporting role rather than a dominant one in the process.

Statistical analysis confirmed that the overall model was highly significant ($p < 0.0001$), with a strong coefficient of determination ($R^2 = 0.95$), indicating that the model effectively explained most of the variability in carotenoid recovery. The adjusted R^2 (0.94) and predicted R^2

(0.93) values further supported the robustness and predictive reliability of the model. However, a statistically significant lack of fit was also observed, suggesting that certain complex interactions or nonlinear behaviors within the system were not fully captured by the model. This indicates that while the model is strong in predictive capability, there remains room for refinement to improve its descriptive accuracy.

Through response surface methodology optimization, the study identified the most favorable extraction conditions as a liquid-to-solid ratio of 40 mL/g and a temperature of 30 °C. Under these conditions, the predicted carotenoid yield was 77.49 ± 13.13 µg β-carotene/g of dried peel. To validate the model, experimental runs were conducted under the same optimized conditions, yielding an actual average value of 76.04 ± 3.71 µg β-carotene/g. The small deviation between predicted and experimental values, reflected in a low relative error, confirmed the reliability and practical applicability of the developed model. This close agreement demonstrates that the optimization approach used in the study is effective for predicting carotenoid extraction performance under varying process conditions.

Further analysis revealed that the liquid-to-solid ratio was the most influential factor affecting carotenoid recovery. This can be attributed to its direct effect on mass transfer dynamics. A higher solvent volume increases the concentration gradient between the solid matrix and the liquid phase, thereby enhancing diffusion rates and preventing early saturation of the solvent. This allows more carotenoids to be released from the plant material into the soybean oil. While temperature also contributes by improving solubility and weakening cellular structures, its effect becomes limited once the system approaches equilibrium or when saturation conditions are reached. In oil-based extraction systems, this suggests that solvent quantity plays a more critical role than thermal energy in determining overall extraction efficiency.

The study also compared its findings with previous research involving other biomass sources such as pumpkin, peach, and carrot byproducts. These comparisons showed consistent trends, where higher solvent-to-solid ratios generally resulted in improved carotenoid recovery.

However, excessive temperature increases were often associated with reduced efficiency due to possible degradation of heat-sensitive carotenoid compounds. This reinforces the idea that moderate temperature conditions combined with optimized solvent ratios are ideal for maximizing yield in green extraction systems.

In addition to extraction efficiency, the study conducted a sustainability evaluation using the Path2Green assessment framework, which measures the environmental performance of extraction processes across 12 standardized principles. These principles include biomass sourcing, transport, pretreatment, solvent selection, scaling, purification, energy use, and waste management. The overall Path2Green score obtained was 0.593 out of a possible maximum of 1.00, indicating a moderately sustainable and environmentally friendly process.

The sustainability assessment revealed that the process performed strongly in several key areas. High scores were recorded for biomass utilization, as the study effectively repurposed agricultural waste rather than relying on freshly cultivated crops. Similarly, solvent selection scored highly due to the use of soybean oil, which is biodegradable, non-toxic, non-volatile, and safe for food-related applications. The process also demonstrated strong performance in scalability, purification, post-treatment, application potential, and waste management, as the extracted product could be directly utilized without requiring extensive purification or solvent removal.

However, certain limitations were identified in the sustainability analysis. Energy consumption received a low score due to the use of freeze-drying, grinding, and controlled temperature water bath systems, all of which require significant energy input. Pretreatment processes also contributed to reduced sustainability performance because mechanical processing steps were necessary to prepare the biomass for extraction. Additionally, yield efficiency was identified as an area for improvement, indicating that the extraction process was not fully exhaustive and that some carotenoids likely remained unextracted in the residual biomass. The study also noted the absence of a closed-loop extraction system, which limited its circular economy potential.

Despite these limitations, the study highlighted that soybean oil-based extraction remains a promising green alternative to conventional solvent systems. The integration of environmentally friendly solvents with waste valorization strategies demonstrates a strong alignment with sustainable development goals and green chemistry principles. Furthermore, the extracted carotenoid-rich oil can potentially be applied directly in food formulations, cosmetics, pharmaceuticals, and nutraceutical products without requiring further purification, thereby reducing processing costs and environmental burden.

In conclusion, Palero et al. (2026) successfully demonstrated that soybean oil is an effective and sustainable solvent for carotenoid extraction from carrot peel waste. The optimized extraction conditions provided high yields with strong model reliability, while the sustainability assessment confirmed its moderate environmental performance. Although improvements are still needed in energy efficiency, extraction completeness, and system integration, the study provides strong evidence that agricultural waste can be effectively transformed into high-value bioactive compounds using green extraction technologies. This research contributes significantly to the

growing field of sustainable bioprocessing and supports the development of eco-friendly industrial practices that prioritize both efficiency and environmental responsibility.

-A study by **Manalo (2021)** discussed the role of Digital Agriculture in supporting the modernization and industrialization goals of the Philippine agricultural sector through the adoption of Information and Communications Technologies (ICTs). The study highlighted how digital technologies have become important tools in improving agricultural productivity, efficiency, and access to information. Various ICT-based innovations have already been utilized in agriculture, including web-based disease diagnosis systems, crop monitoring sensors, mobile applications, and digital trading platforms. In the Philippines, DA-PhilRice developed several ICT-driven applications such as the Leaf Color Chart (LCC) app, e-binhi, e-damuhan, and AgriDoc, which provide farmers with timely and relevant agricultural information. These technologies were designed to support better crop management, improve decision-making, and enhance overall agricultural operations. The study emphasized that digital technologies can help farmers access critical information related to crop production, nutrient management, pest control, and market opportunities, enabling them to make informed decisions and improve farm performance. Through these innovations, Digital Agriculture has been recognized as a significant strategy in transforming traditional farming practices into more efficient, data-driven, and technology-enabled systems.

The study further examined the level of ICT adoption among Filipino farmers and identified several challenges that hinder the effective use of digital technologies. Based on the results of the 2016–2017 Rice-Based Farm Households Survey, approximately 93 percent of

rice-farming households had access to mobile phones, smartphones, and internet services. Similarly, a large percentage of farmers expressed willingness to receive agricultural information through text messaging and internet-based platforms. Despite this high level of access, actual utilization of ICTs for farming activities remained relatively low. The survey revealed that only 31 percent of farmers used text messaging as a source of agricultural information, while only 21 percent actively searched for farming-related information online. These findings suggest that access to technology alone does not guarantee its effective use. The study argued that several factors contribute to this gap, including low levels of digital literacy, limited exposure to technology, lack of confidence in using digital tools, and ICT anxiety among farmers.

According to Manalo (2021), ICT anxiety is one of the major barriers to technology adoption in agriculture. ICT anxiety refers to the discomfort, fear, or uncertainty experienced by individuals when interacting with computers, mobile devices, or other digital technologies. This challenge is particularly common among older populations, including many Filipino farmers. Since a significant portion of the agricultural workforce in the Philippines consists of aging farmers, the adoption of new technologies often becomes difficult. The study explained that limited experience with digital tools and insufficient technical knowledge can discourage farmers from utilizing ICT-based solutions, even when such technologies are readily available. As a result, many farmers continue to rely on traditional methods of obtaining information, such as consulting fellow farmers, agricultural extension workers, and local community networks. This preference for interpersonal communication demonstrates that technological innovations must be accompanied by strategies that address behavioral and educational barriers to ensure successful implementation.

To address these challenges, the study introduced the concept of infomediaries as a means of bridging the gap between technology and farmers. Infomediaries are individuals who facilitate access to information by helping others use digital tools and communication technologies. The study described infomediaries as important agents in reducing the digital divide, particularly among populations that have limited technological skills. The digital divide refers to the disparity between individuals who can effectively access and utilize digital technologies and those who cannot. This divide is influenced by various factors, including educational attainment, socioeconomic status, technological literacy, and access to resources. By mobilizing infomediaries, agricultural organizations can help ensure that the benefits of digital technologies reach a wider population of farmers, including those who may be hesitant or unable to use ICTs independently.

The study highlighted the implementation of the Infomediary Campaign by DA-PhilRice from 2012 to 2017 as an innovative approach to promoting digital inclusion in agriculture. The campaign engaged high school students from rice-farming communities and encouraged them to serve as infomediaries for their parents and local farmers. These students accessed agricultural information through digital platforms and shared the information with farmers who lacked the skills or confidence to use ICT tools themselves. Through this initiative, young people became active participants in agricultural development while helping bridge the information gap within their communities. The campaign demonstrated how youth engagement can contribute to the dissemination of agricultural knowledge and encourage greater acceptance of digital technologies among farmers.

Results of the Infomediary Campaign showed positive outcomes in promoting information access and technology adoption. The study reported that more than 4,000

student-infomediaries submitted nearly 20,000 inquiries to the PhilRice Text Center between 2012 and 2016. These inquiries covered a wide range of agricultural topics, including rice varieties, pest management, nutrient management, crop establishment, and general farming practices. Students not only gathered information but also actively shared their knowledge with their parents, relatives, and other members of the farming community. The study observed that information dissemination often extended beyond the immediate recipients, creating a multiplier effect where agricultural knowledge was continuously shared among farmers. This process contributed to improved awareness of modern farming techniques and increased utilization of available information resources.

Furthermore, the study found evidence that the campaign positively influenced farmers' information-seeking behavior. Farmers who were exposed to the initiative became more likely to use text messaging services and digital communication platforms to obtain agricultural information. In some communities, the habit of consulting digital information sources continued even after the campaign ended. These findings suggest that targeted interventions and support systems can help overcome barriers to technology adoption and encourage the long-term use of ICT-based agricultural solutions. The study emphasized that successful digital transformation requires not only technological infrastructure but also educational programs, capacity-building initiatives, and community support mechanisms that enable users to maximize the benefits of digital tools.

The study concluded that Digital Agriculture can only achieve its full potential if efforts are made to ensure inclusivity and accessibility among all stakeholders. While digital technologies offer significant opportunities for improving productivity, information management, and decision-making, their effectiveness depends largely on the willingness and ability of users

to adopt them. Initiatives such as the Infomediary Campaign demonstrate that human-centered approaches can play a critical role in facilitating technology adoption and reducing barriers associated with digital literacy and ICT anxiety. The study recommended expanding infomediary programs, strengthening ICT training initiatives, integrating agriculture-related digital skills into educational curricula, and providing continuous support to both farmers and agricultural extension workers.

This literature is relevant to the present study as it demonstrates the significant role of digital technologies in improving operational efficiency, information management, and data-driven decision-making. Similar to the proposed system for real-time yield analytics and extraction efficiency monitoring, the study highlights how digital solutions can enhance monitoring processes, facilitate timely access to information, and support more effective management practices. The findings support the development of digital production systems by emphasizing the value of technology in collecting, analyzing, and utilizing data to improve productivity and operational performance. Furthermore, the study reinforces the importance of user adoption and accessibility in ensuring the successful implementation of digital innovations within agricultural and food-processing industries

-A study by **Barroga, Rola, Depositario, Digal, and Pabuayon (2019)** investigated the determinants of the extent of technological innovation adoption among micro, small, and medium food processing enterprises (MSMFEs) in the Davao Region, Philippines. The research focused on enterprises that were beneficiaries of the Department of Science and Technology–Small Enterprises Technology Upgrading Program (DOST-SETUP), a government

initiative that provides technological support, upgraded equipment, and technical assistance to improve productivity and competitiveness in the food processing sector. The study aimed to determine not only whether MSMFEs adopt technological innovations but also the degree or extent to which these innovations are implemented within their operations.

The researchers utilized a census of 52 MSMFEs in Davao Region and employed both quantitative and qualitative methods to analyze the data. Specifically, an ordered logistic regression model was used to identify the factors influencing the level of technological innovation adoption. The study also used index construction to measure the extent of adoption across multiple technological interventions, including food safety training, Good Manufacturing Practices (GMP) assessment, plant layout improvement, packaging and labeling, laboratory testing, product development, energy audit, cleaner production technology (CPT) consultancy, MPEX consultancy, and technology training. These innovations represent a comprehensive package of technical and managerial upgrades introduced through the SETUP program.

Findings of the study showed that MSMFEs in the Davao Region generally exhibited varying levels of technological innovation adoption, ranging from non-adoption to full adoption. A small portion of the enterprises failed to adopt any of the technological innovations provided, while others demonstrated low to moderate adoption levels. However, the majority of MSMFEs were classified as highly adopters, indicating that most enterprises were able to integrate several SETUP innovations into their operations. Only a smaller segment of the sample achieved full adoption, meaning they implemented nearly all of the technological innovations offered by the program.

Among the specific innovations introduced by SETUP, food safety training emerged as the most widely adopted intervention, followed by GMP assessment and MPEX consultancy. These innovations were particularly important for MSMFEs because they directly address compliance with food safety standards and operational efficiency. On the other hand, innovations such as product development and cleaner production technology consultancy were among the least adopted. The study noted that more advanced or resource-demanding innovations tended to have lower adoption rates, especially among smaller enterprises with limited capacity and technical expertise.

The study further highlighted that the adoption of technological innovations does not only depend on whether enterprises accept or reject a particular technology, but also on the depth and breadth of implementation across various business functions. This means that even if MSMFEs adopt certain innovations, the level of integration into their production processes, management systems, and product development activities may differ significantly. Based on this, the researchers categorized MSMFEs into five groups according to their extent of adoption: did not adopt, less adopted, moderately adopted, highly adopted, and fully adopted.

In terms of distribution, most medium-sized enterprises were found to have lower levels of adoption compared to micro and small enterprises. Micro enterprises, despite their limited resources, showed a relatively higher tendency toward adopting multiple technological innovations. Small enterprises were mostly classified as moderate to high adopters, depending on their exposure to export markets and access to external support services. The study suggests that enterprise size alone does not fully determine innovation adoption, but interacts with other organizational and contextual factors.

One of the key findings of the study is that the combination of food safety training, GMP assessment, and MPEX consultancy was the most commonly adopted set of innovations. These three are closely interconnected, as MPEX consultancy often recommends improvements related to GMP compliance and food safety practices. This indicates that MSMFEs tend to adopt innovations in clusters rather than individually, especially when the technologies are complementary and collectively improve production efficiency and compliance with regulatory standards.

The study also revealed that full adoption of technological innovations was relatively limited, with only about 21% of MSMFEs achieving this level. Enterprises under this category were able to implement most or all of the SETUP-supported innovations, demonstrating strong organizational capacity and readiness for technological upgrading. Micro enterprises in this category adopted nearly all available innovations, while small enterprises typically adopted a slightly smaller set, often excluding more advanced innovations such as laboratory analysis or product development.

Highly adopted MSMFEs represented a significant portion of the sample. These enterprises generally implemented key foundational innovations such as GMP assessment and food safety training, along with selected productivity-enhancing interventions. However, they still lacked full integration of more advanced technological services. This suggests that while these enterprises are relatively progressive in adopting innovation, constraints such as cost, technical expertise, and organizational capacity still limit full adoption.

Moderately adopted MSMFEs showed partial integration of SETUP innovations, often focusing on basic operational improvements. These enterprises typically adopted essential

interventions such as food safety training and plant layout improvements but did not fully engage in more complex innovations like laboratory analysis or product development. Their level of adoption indicates an intermediate stage of technological readiness, where awareness and initial implementation exist but full assimilation remains limited.

Less-adopted MSMFEs were characterized by minimal engagement with SETUP innovations, typically adopting only a few interventions such as technology training or food safety training. In this category, medium enterprises were more dominant, suggesting that some established firms may perceive less need for external technological assistance due to existing internal capabilities. This group highlights that adoption is not always higher among larger firms, especially when they already possess in-house technical expertise.

At the lowest level, the study identified a very small number of enterprises that did not adopt any technological innovations at all. These cases were linked to external shocks such as natural disasters and management issues, which disrupted operations and prevented the proper utilization of SETUP assistance. These findings emphasize that external environmental factors and governance issues can also significantly affect innovation adoption outcomes.

The study further examined the determinants of technological innovation adoption using an ordered logistic regression model. The results identified four statistically significant factors influencing the extent of adoption: sex of the owner, educational attainment, business scale, and type of market. These variables were found to consistently influence whether MSMFEs adopt innovations at a higher or lower level.

In terms of sex of the owner, the study found that male-owned enterprises were more likely to achieve higher levels of technological innovation adoption compared to female-owned

enterprises. Although female entrepreneurs represented a larger proportion of MSMFE owners in the sample, male owners were more likely to engage in extensive adoption. This was attributed to differences in mobility, access to training opportunities, and exposure to external information networks, which may influence innovation-related decision-making.

Educational attainment was also found to have a positive relationship with the extent of adoption. MSMFE owners with higher educational levels were more likely to adopt a wider range of technological innovations. This is because education enhances the ability to understand technical information, evaluate innovation benefits, and manage the risks associated with technological change. As a result, more educated owners tend to be more open to upgrading their production systems and integrating new technologies into their operations.

Business scale, on the other hand, showed a negative relationship with adoption extent, indicating that micro enterprises were more likely to adopt innovations compared to small and medium enterprises. This finding suggests that smaller firms are more flexible and can implement changes more quickly due to simpler organizational structures and centralized decision-making. In contrast, larger firms may experience more complex internal processes that slow down innovation adoption.

The type of market was also identified as a significant determinant. MSMFEs that cater to both domestic and export markets were more likely to adopt technological innovations at higher levels. This is because export-oriented firms are required to comply with stricter international standards, which encourages them to adopt advanced technologies and quality control systems to remain competitive in global markets.

Other variables such as age, years of experience, organizational type, and sources of information were found to be statistically insignificant in influencing the extent of technological innovation adoption. This suggests that while these factors may have some descriptive influence, they do not significantly determine the level of adoption when analyzed together with stronger predictors such as education, gender, business scale, and market orientation.

Overall, the study concluded that technological innovation adoption among MSMFEs in the Davao Region is a multi-dimensional process influenced by both individual and organizational factors. The findings highlight that innovation adoption is not simply a matter of access to technology, but also depends on the characteristics of decision-makers and the market environment in which enterprises operate.

The study is relevant to the present research because it provides empirical evidence on how personal demographics and organizational characteristics influence the extent of technological innovation adoption in food processing enterprises. It also highlights the importance of government support programs such as DOST-SETUP in promoting innovation among MSMEs, particularly in developing regions.

A study by **Ang, Amongo, Paras Jr., Suministrado, and Peralta (2022)**, entitled *Development of Banana Peduncle Juice Extractor for Ethanol and Fiber Production*, focused on the conceptualization, design, fabrication, and performance evaluation of a specialized mechanical system intended for the efficient processing of banana (*Musa acuminata* C.) peduncles. The research was motivated by the increasing concern regarding agricultural waste management in the Philippines, particularly within large-scale banana plantations where peduncles are typically discarded despite their significant potential as raw materials for

bioethanol production and natural fiber utilization. The authors emphasized that although banana peduncles contain valuable lignocellulosic components and fermentable sugars, their utilization remains limited due to the lack of appropriate extraction technology that can simultaneously maximize juice recovery while preserving fiber integrity. Conventional extraction machines, originally designed for other agricultural commodities such as sugarcane, were found to be inefficient when applied to banana peduncles due to differences in material structure, density, and mechanical resistance. This gap in technology prompted the development of a dedicated juice extraction system tailored specifically for banana peduncle processing.

The developed machine was designed using engineering principles that prioritized functionality, manufacturability, safety, and cost-effectiveness. The researchers deliberately utilized locally available materials in order to ensure affordability and ease of replication, particularly for potential use in rural agricultural communities and small-scale processing facilities. The system employed a crushing roll mechanism as the primary extraction component, consisting of a main roll and multiple feeder rolls arranged in a configuration that enables controlled compression of the peduncle material. This configuration was intended to optimize juice release while minimizing excessive mechanical damage to the fiber structure, thereby allowing the biomass to be utilized for both ethanol fermentation and fiber recovery applications. The structural design also incorporated protective enclosures and adjustable components to enhance operational safety and accommodate varying operator conditions. In addition, the transmission system was engineered using a combination of engine-driven pulleys, gear reducers, and chain-sprocket assemblies, allowing controlled variation of rotational speed to evaluate performance under different operational settings.

To assess the effectiveness of the developed machine, the researchers employed a systematic experimental design using a two-factor, three-level full factorial approach. The two independent variables considered were roll rotational speed (50 rpm, 70 rpm, and 90 rpm) and crushing load (0.8 kN, 1.2 kN, and 1.6 kN). A total of 27 experimental runs were conducted to determine the influence of these variables on machine performance. Each trial utilized approximately 40 kilograms of freshly harvested banana peduncle samples, which were processed individually to ensure accurate measurement of performance indicators. The study followed standardized testing procedures based on established agricultural machinery evaluation protocols, specifically PAES 235:2008 for multi-crop juice extractors and PAES 229:2005 for fiber decorticators. These standards ensured that the evaluation process was consistent, reliable, and comparable to other agricultural machinery studies.

The performance of the machine was evaluated using multiple response variables, including input capacity, juice extraction rate, juice recovery, extraction losses, extraction efficiency, fuel consumption, fiber extraction rate, fiber recovery, and fiber tensile strength. In addition, the °Brix level of the extracted juice was measured to determine sugar concentration, which is an important factor in ethanol production. The inclusion of both mechanical and biochemical indicators allowed for a comprehensive assessment of the system, covering not only productivity but also product quality. The tensile strength of the extracted fiber was also analyzed using a universal testing machine following ASTM D3822-01 standards to determine whether mechanical processing had adversely affected fiber integrity.

Results of the study showed that both roll rotational speed and crushing load significantly influenced machine performance, although their effects varied depending on the specific output variable. Higher roll rotational speeds generally resulted in increased input capacity, juice

extraction rate, and fiber extraction rate due to reduced processing time and faster material throughput within the crushing chamber. Conversely, lower rotational speeds tended to reduce throughput but improved certain recovery metrics due to longer retention time within the extraction system. Crushing load, on the other hand, exhibited a dual effect depending on the response variable. Lower crushing loads facilitated easier passage of peduncle material through the rolls, thereby increasing processing speed, while higher crushing loads enhanced the mechanical compression of the material, resulting in improved juice recovery and extraction efficiency.

The interaction between roll speed and crushing load revealed important trade-offs in system performance. For instance, although higher crushing loads improved juice recovery, they also increased resistance within the system, which reduced processing speed and increased fuel consumption. Similarly, higher rotational speeds improved throughput but contributed to higher extraction losses, as some juice was not fully recovered before the material exited the machine. These findings highlight the complexity of optimizing mechanical systems for biomass processing, where improvements in one performance aspect may negatively affect another. As such, the study emphasized the importance of multi-response optimization techniques to determine the most balanced operating conditions.

Using response surface methodology (RSM), the researchers performed a multi-objective optimization analysis to identify the most desirable combination of operating parameters. The optimization process considered multiple performance goals simultaneously, including maximizing input capacity, juice recovery, extraction efficiency, fiber extraction rate, and fiber recovery, while minimizing extraction losses and fuel consumption. Based on desirability function analysis, the optimal operating condition was identified at approximately 71.378 rpm

and 0.8 kN crushing load. At this optimal setting, the machine achieved an input capacity of about 307.24 kg/h, a juice extraction rate of 82.92 kg/h, juice recovery of 26.08%, extraction efficiency of 18.46%, fiber extraction rate of 54.83 kg/h, fiber recovery of 42.29%, and fuel consumption of 0.391 L/h. These results demonstrate that the system can achieve a balanced level of efficiency while maintaining both juice and fiber output quality when operated under optimized conditions.

Further analysis revealed that neither roll rotational speed nor crushing load significantly affected the °Brix content of the extracted juice or the tensile strength of the recovered fiber. This indicates that while the mechanical parameters influence the quantity of juice and fiber produced, they do not significantly alter the chemical composition of the juice or the structural strength of the fiber. This finding is particularly important because it suggests that the quality of the end products remains stable regardless of operational variations, making the system more reliable for industrial applications where consistency is essential.

In terms of fermentation performance, the extracted juice was subjected to ethanol production analysis using *Saccharomyces cerevisiae* as the fermentation agent. The results showed that the fermented slurry contained approximately 10–11% ethanol by volume across all treatment combinations. Statistical analysis indicated no significant differences in ethanol content among the different mechanical settings, reinforcing the observation that sugar concentration, rather than mechanical extraction conditions, is the primary determinant of ethanol yield. This further supports the conclusion that the machine primarily influences extraction efficiency and throughput rather than biochemical conversion outcomes.

In conclusion, the study demonstrated that the developed banana peduncle juice extractor is an effective machine for converting agricultural waste into valuable products such as bioethanol feedstock and natural fiber. The system successfully addressed the limitations of conventional extraction machines by introducing a design specifically optimized for banana peduncle characteristics. The findings highlight the importance of mechanical optimization in improving productivity while maintaining product quality. Moreover, the study underscores the potential of banana peduncle as a sustainable raw material for bioenergy and fiber-based industries, contributing to waste reduction, renewable energy development, and value-added agricultural processing. The authors further recommended future improvements such as modifying roll geometry, improving transmission efficiency, experimenting with alternative crushing mechanisms, and testing additional banana varieties to enhance the versatility and efficiency of the system.

-A **study by Jiang and Po (2023)** investigated the influence of human resource practices on employee retention in the food manufacturing industry in the Philippines and examined the moderating role of the work environment. The researchers focused on five key human resource practices: training and development, employee participation in decision-making, compensation, job security, and performance appraisal. Grounded in Social Exchange Theory, the study proposed that employees are more likely to remain with an organization when they perceive that the organization provides adequate support, fair treatment, and opportunities for growth. In return, employees reciprocate through commitment, loyalty, and long-term retention.

The study employed a quantitative research design using a cross-sectional survey method. Data were collected from 206 employees working in various food manufacturing companies in the Philippines through both online and offline questionnaires. The researchers used a five-point Likert scale to measure employee perceptions regarding human resource practices, work environment, and employee retention. Statistical analyses, including regression and moderation analysis, were conducted using JAMOV software to determine the relationships among the variables.

The findings revealed that employee participation in decision-making, compensation, job security, and performance appraisal significantly and positively influenced employee retention. Employees who were given opportunities to participate in organizational decisions tended to develop a stronger sense of belonging and commitment to their companies. Likewise, fair compensation and benefits were found to be major factors that encouraged employees to remain in their organizations. The results further indicated that job security contributed positively to retention because employees preferred stable employment arrangements that reduced uncertainty regarding their future careers. Performance appraisal was also shown to strengthen employee retention by providing feedback, recognition, and opportunities for professional improvement.

Among the identified human resource practices, compensation emerged as the strongest predictor of employee retention. The researchers found that employees were more likely to remain in organizations when they perceived that they were fairly compensated for their work and contributions. This finding emphasizes the importance of providing competitive salaries and benefits in order to attract and retain skilled employees in the food manufacturing sector. The study concluded that compensation plays a critical role in maintaining workforce stability and reducing employee turnover.

In contrast, the study found that training and development did not significantly influence employee retention. This result differed from several previous studies that suggested training opportunities increase employee loyalty and commitment. The researchers proposed that training programs may not automatically lead to retention unless employees perceive them as valuable, relevant, and beneficial to their career advancement. This finding highlights the need for organizations to carefully evaluate the effectiveness of their training and development initiatives and ensure that such programs align with employee needs and organizational objectives.

Another significant contribution of the study was its examination of the work environment as a moderating variable. The findings showed that the work environment strengthened the relationship between employee participation in decision-making, compensation, job security, performance appraisal, and employee retention. Employees working in supportive environments were more likely to respond positively to human resource practices and demonstrate stronger intentions to remain in the organization. The study emphasized that a positive work environment includes both physical and non-physical factors, such as workplace safety, adequate facilities, healthy interpersonal relationships, effective communication, and a supportive organizational culture.

The researchers further explained that improvements in the physical work environment, including proper ventilation, safe production equipment, reduced workplace noise, and clean facilities, can increase employee satisfaction and productivity. At the same time, non-physical aspects such as teamwork, leadership support, organizational structure, and employee relationships contribute significantly to employee motivation and retention. These findings suggest that organizations should not only focus on implementing effective human resource

policies but also invest in creating a work environment that promotes employee well-being and engagement.

The study has important implications for food manufacturing companies in the Philippines. It suggests that organizations should prioritize compensation systems, employee involvement, job security measures, and performance evaluation mechanisms to improve retention rates. Furthermore, managers should recognize the importance of maintaining a positive work environment to maximize the effectiveness of these human resource practices. By doing so, organizations can reduce employee turnover, improve workforce stability, and enhance overall organizational performance.

This literature is relevant to the present study because it demonstrates how organizational practices, performance monitoring, and workplace conditions influence efficiency, productivity, and long-term organizational success in the food manufacturing industry. The findings support the development of digital systems that provide real-time monitoring, performance tracking, and data-driven decision-making capabilities. Similar to the proposed system for real-time yield analytics and extraction efficiency monitoring, the study highlights the importance of accurate information, continuous monitoring, and effective management practices in improving operational performance and achieving organizational goals.

2.3 Related Literature

A study by Somsen, Capelle, and Tramper (2004) introduced Production Yield Analysis (PYA), a systematic method used to evaluate raw material efficiency in food processing. The study developed a “Yield Index” to distinguish between necessary losses and avoidable losses, helping

identify inefficiencies in production systems. The authors emphasized that relying solely on historical benchmarks may conceal actual waste, highlighting the importance of model-based analysis. This literature is relevant to the present study as it supports the development of a digital system capable of identifying and analyzing production losses in real time.

[6]A study by **Charbonnier, Béranger, and Cognon (2005)** investigated the application of trend extraction and analysis techniques for complex system monitoring and decision support in industrial and medical environments. The researchers focused on developing an online monitoring methodology capable of analyzing continuously changing data from highly complex systems such as industrial automation processes, food manufacturing operations, and intensive care monitoring systems. According to the researchers, modern industrial environments generate large volumes of operational data that are difficult for human operators to interpret manually, especially in situations where process conditions rapidly change and immediate decisions are required. The study emphasized that traditional monitoring approaches often rely heavily on manual observation and operator expertise, which may result in delayed responses, reduced efficiency, and increased risk of operational errors during abnormal conditions.

To address these challenges, the researchers proposed a real-time trend extraction methodology designed to analyze univariate time-series data using semi-qualitative representations. The system utilized three primary trend primitives: **Increasing, Decreasing, and Steady**. These primitives were used to represent operational behavior over specific time intervals, allowing users to interpret process conditions more effectively. According to the study, converting numerical signals into symbolic trend representations helps reduce cognitive overload and improves operator understanding of complex operational situations.

The methodology involved several stages. First, continuously generated process data were segmented into **linear intervals**. These intervals were then classified into temporal patterns such as increasing trends, decreasing trends, stable conditions, transient variations, and step changes. After classification, the detected temporal patterns were transformed into semi-quantitative episodes describing the behavior of operational variables over time. The researchers explained that this approach enabled the monitoring system to identify abnormal process behavior without requiring highly complex mathematical models of the entire industrial system.

The study also explored the application of trend extraction for predictive maintenance and operational supervision in industrial systems. The researchers explained that highly automated production systems often operate under demanding conditions to maximize productivity and profitability, increasing the likelihood of equipment degradation and operational faults. By continuously monitoring operational variables and analyzing abnormal trend patterns, the proposed methodology could assist in identifying equipment deterioration before critical failures occurred. The researchers concluded that predictive monitoring capabilities support preventive maintenance strategies, reduce operational downtime, and improve overall equipment reliability.

[7]A study by **Davies, Fried, and Gather (2003)** entitled “Robust Signal Extraction for On-line Monitoring Data” examined statistical methods for extracting meaningful information from continuously generated time-series data in real-time monitoring environments. The researchers focused on developing and evaluating **robust techniques** that can effectively separate true underlying signals from noise, outliers, and irregular fluctuations. The study was motivated by the increasing demand for reliable monitoring systems in critical environments such as

intensive care units and industrial process control systems, where fast and accurate interpretation of data is essential for decision-making.

The researchers explained that modern monitoring systems generate large volumes of data due to continuous recording of physiological or operational variables. In healthcare settings, for example, patient data such as heart rate, blood pressure, and oxygen levels are recorded at very short intervals, producing complex datasets that must be analyzed immediately. Similarly, industrial systems also generate high-frequency operational data that reflect machine performance, process efficiency, and production status. However, these datasets are often affected by noise, measurement errors, and external disturbances, which make it difficult to identify meaningful patterns using traditional analysis methods.

According to the study, one of the major challenges in online monitoring is distinguishing clinically or operationally relevant changes from irrelevant noise and outliers. The researchers emphasized that false alarms in monitoring systems are a significant problem, particularly in intensive care environments where excessive alarms may lead to alarm fatigue among medical staff. This reduces trust in automated systems and may cause important warnings to be ignored. In industrial environments, inaccurate signal interpretation may also lead to unnecessary maintenance actions or failure to detect real system faults. Because of these issues, there is a strong need for robust statistical methods that can reliably extract true signals from noisy data in real time.

Another important contribution of the study is its discussion on the trade-off between robustness and computational efficiency. The researchers emphasized that while highly robust methods provide better resistance to data contamination, they may not always be suitable for

real-time applications due to processing delays. Therefore, selecting an appropriate signal extraction method requires balancing accuracy, stability, and computational cost depending on the system requirements.

In conclusion, the study by Davies, Fried, and Gather (2003) provides valuable insights into the importance of robust statistical methods in real-time monitoring systems. Its emphasis on noise reduction, trend extraction, and reliable signal interpretation makes it highly applicable to modern industrial systems, particularly in the context of automated food production and digital manufacturing technologies.

[8]A study conducted by **Charmine Sheena R. Saflor and Marvin I. Noroña (2021)** explored the implementation of Total Productive Maintenance (TPM) within the Philippine rice production sector. The research focused on addressing the increasing operational and maintenance challenges experienced by farmers and agricultural cooperatives due to the growing use of mechanized farming equipment. The authors emphasized that many rice production operators in the Philippines still rely on reactive maintenance practices, where machines are repaired only after breakdowns occur. This approach often results in frequent equipment failures, increased maintenance expenses, production delays, and reduced overall productivity. The study highlighted that inadequate maintenance management can shorten the lifespan of agricultural machinery and negatively affect the efficiency of rice production operations.

The study is relevant to the present research because it demonstrates the importance of monitoring equipment performance and maintenance activities to improve operational efficiency and reduce production losses. It supports the idea that implementing a digital and data-driven

monitoring system can help organizations identify inefficiencies, manage maintenance schedules, and enhance productivity in real time. Similar to the proposed study on real-time yield analytics and extraction efficiency monitoring, the research emphasizes the value of systematic analysis and continuous monitoring in optimizing production processes, minimizing waste, and improving decision-making within industrial and agricultural operations.

[9]A study conducted by **Desai Shivani Sarjerao (2020)**, Design, Development and Performance Evaluation of Juice Extractor, presented the development of a hydraulically operated juice extraction machine intended for processing pomegranate arils and bottle gourd.

The experimental evaluation focused on three main independent variables: size reduction (shred size), applied hydraulic pressure, and filter bag pore size. These were tested in different combinations to determine their effects on several dependent performance indicators, namely juice yield, juice extraction efficiency, extraction losses, machine capacity, and sedimentation percentage. A series of experimental trials were conducted for both pomegranate arils and bottle gourd, with repeated measurements used to ensure reliability of results. Statistical analysis, including **ANOVA**, was applied to determine the significance of main effects and interaction effects among the variables.

In terms of juice extraction efficiency, the study reported maximum values of 94.30% for pomegranate and 80.27% for bottle gourd under optimal operating conditions. The results also showed a consistent decrease in extraction losses as pressure increased and shred size decreased, demonstrating improved material utilization and reduced wastage. Although most interaction

effects between variables were statistically non-significant, individual factors such as pressure and filter bag pore size were found to have a significant impact on extraction performance.

Overall, the study emphasizes that mechanical design parameters and operating conditions play a critical role in determining the efficiency of juice extraction systems. It further suggests that optimizing these variables can significantly enhance juice recovery, reduce processing losses, and improve machine productivity. This research is relevant to the present study as it provides an engineering-based framework for evaluating extraction systems and supports the development of optimized, efficiency-driven food processing technologies in agricultural production systems.

A study by **Bartoszewicz and Wdowicz (2019)** presented an automated reporting approach that integrates the SAP ERP Production Planning (PP) module with Microsoft Excel and Visual Basic for Applications (VBA). The researchers observed that while SAP ERP provides extensive support for production planning and operations, its standard reporting features may not always satisfy the analytical and reporting needs of production managers. To address this limitation, they developed the **Follow the Plan (FTP) Report**, an application that automatically retrieves production data from SAP ERP, processes the information, and generates reports comparing planned production activities with their actual execution. This automation reduces the need for manual data handling while improving the speed and consistency of production reporting.

The proposed system gathers production information from the **COOISPI** and **MB51** transactions within SAP ERP. These transactions contain data on production orders, work

centers, planned schedules, completed orders, and material movements. After extracting the required information, the application organizes and analyzes the data in Microsoft Excel, allowing users to determine whether production orders were completed as planned. The system also identifies production deviations, such as orders that were not started, orders produced beyond planned quantities, and production results that fall outside the accepted tolerance. In addition, users can record standardized root causes, including raw material shortages, equipment failures, low machine performance, quality issues, and inaccurate system data, to support further investigation and continuous process improvement.

The researchers found that automating the reporting process significantly improved operational efficiency. Preparing production compliance reports manually previously required approximately two hours because users had to retrieve, organize, and validate data from multiple SAP transactions. With the FTP Report application, the same process could be completed in about five minutes, while also reducing the possibility of human error. The study concluded that integrating ERP systems with automated reporting tools enhances production monitoring, supports faster managerial decision-making, and provides more reliable information for evaluating production performance. This literature is relevant to the present study because it demonstrates how digital technologies can automate production monitoring and reporting, supporting real-time analysis and more effective evaluation of production efficiency, which aligns with the objectives of developing a real-time yield analytics and extraction efficiency monitoring system for ALDi Foods Inc.

2.4 Synthesis of the Review Literature

The reviewed foreign and local literature consistently demonstrates that production systems in agriculture and food manufacturing are undergoing a significant transformation from traditional manual operations toward digitally enabled, data-driven, and automated environments. This transformation is largely driven by rapid advancements in Information and Communications Technology (ICT), artificial intelligence (AI), machine learning, big data analytics, and real-time monitoring systems. Collectively, these technologies enhance operational efficiency, improve production accuracy, and support faster, evidence-based decision-making. Across the studies reviewed, there is a clear consensus that digital integration is no longer optional but a necessary component for modern, competitive, and efficient production systems, particularly in industries involving food processing and extraction operations.

One of the most consistent findings across the literature is the growing importance of real-time monitoring systems and centralized digital platforms in production environments. Studies such as Verhoef et al. (2021), Makhmudah et al. (2025), and Bartoszewicz and Wdowicz (2019) emphasize that traditional paper-based recording systems and manual reporting processes are highly inefficient due to delays, data inconsistencies, and lack of immediate accessibility. These limitations prevent production managers from identifying inefficiencies, equipment malfunctions, and yield losses at the moment they occur. In contrast, digital dashboards, ERP-integrated systems, and automated reporting tools provide real-time visibility of production activities, allowing faster response to operational issues. This shift highlights that real-time data accessibility is a critical requirement for effective production monitoring systems such as the proposed yield and extraction efficiency monitoring system for ALDi Foods Inc.

Another major finding is the increasing role of data analytics, artificial intelligence, and intelligent computing systems in optimizing production processes. Studies by Siddique et al. (2025) and Sharma et al. (2021) show that AI-powered systems are capable of processing large and complex datasets to identify operational patterns, predict production outcomes, and optimize resource allocation. These systems support predictive, descriptive, and prescriptive analytics, enabling manufacturers to minimize waste, improve productivity, and enhance decision-making accuracy. Furthermore, technologies such as machine learning and computer vision are increasingly used to automate inspection, monitor production flow, and detect irregularities, reducing reliance on manual observation and improving operational reliability in industrial environments.

The literature also highlights the critical importance of production efficiency metrics, yield analysis, and performance evaluation systems in modern manufacturing. Mansour and Al-Hamdani (2024) introduced internal key performance indicators (KPIs) as tools for identifying production inefficiencies and defects, while Somsen et al. (2004) emphasized Production Yield Analysis (PYA) as a structured method for distinguishing between avoidable and unavoidable production losses. Similarly, Dora et al. (2020) found that most food losses occur during the production stage due to machine inefficiencies, human errors, poor process control, and inadequate operational management. These findings strongly indicate that continuous monitoring of yield and efficiency is essential in minimizing production losses, improving resource utilization, and ensuring consistent operational performance—core objectives of real-time production monitoring systems.

In food extraction and processing systems, multiple studies such as Desai (2020), Mphahlele et al. (2016), Aviara et al. (2020), and Ang et al. (2022) demonstrate that mechanical and

operational parameters significantly influence extraction yield, product quality, and production efficiency. Variables such as pressure, rotational speed, feed condition, and extraction technique directly affect both the quantity and quality of output. In addition, Mphahlele et al. (2016) further explain that extraction methods not only influence yield but also affect chemical composition, antioxidant content, aroma compounds, and overall product quality. These findings reinforce the idea that monitoring extraction processes is necessary not only for improving efficiency but also for ensuring product consistency and quality control in food manufacturing systems.

The reviewed literature also shows a growing adoption of automation technologies and intelligent monitoring systems in industrial environments. Sabih et al. (2023) demonstrated that computer vision and machine learning can effectively automate material flow monitoring, improving accuracy while reducing manual intervention. Similarly, ERP-based systems studied by Bartoszewicz and Wdowicz (2019) show that automated reporting systems significantly enhance production tracking efficiency, reduce human error, and support faster managerial decision-making. These studies collectively indicate a global shift toward integrated digital production systems that enable real-time monitoring, automated computation, and improved operational transparency.

Local studies further confirm the relevance and urgency of digital transformation in the Philippine context, particularly in agriculture and food manufacturing industries. Bagaforo and Albason (2024) and Barroga et al. (2019) found that while MSMEs benefit from technological adoption, they continue to face challenges such as limited technical expertise, financial constraints, and inadequate infrastructure. De Leon et al. (2017) emphasized that centralized digital systems significantly improve data reliability, production tracking, and operational efficiency in local enterprises. Meanwhile, Manalo (2021) highlighted that although ICT access

is widespread among Filipino farmers, actual utilization remains low due to digital literacy gaps, ICT anxiety, and lack of technical support. These findings suggest that the success of digital systems depends not only on availability but also on usability, accessibility, and user capability.

Collectively, the literature establishes that digital transformation, real-time monitoring, and data-driven analytics are essential components of modern production systems. This is particularly significant in food processing and extraction industries where efficiency, yield optimization, and waste reduction directly affect productivity and profitability. The findings strongly support the development of integrated systems that provide real-time data capture, automated yield computation, and extraction efficiency monitoring—such as the proposed system for AIDi Foods Inc.—to enhance operational performance and support data-driven decision-making in production environments.

Research Gaps Identified

Despite the extensive body of literature on digital transformation, production optimization, and intelligent manufacturing systems, several important gaps remain evident, particularly when applied to food processing and juice extraction operations.

First, there is a noticeable lack of research and system development focused specifically on real-time monitoring systems designed for juice extraction and similar food processing operations. Most existing studies concentrate on broader industrial manufacturing environments, agricultural ICT applications, or general automation systems. While these studies provide valuable insights into production optimization, they do not sufficiently address the unique requirements of extraction-based processes, where continuous measurement of input materials, juice yield, and extraction efficiency is critical. As a result, there is limited exploration of

systems tailored to monitor and evaluate performance in real-time within juice production environments such as ALDi Foods Inc.

Second, the current literature often treats key production functions—such as production monitoring, yield analysis, quality control, and reporting—as separate and disconnected systems. This fragmented approach creates inefficiencies in data handling and delays in decision-making, as information must be processed across multiple platforms or manual procedures. There is a clear absence of a unified and integrated system that consolidates these functions into a single real-time platform capable of automatically computing, analyzing, and presenting production performance indicators in a cohesive manner.

Third, many of the existing advanced solutions in production monitoring rely heavily on high-cost and technically complex technologies, including IoT-based sensor networks, machine learning algorithms, cloud computing infrastructures, and industrial automation systems. While these technologies are highly effective in large-scale manufacturing environments, they are often impractical for small and medium-sized enterprises (SMEs) due to financial constraints, lack of technical expertise, and limited infrastructure support. This creates a technological gap where SMEs in the food processing sector are unable to fully benefit from advanced production monitoring innovations.

Fourth, although local studies in the Philippines confirm the advantages of digital transformation in agriculture and food manufacturing, there is still a lack of localized, real-time production monitoring systems specifically developed for juice extraction industries. Most available systems or studies focus on agricultural production, MSME business processes, or general ICT adoption rather than actual operational monitoring within food processing plants. This gap is particularly

evident in companies such as ALDi Foods Inc., where production tracking is still largely dependent on manual or semi-digital recording systems.

Fifth, while modern production environments generate large volumes of operational data, there remains a significant gap in transforming raw production data into real-time, actionable insights. Many systems are capable of collecting data but lack intelligent mechanisms for immediate interpretation, such as automated yield computation, extraction efficiency evaluation, and instant performance feedback. This limitation reduces the usefulness of collected data and prevents production managers from making timely, data-driven decisions during ongoing operations.

Lastly, there is still limited research addressing the systematic transition from traditional manual recording systems to fully digital, real-time production monitoring systems, particularly in food processing industries. Many facilities continue to rely on paper-based logs, spreadsheets, or semi-manual recording methods, which are prone to human error, data loss, and delayed reporting. The literature lacks practical frameworks and implemented systems that guide SMEs in shifting toward fully integrated digital production monitoring environments without requiring high-cost industrial infrastructure.

Overall, these gaps highlight the need for a cost-effective, integrated, and real-time production monitoring system that is specifically designed for juice extraction processes. The proposed study, *“Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc.”*, directly addresses these gaps by developing a centralized system that enables real-time tracking of production input, output, yield computation, and extraction efficiency analysis, thereby supporting improved operational efficiency and data-driven decision-making in food processing operations.

How the Current Project Addresses the Gaps

The proposed study, “*Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AIDi Foods Inc.*”, directly addresses the research gaps identified in the reviewed literature through the development of a centralized, digital, and real-time production monitoring system specifically designed for juice extraction operations. The system aims to replace traditional paper-based production recording with a more efficient digital platform that enables accurate data collection, automated computation, and timely access to production information. By providing production managers with organized and up-to-date operational data, the system supports more effective production management and data-driven decision-making.

First, the proposed system specifically addresses the lack of real-time monitoring solutions developed for juice extraction and food processing operations. While many existing studies focus on general manufacturing automation, agricultural information systems, or industrial monitoring applications, few concentrate on the unique production requirements of juice extraction. The proposed system is designed to monitor critical production variables such as raw material input, extracted juice output, packed yield, yield percentage, and extraction efficiency. These indicators are essential in evaluating production performance and identifying areas where operational improvements can be made. By focusing on extraction-specific performance measures, the study contributes practical knowledge to food processing research while addressing the operational needs of AIDi Foods Inc.

Second, the proposed system addresses the fragmented nature of existing production monitoring approaches by integrating multiple production management functions into a single centralized platform. Instead of maintaining separate records for production monitoring, yield computation,

efficiency analysis, and report generation, the system consolidates these processes into one digital environment. Production data entered into the system are automatically processed to generate yield percentages, extraction efficiency values, and production summaries. This integration reduces redundant work, minimizes inconsistencies caused by manual computation, and provides production managers with a comprehensive view of operational performance through a single interface.

Third, the proposed study offers a cost-effective and practical digital solution that is suitable for small and medium-sized food manufacturing enterprises. Unlike many advanced production monitoring systems discussed in the literature that require IoT devices, sophisticated sensors, machine learning algorithms, or expensive industrial automation equipment, the proposed system emphasizes centralized data management, automated calculations, and real-time reporting using readily available digital technologies. This approach makes the system more feasible for organizations with limited financial and technical resources while still providing significant improvements in production monitoring and operational efficiency.

Fourth, the proposed system improves the utilization of production data by transforming manually recorded operational information into real-time analytical insights. Rather than simply storing production records for documentation purposes, the system automatically computes key production indicators and presents them through organized reports and dashboards. This allows production managers to monitor production progress continuously, identify abnormal production trends, detect extraction losses, and evaluate operational performance while production activities are ongoing. As a result, corrective actions can be implemented more quickly, reducing delays associated with end-of-day or end-of-batch manual reporting.

Fifth, the study contributes to the digital transformation of food manufacturing operations within the Philippine context. Although previous local studies have demonstrated the importance of digital technologies in agriculture and manufacturing, relatively few have developed practical production monitoring systems specifically intended for food processing companies. By implementing a real-time yield analytics and extraction efficiency monitoring system at ALDi Foods Inc., the study provides a localized example of how digital technologies can improve production management, support operational transparency, and enhance overall manufacturing efficiency. The proposed system may also serve as a reference model for other small and medium-sized food processing enterprises that seek to modernize their production monitoring practices.

Lastly, the proposed system converts raw production data into meaningful and actionable information that supports continuous process improvement. Through automated computation of yield percentages, extraction efficiency, raw material utilization, and production output, the system enables managers to evaluate production performance objectively using measurable indicators. These real-time analytics facilitate evidence-based decision-making, improve resource utilization, reduce production losses, and enhance overall operational efficiency. By providing timely and accurate production information, the system supports continuous monitoring and process optimization, ultimately contributing to improved productivity, better product consistency, and more effective management of juice extraction operations at ALDi Foods Inc.

Overall, the proposed study responds directly to the limitations identified in the reviewed literature by developing an integrated, practical, and real-time production monitoring system specifically tailored to the operational requirements of juice extraction. Through digitalization,

automated yield analytics, extraction efficiency computation, and centralized reporting, the system supports the modernization of production management practices while promoting greater operational efficiency, improved resource utilization, and informed decision-making in food processing environments.

A study by Mphahlele, Fawole, Mkwena, and Opara (2016) investigated the influence of different extraction methods on the chemical composition, bioactive compounds, volatile compounds, antioxidant properties, and overall quality of pomegranate (*Punica granatum* L. cv. Wonderful) juice. The researchers emphasized that the quality of fruit juice is not determined solely by the characteristics of the fruit itself but is also greatly affected by the method used during juice extraction. Since consumers increasingly prefer fruit juices that retain their natural nutritional value and health-promoting compounds, identifying extraction methods that maximize juice yield while preserving important chemical and nutritional properties has become an important area of research. The study was conducted to determine how different extraction techniques influence the final composition and quality of pomegranate juice and to provide scientific evidence that could assist food processors in selecting the most appropriate extraction method.

To achieve this objective, the researchers collected commercially harvested pomegranate fruits of the cultivar 'Wonderful' from South Africa and processed them using four different extraction methods. These methods included extracting juice from the arils without crushing the seeds, blending the arils together with the seeds, pressing the whole fruit, and extracting juice from

halved fruits using a commercial hand press. The resulting juice samples were analyzed for several quality parameters, including total soluble solids, titratable acidity, pH, juice yield, sugar composition, organic acid content, phenolic compounds, flavonoids, anthocyanins, antioxidant activity, and volatile organic compounds responsible for aroma. Statistical analyses, including analysis of variance and principal component analysis, were also performed to determine whether significant differences existed among the extraction methods and to identify which technique produced the most desirable juice characteristics.

The results demonstrated that the extraction method significantly influenced many of the physicochemical properties of pomegranate juice. Although the total soluble solids remained relatively similar across all extraction methods, noticeable differences were observed in acidity, pH, juice yield, and color. Juice extracted from halved fruits exhibited higher titratable acidity and lower pH, indicating a more acidic product. The researchers explained that lower pH levels may contribute to better microbial stability because acidic environments inhibit the growth of many spoilage microorganisms. Differences were also observed in juice yield, with extraction methods producing varying amounts of juice depending on how effectively the fruit tissues were disrupted during processing. Juice extracted from halved fruits produced the highest juice yield, while pressing whole fruits resulted in the lowest yield because some juice remained trapped inside the fruit tissues during extraction. These findings demonstrated that extraction efficiency is directly affected by the extraction technique employed during processing.

The researchers also examined the sugar and organic acid composition of the juice produced by each extraction method. Glucose and fructose were identified as the major sugars present in all samples, while citric acid was found to be the predominant organic acid. Although sugar concentrations showed only minor differences among the extraction methods, variations in

organic acid content were more pronounced. Juice obtained from the arils contained relatively higher citric acid levels, whereas blending the arils with the seeds produced lower concentrations because the seed components diluted the juice composition. The study emphasized that organic acids play an important role in determining the taste, flavor, and stability of fruit juice, making their accurate evaluation essential for maintaining product quality.

In addition to evaluating basic chemical properties, the researchers investigated the effects of extraction methods on bioactive compounds that contribute to the nutritional value of pomegranate juice. Several flavonoids, including catechin, epicatechin, and rutin, together with gallic acid, were identified in all juice samples. Catechin was the most abundant flavonoid regardless of the extraction method, while gallic acid was the predominant phenolic acid. However, the concentration of these compounds varied depending on the extraction technique. Blending the arils together with the seeds produced higher catechin concentrations because phenolic compounds were also extracted from the seeds. On the other hand, juice extracted from halved fruits contained higher concentrations of several beneficial phenolic compounds because the pressing process allowed compounds from the peel and internal membranes to become incorporated into the juice. These findings demonstrate that the fruit fractions included during extraction significantly influence the nutritional composition of the final product.

The study further analyzed anthocyanins, which are natural pigments responsible for the characteristic red color of pomegranate juice. Eight individual anthocyanin compounds were identified, with cyanidin-3,5-diglucoside representing the dominant pigment in all juice samples. Although the same anthocyanin compounds were detected across all extraction methods, their concentrations differed depending on the extraction technique. Juice extracted from whole fruits exhibited relatively higher total anthocyanin content than several of the other extraction methods.

According to the researchers, extraction conditions such as acidity and pressing pressure may influence anthocyanin stability and concentration, ultimately affecting the visual appearance and consumer acceptability of the juice.

The antioxidant properties of the juice were also evaluated using radical scavenging activity and ferric reducing antioxidant power assays. The findings revealed that juice extracted from halved fruits consistently exhibited the highest antioxidant activity among all extraction methods. This increase was closely associated with higher concentrations of total phenolic compounds, tannins, and vitamin C present in the juice. Statistical analysis showed a positive relationship between phenolic content and antioxidant activity, indicating that phenolic compounds are major contributors to the antioxidant capacity of pomegranate juice. These results suggest that extraction methods capable of incorporating beneficial compounds from additional fruit tissues may improve the nutritional value and functional properties of the final product.

Furthermore, the researchers investigated the volatile organic compounds responsible for the aroma and flavor of pomegranate juice. Ten volatile compounds belonging to different chemical groups, including esters, ketones, alcohols, and terpenes, were identified. Ethyl acetate and 3-octanone were the dominant volatile compounds detected in all juice samples. However, their relative concentrations differed according to the extraction method, demonstrating that processing conditions influence aroma development as well as chemical composition. Juice produced by blending arils together with seeds contained the highest overall concentration of volatile compounds, suggesting that including additional fruit tissues during extraction may enhance aroma intensity. Since aroma is one of the primary quality attributes evaluated by consumers, these findings further highlight the importance of selecting appropriate extraction techniques during juice processing.

To better understand the relationships among the measured quality parameters, the researchers performed principal component analysis. The statistical analysis successfully distinguished the juice samples according to their extraction methods based on differences in chemical composition, antioxidant activity, bioactive compounds, sugars, organic acids, and volatile compounds. This finding confirmed that each extraction technique produced a unique quality profile. The researchers concluded that extraction methods influence multiple quality characteristics simultaneously rather than affecting only a single property, emphasizing the importance of considering overall product quality when selecting processing techniques.

Overall, the study concluded that extraction method is a critical factor influencing juice yield, chemical composition, nutritional value, antioxidant capacity, aroma, and overall product quality. The findings demonstrated that selecting an appropriate extraction technique can improve production efficiency while preserving valuable bioactive compounds that contribute to the health benefits and sensory characteristics of pomegranate juice. The study also showed that monitoring extraction conditions allows food manufacturers to optimize both product quality and processing performance.

This literature is relevant to the present study because it demonstrates that extraction methods directly affect production yield, extraction efficiency, and product quality. The findings support the importance of continuously monitoring production parameters to identify variations that influence output and resource utilization. Likewise, the proposed system, *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc.*, aims to provide real-time monitoring and analysis of production yield and extraction efficiency, enabling manufacturers to evaluate processing performance, minimize production losses, improve

operational efficiency, and make data-driven decisions that enhance overall productivity and product quality.

A study by **Aviara, Lawal, Nyam, and Bamisaye (2020)** focused on the design, construction, and performance evaluation of a multi-fruit juice extractor intended for small- and medium-scale fruit processing operations. The researchers developed the machine to overcome the limitations of traditional juice extraction methods, which are often labor-intensive, time-consuming, unhygienic, and inefficient. The extractor was designed to operate using the combined principles of compression and shear force through a screw conveyor mechanism that gradually squeezes juice from fruits as they move through the extraction chamber. The machine was constructed using locally available materials to ensure affordability, ease of maintenance, and suitability for local fruit juice processors.

To evaluate the performance of the machine, the researchers conducted experimental tests using pineapple, orange, and watermelon in both peeled and unpeeled conditions. The study measured three primary performance indicators: percentage juice yield, extraction efficiency, and extraction loss. Each fruit sample underwent repeated testing to ensure the reliability of the results, and the collected data were analyzed using Analysis of Variance (ANOVA) to determine the effects of fruit type and peel condition on machine performance. The researchers found that both the type of fruit and whether it was peeled significantly influenced the juice extraction results, while variations in feed rate had little effect on overall machine performance.

The findings revealed that peeled fruits generally produced higher juice yields and extraction efficiencies than unpeeled fruits because the removal of the peel reduced resistance during the extraction process. Watermelon recorded the highest juice yield at approximately 89.5% for peeled samples, while peeled pineapple achieved the highest extraction efficiency of about 96.9%. The study also found relatively low extraction losses across all tested fruits, indicating that the machine effectively recovered a large proportion of the available juice. These results demonstrated that the extractor could consistently produce high-quality output while minimizing waste during juice processing.

Furthermore, the researchers concluded that the machine was simple to operate, economically feasible, and capable of supporting small-scale fruit juice production using locally sourced materials. Its efficient design provides an alternative to conventional manual extraction methods by improving productivity, reducing processing time, and increasing juice recovery. The study emphasized that the machine can contribute to reducing post-harvest losses of tropical fruits while promoting value-added processing within local communities.

This literature is relevant to the present study because it highlights the importance of evaluating production performance through measurable indicators such as juice yield, extraction efficiency, and extraction loss. These performance metrics demonstrate how production processes can be monitored to identify inefficiencies and improve operational outcomes. Likewise, the proposed IoT-based production management system aims to digitally record production data, monitor raw material utilization and packed yield, and generate real-time production reports. By providing accurate and organized production information, the system can assist manufacturers in analyzing production efficiency, minimizing losses, and supporting data-driven decision-making throughout the juice production process.

A study by **Dora et al. (2020)** examined the role of sustainable operations in minimizing food loss within the Belgian food processing industry. The researchers recognized that although food waste at the consumer level has been extensively studied, there is limited research focusing on food losses that occur during industrial food processing. To address this gap, they investigated food loss across different production stages in food processing companies to determine where losses most frequently occur and what operational factors contribute to them. The study aimed to identify food loss hotspots and evaluate how sustainable operational practices can help reduce waste and improve production efficiency.

To conduct the research, the authors employed a multiple case study approach involving 47 food processing companies from various sectors in Belgium, including fruit and vegetable processing, dairy, beverages, bakery, chocolate, grain milling, sauces, ready meals, and meat processing. Primary data were collected through a semi-structured questionnaire, company interviews, on-site observations, and documentary analysis. The questionnaire gathered information about company operations, production capacity, food loss at different stages of processing, and the perceived causes of these losses. The collected quantitative data were analyzed using Microsoft Excel and SPSS statistical software, while qualitative observations were used to support and validate the findings.

The findings revealed that food loss accounted for an average of 2.1% of total production among the participating companies. Among the three production stages examined—before production, during production, and after production—the production stage, particularly the processing phase, was identified as the most significant hotspot for food loss. Statistical analysis showed that

increases in operational problems during processing were directly associated with higher food loss. The researchers identified several major causes of food loss, including human errors, interrupted production, transportation problems, machine inefficiencies, product changes during production, product defects, packaging errors, poor inventory management, and buyer contract issues. Companies involved in fruit and vegetable processing experienced the highest percentage of food loss, while chocolate and ready-meal manufacturers incurred the greatest financial losses. The study also found that smaller food processing companies generally experienced higher levels of food loss than larger companies.

Based on these findings, the researchers emphasized that sustainable operational practices play a critical role in reducing food loss throughout food processing operations. They recommended improving production planning, inventory management, equipment maintenance, employee training, and quality control procedures to minimize operational inefficiencies. The study also highlighted the importance of implementing management approaches such as Lean Manufacturing, Lean Six Sigma, effective scheduling, and the use of Key Performance Indicators (KPIs) to continuously monitor and reduce food loss. Furthermore, the researchers stressed that systematic measurement of food loss enables companies to identify problem areas, develop targeted improvement strategies, and enhance overall production efficiency. These findings demonstrate that effective operational management not only reduces food waste but also improves productivity, sustainability, and cost efficiency within the food processing industry.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study employed a Developmental Research Design as the primary research methodology to guide the planning, design, development, implementation, testing, and evaluation of the system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc."* Developmental research was widely used in information technology and software engineering because it focused on designing and developing innovative technological solutions that addressed existing organizational problems while ensuring that the developed product met user requirements and operational objectives.

The primary purpose of this research was to develop a digital production monitoring platform that would replace the company's existing manual, paper-based production recording and monitoring process. The previous production monitoring procedure required production personnel to manually record raw material usage, extraction output, packed yield, and production data using logbooks and spreadsheets. This traditional method increased the possibility of recording errors, delayed report generation, inconsistent data, and difficulty in tracking production performance over time. These challenges hindered management from obtaining accurate and timely information needed for effective operational decision-making.

To address these issues, the proposed system automated the recording, monitoring, computation, and reporting of production data through a centralized digital platform. The system enabled production personnel to record production information electronically while automatically calculating yield percentages, extraction efficiency, production losses, and production stability indicators. By digitizing these processes, the organization significantly reduced manual

computation, improved data accuracy, and enhanced the overall efficiency of production monitoring.

Developmental research followed a systematic process that began with identifying the existing problems within the organization. To accomplish this, the researchers conducted data-gathering activities such as interviews with production supervisors and management personnel, direct observation of the existing production workflow, and a review of existing production records and company documents. These activities provided a comprehensive understanding of the organization's operational processes, existing challenges, and information requirements. The collected information served as the foundation for identifying system requirements and designing appropriate technological solutions.

Following the requirements gathering phase, the researchers proceeded with system analysis and design. During this stage, the functional and non-functional requirements of the proposed system were identified and documented. Process models, database structures, system architecture, user interface layouts, and workflow diagrams were prepared to ensure that every component of the system aligned with the operational processes of AIDi Foods Inc. The design phase also aimed to create a user-friendly interface that enabled production personnel and management to efficiently perform their respective tasks with minimal technical difficulty.

The development phase involved the actual construction of the proposed production monitoring system based on the approved system specifications. Various software development tools, programming languages, database management systems, and web technologies were utilized to implement the required functionalities. Core modules of the system included user management, production data entry, raw material monitoring, extraction monitoring, packed yield recording,

automated computation of yield percentages, extraction efficiency analysis, production stability monitoring, dashboard visualization, and report generation.

A significant characteristic of developmental research was its iterative nature. Throughout the software development process, the researchers continuously conducted testing, validation, debugging, and refinement of individual system components. Feedback from the company representatives and intended users was incorporated during each stage of development to ensure that the proposed system effectively addressed operational requirements. This iterative approach allowed improvements to be implemented before the final deployment of the system, thereby increasing its usability, reliability, and overall quality.

The proposed system specifically monitored critical production variables that directly influenced manufacturing efficiency. These included the quantity of raw materials used during production, extracted product output, packed yield records, computed yield percentages, extraction efficiency indicators, production losses, and production stability status. The system automatically performed mathematical computations based on production inputs, eliminating the need for manual calculations and minimizing computational errors.

In addition to real-time production monitoring, the proposed system generated comprehensive analytical reports that summarized production performance over different reporting periods. Reports included daily, weekly, monthly, quarterly, and yearly summaries of raw material utilization, extraction performance, production yield trends, efficiency measurements, and production stability analyses. These reports enabled management to evaluate operational performance, identify production inefficiencies, monitor resource utilization, and support evidence-based decision-making for continuous process improvement.

The proposed platform also featured a centralized database that securely stored historical production records. This capability allowed authorized personnel to retrieve previous production information, compare production performance across different periods, identify long-term operational trends, and generate historical analytics that were previously difficult to obtain using manual records. Data security measures, user authentication, and controlled system access were likewise incorporated to ensure the confidentiality and integrity of production information.

Furthermore, the effectiveness of the developed system was evaluated through system testing and user acceptance evaluation. Functional testing verified whether every module performed according to the specified requirements, while usability evaluation determined whether the intended users could efficiently operate the system. User feedback regarding functionality, usability, reliability, accuracy, efficiency, and overall satisfaction was collected and analyzed to assess the quality and acceptability of the proposed solution.

Through the application of the Developmental Research Design, this study produced a fully functional, reliable, and user-centered digital production monitoring platform that addressed the operational needs of ALDi Foods Inc. The completed system improved production monitoring efficiency, reduced manual recording errors, automated production analytics, provided accurate real-time information, enhanced reporting capabilities, optimized raw material utilization, supported managerial decision-making, and contributed to the company's continuous improvement initiatives and digital transformation efforts.

3.2 System Development Methodology

The study utilized the Agile Development Model as the primary software development methodology in designing, developing, testing, and implementing the proposed system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc."* Agile was chosen as an iterative and flexible software development methodology that emphasized collaboration, continuous improvement, customer involvement, rapid delivery of working software, and adaptability to changing requirements. Compared to traditional software development approaches, Agile allowed developers to respond quickly to feedback and make incremental improvements throughout the development process, resulting in a more reliable and user-centered information system.

The selection of the Agile Development Model was appropriate for this study because the proposed production monitoring system consisted of several interconnected modules that could be developed and evaluated incrementally. Since production requirements evolved during system development, Agile provided the flexibility needed to modify features, improve functionality, and accommodate user suggestions without significantly affecting the overall project timeline. Continuous communication with the management and production personnel of ALDi Foods Inc. ensured that the developed system remained aligned with the organization's operational requirements and production monitoring objectives.

Unlike the traditional Waterfall Model, which followed a strictly sequential process, Agile divided the project into smaller and manageable development cycles called sprints. Each sprint focused on a specific set of system functionalities that were planned, designed, developed, tested,

evaluated, and refined before moving to the succeeding iteration. At the end of every sprint, a functional component of the system was produced, allowing stakeholders to review its performance and recommend improvements. This iterative approach minimized development risks, facilitated early detection of errors, and promoted continuous enhancement of the proposed system.

The Agile Development Model adopted in this study consisted of the following phases:

Requirement Analysis

The first phase of the Agile methodology was the Requirement Analysis phase. This stage focused on understanding the existing production monitoring process practiced by ALDi Foods Inc. The researchers gathered comprehensive information regarding the company's operational workflow through various data collection techniques, including interviews with company management, production supervisors, and production personnel. Direct observation of daily production activities was also conducted to understand how production data was recorded, monitored, and reported. In addition, existing production forms, manual logbooks, spreadsheets, and production reports were reviewed to identify the strengths and limitations of the system.

The primary objective of this phase was to determine both the functional and non-functional requirements of the proposed digital production monitoring platform. Functional requirements included identifying necessary features such as production data entry, automated yield computation, extraction efficiency monitoring, production stability interpretation, dashboard visualization, and report generation. Non-functional requirements included system usability, security, reliability, performance, accessibility, and maintainability.

Special emphasis was placed on identifying operational challenges associated with the manual process, including delayed report preparation, repetitive manual computations, inconsistent data recording, difficulty in retrieving historical records, and limited visibility of real-time production performance. The information gathered during this phase served as the foundation for designing a system that effectively addressed these operational issues while supporting the company's digital transformation initiatives.

System Design

After completing the requirements analysis, the researchers proceeded to the System Design phase. This phase involved translating the identified requirements into detailed technical specifications that served as the blueprint for developing the proposed system.

The system architecture was designed to support centralized production data management, secure user authentication, automated processing, and real-time analytics. Database design activities involved creating relational database structures capable of storing and managing production information efficiently. Database tables included user accounts, production batches, raw material records, extraction records, packed yield data, computed yield percentages, extraction efficiency results, production stability classifications, historical production records, and generated reports.

The user interface was carefully designed to provide an intuitive, responsive, and user-friendly experience for production personnel and management. Dashboard layouts were organized to display production information clearly through tables, summary cards, charts, and graphical indicators, enabling users to monitor production performance efficiently.

System process flow diagrams, use case diagrams, activity diagrams, and navigation structures were also prepared during this phase to define how users interacted with the system and how information flowed between various modules. Automated workflows were incorporated to support real-time monitoring, automatic computations, report generation, historical data retrieval, and dashboard visualization.

The overall design prioritized simplicity, accuracy, efficiency, scalability, and ease of maintenance to ensure that the proposed platform remained effective for future operational requirements.

Development

During the Development phase, the researchers constructed the proposed production monitoring system based on the approved system design. The development process followed Agile principles by implementing system modules incrementally, allowing each component to be tested and refined before integrating it into the complete system.

The web-based application was developed using HTML and CSS for the front-end user interface to provide an accessible and responsive design. PHP was utilized as the server-side programming language responsible for implementing the application's business logic, processing production data, managing user authentication, and performing automated computations. MySQL served as the centralized relational database management system that securely stored all production-related information. Chart.js was integrated into the system to generate interactive charts, graphs, and dashboard visualizations that enabled users to analyze production trends and performance effectively.

The development activities focused on implementing the following major modules:

- User Authentication and Access Management Module
- Production Management Module
- Raw Material Recording Module
- Extraction Monitoring Module
- Packed Yield Monitoring Module
- Real-Time Yield Analytics Module
- Extraction Efficiency Monitoring Module
- Production Stability Interpretation Module
- Historical Data Management Module
- Dashboard and Data Visualization Module
- Automated Report Generation Module

The User Authentication Module ensured that only authorized personnel could access the system based on their assigned user roles and permissions.

The Production Management Module managed production batches and organized production records according to production schedules.

The Raw Material Recording Module enabled production personnel to digitally record the quantity and type of raw materials utilized during production.

The Extraction Monitoring Module recorded extracted product quantities while automatically linking extraction outputs to their corresponding production batches.

The Packed Yield Monitoring Module monitored the quantity of finished products obtained after packaging and served as the basis for yield computation.

The Real-Time Yield Analytics Module automatically calculated production yield percentages immediately after production data was entered, eliminating manual computation and providing management with up-to-date production performance indicators.

The Extraction Efficiency Monitoring Module computed extraction efficiency by comparing raw material inputs with extraction outputs. The generated efficiency values allowed management to evaluate how effectively raw materials were converted into usable products.

The Production Stability Interpretation Module automatically interpreted production performance by comparing calculated efficiency values against predefined organizational thresholds. Based on these standards, production batches were classified as either Stable or Not Stable, enabling management to quickly identify production inconsistencies and operational concerns.

The Historical Data Management Module securely archived production records, allowing authorized users to retrieve historical information for trend analysis, comparison of production periods, and long-term operational evaluation.

The Dashboard and Data Visualization Module presented production information through interactive charts, summary indicators, graphs, and analytical dashboards that provided management with a comprehensive overview of production activities.

Finally, the Automated Report Generation Module generated production reports categorized into daily, weekly, monthly, quarterly, and yearly summaries. These reports included raw material

utilization, extraction output, packed yield, yield percentages, extraction efficiency, production stability results, and historical production performance. The generated reports supported production planning, resource allocation, operational monitoring, and managerial decision-making.

Testing

Following system development, comprehensive testing procedures were conducted to ensure that all modules performed according to the specified requirements. Testing was an essential component of the Agile methodology because it allowed developers to identify defects early and continuously improve system quality throughout development.

The testing process included several levels of evaluation.

Unit Testing verified the functionality of individual modules independently to ensure that each feature performed accurately according to its intended purpose.

Integration Testing evaluated the interaction between interconnected modules to confirm that information was transferred correctly throughout the system.

System Testing assessed the complete application as a fully integrated system. This phase ensured that all functionalities operated together without conflicts and that the overall performance met the project's objectives.

Finally, User Acceptance Testing (UAT) was conducted with selected personnel from AlDi Foods Inc. End users evaluated the system's functionality, usability, accuracy, efficiency,

reliability, and overall effectiveness in supporting production monitoring activities. Their feedback was documented and incorporated into subsequent refinements of the system.

Particular attention was devoted to validating the accuracy of automated yield computations, extraction efficiency calculations, production stability interpretations, dashboard analytics, historical record retrieval, and automated report generation. This ensured that the system produced reliable information for operational and managerial decision-making.

Deployment (Prototype)

Upon successful completion of system testing and refinement, the proposed production monitoring platform was deployed as a prototype within a controlled environment at AIDi Foods Inc. Prototype deployment allowed the organization to evaluate the functionality and practicality of the developed system under actual operational conditions without immediately replacing the existing production monitoring procedures.

During this phase, production personnel and management used the prototype to perform actual production recording, monitor real-time production activities, generate analytical dashboards, calculate yield percentages, evaluate extraction efficiency, classify production stability, and produce automated reports. This practical implementation enabled stakeholders to assess how effectively the proposed system addressed the operational limitations of the manual process.

The deployment phase also provided an opportunity to collect user feedback regarding system usability, functionality, interface design, report accuracy, processing speed, and overall user experience. Suggestions gathered from stakeholders were carefully analyzed and used to identify

additional improvements, optimize existing features, and refine system performance before any future full-scale implementation.

By following the Agile Development Model throughout the project lifecycle, the researchers aimed to produce a reliable, efficient, and user-centered production monitoring platform that supported the operational objectives of ALDi Foods Inc. The iterative nature of Agile ensured continuous system enhancement while enabling the organization to benefit from improved production monitoring, accurate real-time analytics, efficient resource utilization, and informed managerial decision-making.

3.3 System Architecture

The proposed *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc.* adopted a three-tier system architecture consisting of the Client Layer, Application Layer, and Database Layer. This architectural design provided a structured framework that separated the user interface, application processing, and data management components of the system. By separating these layers, the proposed platform became more organized, maintainable, scalable, and secure while ensuring efficient communication among system components.

The three-tier architecture allowed each layer to perform specialized responsibilities independently while continuously exchanging information with the other layers. This separation improved system reliability, simplified maintenance, facilitated future enhancements, and

enabled efficient processing of production information. It also minimized system complexity by organizing operational tasks according to their specific functions.

Within the proposed production monitoring platform, users interacted only with the Client Layer. The Client Layer communicated with the Application Layer, which processed requests, performed computations, and retrieved or stored data within the Database Layer. Once processing was completed, the Application Layer returned the processed information back to the Client Layer, allowing users to immediately view updated production analytics, reports, and dashboard visualizations.

Client Layer

The Client Layer served as the presentation layer and acted as the primary interface between users and the proposed production monitoring system. It provided an accessible web-based environment where authorized users interacted with the application using computers or other compatible devices connected to the organization's local network or internet connection.

The primary users of the system included production personnel, supervisors, and system administrators. Each authorized user was provided with secure login credentials that determined the functions accessible according to their assigned role. Through this interface, production personnel could efficiently perform daily production recording activities while supervisors and management personnel could monitor operational performance using analytical dashboards and generated reports.

The Client Layer enabled users to digitally record all production-related information required throughout the manufacturing process. These records included production batch information,

quantities of raw materials utilized, extraction output measurements, packed yield records, production dates, operator information, and other relevant production details. Replacing manual recording with electronic data entry significantly minimized clerical errors, eliminated redundant paperwork, and ensured that production records were stored in a consistent format.

After production information was submitted, users were immediately provided with updated production results through an interactive dashboard. The dashboard presented key performance indicators using summary cards, tables, graphical charts, and visual analytics. Users could instantly monitor production activities, compare production batches, and evaluate operational performance without manually computing production statistics.

The Client Layer also provides real-time visualization of important production indicators such as:

- Raw material consumption
- Extraction output
- Packed yield
- Yield percentage
- Extraction efficiency
- Production stability status
- Historical production trends
- Periodic production summaries

One of the major advantages of the dashboard is its ability to automatically display the operational status of each production batch. Once production information has been processed, the dashboard immediately indicates whether production performance is classified as **Stable** or **Not**

Stable, allowing production personnel and supervisors to identify operational concerns without waiting for manual report preparation.

To improve usability, the user interface is designed following principles of simplicity, consistency, responsiveness, and accessibility. Navigation menus are organized logically to minimize user confusion, while input forms are structured to reduce encoding errors and improve efficiency. Dashboard components are arranged to provide clear visualization of production performance while requiring minimal technical expertise from users.

By providing immediate access to production information, analytical dashboards, and automated reports, the Client Layer significantly improves operational visibility and enables management to make faster and more informed production decisions.

Application Layer

The **Application Layer** serves as the central processing component and core engine of the proposed production monitoring platform. It functions as the intermediary between the Client Layer and the Database Layer by receiving production information from users, processing the submitted data according to predefined business rules, performing automated analytical computations, and returning meaningful information back to the user interface.

All major operational logic of the proposed system is implemented within this layer. Every production transaction initiated by users undergoes validation to ensure that required information is complete, accurate, and logically consistent before being stored in the database. Validation procedures help minimize data entry errors and improve the overall quality of production records.

One of the primary responsibilities of the Application Layer is performing automated production calculations. Instead of requiring production personnel to manually compute operational statistics, the system automatically calculates important production indicators immediately after production data has been entered.

These automated computations include:

- Yield percentage calculation
- Extraction efficiency calculation
- Packed yield computation
- Raw material utilization analysis
- Production loss monitoring
- Historical production comparison

The automated computation process significantly reduced computational errors, eliminated repetitive manual calculations, and provided management with accurate production information in real time.

Another important function of the Application Layer was the implementation of the Production Stability Interpretation Module. This module continuously evaluated production performance using predefined organizational standards and operational thresholds established by AIDi Foods Inc. The system compared computed yield percentages and extraction efficiency values against these predefined criteria to determine whether production performance met acceptable operational standards.

Based on the computed analytical results, the system automatically classified each production batch into one of two operational categories:

- **Stable**
- **Not Stable**

This automated interpretation assisted management in quickly identifying production inconsistencies, abnormal production behavior, excessive production losses, and potential operational issues that required immediate corrective action. Instead of manually reviewing numerous production records, supervisors could immediately determine production status through dashboard indicators generated by the system. The Application Layer also managed the Real-Time Yield Analytics Module, which continuously updated production performance indicators whenever new production information was recorded. This enabled management to monitor production progress as manufacturing activities occurred rather than waiting until the completion of production reports.

In addition, the Application Layer supported the Extraction Efficiency Monitoring Module, which evaluated how efficiently raw materials were converted into extracted products. By continuously monitoring extraction efficiency, management could identify production batches with unusually low extraction performance and investigate possible operational causes such as equipment performance, processing methods, or raw material quality. The system also incorporated an Automated Report Generation Module, which prepared production reports without requiring manual encoding or spreadsheet calculations. Using production information stored in the database, the Application Layer generated comprehensive reports that summarized operational performance according to various reporting periods.

Generated reports include:

- Daily production summaries
- Weekly raw material utilization reports
- Monthly production performance reports
- Quarterly extraction efficiency reports
- Yearly production analytics
- Historical production trend reports

These reports enabled management to monitor operational efficiency, evaluate resource utilization, identify production trends, and support long-term strategic planning.

The Application Layer further managed user authentication, role-based access control, session management, historical data retrieval, dashboard visualization, and communication between the user interface and the database. Centralizing these responsibilities ensured consistency, improved processing efficiency, enhanced system reliability, and maintained data integrity throughout the entire application.

Database Layer

The Database Layer functioned as the centralized repository of all information generated and processed by the proposed production monitoring platform. This layer was responsible for securely storing, organizing, maintaining, retrieving, and managing production records throughout the system's operation. The proposed system utilized a relational database to ensure that production information was properly structured and easily accessible. By maintaining a centralized database, the organization eliminated dependence on manual record books,

spreadsheets, and scattered paper documents, thereby improving data consistency, security, and accessibility.

The database stored multiple categories of operational information, including:

- User roles and permissions
- Production batch records
- Raw material input records
- Extraction output records
- Packed yield information
- Yield percentage calculations
- Extraction efficiency results
- Production stability classifications
- Historical production records
- Dashboard summary information
- Generated production reports
- System activity logs

Every production transaction submitted by authorized users was securely stored within the database and became immediately available for future retrieval and analysis. Historical records remained preserved, enabling management to compare production performance across different operational periods and identify long-term production trends.

The centralized database significantly improved information management by eliminating problems commonly associated with paper-based record keeping, including misplaced documents, duplicate records, inconsistent information, delayed retrieval, and accidental data

loss. To maintain data security and integrity, access to stored information was controlled through authenticated user accounts and predefined user permissions. Only authorized personnel were permitted to view, edit, or manage specific production records according to their assigned responsibilities.

Furthermore, the Database Layer served as the primary source of information used by the reporting and analytics modules. Every dashboard visualization, graphical chart, analytical report, production summary, yield computation, extraction efficiency analysis, and production stability interpretation was generated using structured data retrieved from the centralized database. Because all production information was stored within a single integrated repository, the organization could efficiently retrieve historical production records, monitor operational performance, evaluate production efficiency, and generate comprehensive management reports whenever needed.

System Data Flow

The overall operation of the proposed production monitoring platform followed a continuous flow of information among the three architectural layers. The process began when authorized production personnel logged into the system through the Client Layer using valid authentication credentials. Once access was granted, the user recorded production information such as production batch details, raw material quantities, extraction outputs, packed yield measurements, and other required production data.

The entered information was transmitted to the Application Layer, where the system first validated the submitted data to ensure completeness and accuracy. After successful validation, the Application Layer automatically performed production-related computations, including yield percentage calculation, extraction efficiency analysis, raw material utilization computation, and production stability interpretation. Once processing was completed, all production information and computed analytical results were securely stored within the Database Layer. The database immediately updated historical production records, ensuring that newly entered information became available for future retrieval, reporting, and trend analysis. Whenever users accessed the dashboard or requested analytical reports, the Application Layer retrieved the required production information from the Database Layer, processed the requested data, generated graphical visualizations and analytical summaries, and returned the processed results to the Client Layer.

The Client Layer then displayed updated dashboards containing real-time production statistics, extraction efficiency indicators, yield percentages, production stability classifications, historical production records, and periodic raw material utilization reports. Because the information was processed automatically, users received immediate feedback regarding production performance without performing manual calculations. This continuous exchange of information among the Client Layer, Application Layer, and Database Layer enabled the proposed system to provide accurate real-time monitoring, centralized information management, automated analytical processing, efficient report generation, and timely decision support. Through this architecture, ALDi Foods Inc. was able to improve operational efficiency, reduce manual recording errors, enhance production visibility, and support data-driven decision-making across its manufacturing operations.

3.4 Tools and Technologies Used

The development of the proposed system, "Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.," required the integration of various software technologies and development tools that worked together to create an efficient, reliable, and user-friendly web-based production monitoring platform. The selection of these technologies played a significant role in ensuring that the system could accurately record production data, perform automated analytical computations, generate real-time dashboards, and produce comprehensive production reports that supported the operational needs of the organization.

The proposed system was designed to replace the existing manual production monitoring process with a centralized digital platform. To achieve this objective, the researchers selected technologies that were widely recognized for their stability, compatibility, flexibility, and ease of implementation. These technologies enabled the system to automate production monitoring activities, reduce manual computations, improve data accuracy, and provide management with timely information for decision-making.

Each technology was carefully evaluated based on several criteria, including compatibility with other development tools, processing efficiency, security, maintainability, accessibility, scalability, and suitability for web-based information systems. Since the proposed platform consisted of multiple interconnected modules—including production recording, yield analytics, extraction efficiency monitoring, production stability interpretation, dashboard visualization, and report generation—the selected technologies had to work seamlessly together to ensure smooth system operation.

Another important consideration in selecting these technologies was their ability to support centralized information management. Rather than storing production records in paper logbooks or disconnected spreadsheet files, the proposed system consolidated all production information into a centralized database that could be accessed by authorized personnel. This approach improved information consistency, minimized duplication of records, enhanced data security, and enabled faster retrieval of historical production information whenever required.

The selected development tools also supported the implementation of real-time monitoring features. As production personnel recorded operational data into the system, the application immediately processed the information, performed the required analytical computations, updated production dashboards, and stored records in the centralized database. This automation significantly reduced delays associated with manual report preparation while allowing management to monitor production performance more effectively.

Furthermore, the chosen technologies were widely used in both academic and professional software development environments. Their extensive documentation, active developer communities, and long-term support made them practical choices for developing an information system that could be maintained, improved, and expanded in the future as organizational requirements evolved.

Frontend Technologies

HTML (HyperText Markup Language)

HTML (HyperText Markup Language) served as the primary markup language used to construct the structural components of the proposed web-based production monitoring system. It provided the basic framework upon which all web pages, forms, navigation menus, tables,

dashboards, and other interface components were built. HTML defined how information was organized and displayed, ensuring that users could interact efficiently with the different modules of the application.

Within the proposed system, HTML was used to develop user interfaces for production recording, raw material encoding, extraction monitoring, packed yield recording, production dashboards, historical record viewing, report generation, and administrative functions. Production personnel used HTML-based forms to enter production information, while supervisors and management personnel accessed analytical dashboards and production reports through organized web pages.

The flexibility of HTML allowed the proposed application to maintain compatibility across modern web browsers, ensuring that authorized users could access the system without installing specialized software. Because the platform was web-based, users could conveniently perform production monitoring activities through standard computers connected to the organization's network.

HTML also supported integration with CSS, PHP, JavaScript libraries, and database-driven applications, making it an appropriate foundation for developing an interactive production monitoring system that met the operational requirements of ALDi Foods Inc.

CSS (Cascading Style Sheets)

CSS (Cascading Style Sheets) was utilized to improve the visual appearance, usability, responsiveness, and overall user experience of the proposed system. While HTML defined the structural layout of the application, CSS controlled the presentation of system components by

specifying colors, fonts, spacing, positioning, menus, buttons, tables, dashboards, and graphical layouts.

The implementation of CSS enabled the researchers to design a clean, organized, and professional user interface that minimized user confusion and improved operational efficiency. Proper visual organization was particularly important because production personnel frequently interacted with production records, dashboard summaries, and analytical reports during daily operations.

CSS was also used to implement responsive design techniques that allowed the application to adapt to different screen sizes and display resolutions. Whether the system was accessed using desktop computers or laptops within the organization, the interface remained consistent, readable, and easy to navigate.

In addition, CSS enhanced the dashboard by improving the presentation of production statistics, graphical indicators, report summaries, and navigation menus. These improvements enabled users to quickly identify important production information and supported faster operational decision-making.

Backend Technology

PHP (Hypertext Preprocessor)

PHP (Hypertext Preprocessor) served as the primary server-side programming language responsible for implementing the core functionality and business logic of the proposed

production monitoring platform. PHP processed requests submitted by users, communicated with the database, performed automated computations, validated production information, and generated dynamic web pages.

As the central processing component of the system, PHP performs numerous operational tasks, including:

- User authentication and login verification
- User account and access control management
- Production batch processing
- Raw material recording
- Extraction monitoring
- Packed yield recording
- Yield percentage calculation
- Extraction efficiency computation
- Production stability interpretation
- Historical record retrieval
- Dashboard data processing
- Automated report generation

One of PHP's most important responsibilities within the proposed system was performing automated analytical computations immediately after production data had been entered. Rather than requiring production personnel to manually calculate yield percentages and extraction efficiency values, PHP executed these calculations automatically, reducing computational errors while improving processing speed.

PHP also managed communication between the user interface and the MySQL database, ensuring that production records were stored securely and retrieved accurately whenever requested by users. Because PHP integrated efficiently with MySQL, it was well suited for developing database-driven information systems such as the proposed production monitoring platform.

Analytics Processing Technology

[Chart.js](#)

Chart.js was utilized as the primary data visualization library for displaying production analytics and operational reports. The library provided interactive graphical representations of production data, enabling users to understand performance trends quickly and effectively.

The system will use Chart.js to visualize:

- Current Yield Percentage
- Extraction Efficiency Performance
- Stable and Not Stable Production Trends
- Raw Material Consumption Analysis
- Production Output Trends
- Weekly Usage Reports
- Monthly Usage Reports
- Quarterly Usage Reports
- Yearly Usage Reports

By converting raw numerical information into visual charts and graphs, Chart.js improved data interpretation and supported data-driven decision-making within ALDi Foods Inc.

Database Technology

MySQL

MySQL served as the centralized relational database management system for the proposed production monitoring platform. It was responsible for securely storing, organizing, retrieving, updating, and managing all production-related information generated throughout system operation.

The centralized database served as the foundation of the entire application by ensuring that production records remained accurate, organized, and readily accessible to authorized users. Unlike traditional paper-based recording methods, MySQL enabled efficient information management while minimizing duplication, inconsistency, and loss of production data.

The database will store various categories of operational information, including:

- User Accounts
- User Roles and Permissions
- Production Batches
- Raw Material Inputs
- Extraction Outputs
- Packed Yield Records
- Yield Percentage Results
- Extraction Efficiency Results

- Production Stability Status
- Historical Production Records
- Dashboard Summary Data
- Generated Reports

Historical production information stored within MySQL enabled management to perform long-term production analysis, compare operational performance across different reporting periods, identify production trends, and generate comprehensive management reports.

The database also supported automated report generation and dashboard visualization by providing structured production information that could be processed efficiently by PHP and displayed through [Chart.js](#).

Development Environment

Visual Studio Code

Visual Studio Code (VS Code) was utilized as the primary Integrated Development Environment (IDE) during the development of the proposed production monitoring platform. VS Code provided a lightweight yet powerful environment that supported both frontend and backend development, enabling researchers to efficiently create, edit, test, and maintain the application's source code.

Several built-in and extensible features contributed to improving development productivity, including:

- Syntax highlighting
- Intelligent code completion
- Error detection
- Debugging support
- Extension marketplace
- Integrated terminal
- Version control integration
- File management tools

These capabilities enabled the researchers to organize project files efficiently, detect programming errors early, simplify debugging activities, and maintain code quality throughout the development lifecycle.

Local Development Server

XAMPP

XAMPP was used as the local development server environment during system development and testing. It provided the necessary components required for executing PHP applications and managing MySQL databases within a local machine.

XAMPP includes:

- Apache Web Server
- MySQL Database Server
- PHP Interpreter

- phpMyAdmin Database Management Tool

The use of XAMPP allows the researchers to develop, test, and validate system functionalities before deployment.

Technology Integration

The proposed system operated through the integration of these technologies. HTML and CSS provided the user interface through which production personnel interacted with the system. PHP processed incoming production data and performed calculations related to yield analytics and extraction efficiency monitoring. MySQL stored production records and historical information, while Chart.js converted processed data into graphical dashboards and analytical reports. Visual Studio Code and XAMPP served as the development and testing environment that supported the creation and validation of the entire application.

Together, these technologies provided the technical foundation necessary to implement a digital production monitoring platform capable of supporting real-time yield analytics, extraction efficiency monitoring, production stability interpretation, and periodic raw material usage reporting for ALDI Foods Inc.

3.5 Data Gathering Techniques

To obtain the necessary information for the design, development, and implementation of the proposed system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction*

Efficiency Monitoring for ALDi Foods Inc.," the researchers employed a combination of carefully selected data gathering techniques. Gathering accurate and comprehensive information was an essential phase of the system development process because it provided the researchers with a clear understanding of the organization's existing production monitoring practices, operational requirements, and the challenges associated with its current manual procedures.

Rather than relying on a single source of information, the study adopted a multi-method approach that combined different data gathering techniques to obtain information from various operational perspectives. This approach improved the reliability and completeness of the collected data by allowing the researchers to compare information obtained from different sources and verify its consistency. The integration of multiple techniques also enabled the researchers to develop a more comprehensive understanding of how production information flowed throughout the manufacturing process and how different personnel performed their respective responsibilities.

The primary objective of the data gathering process was to examine the current production monitoring procedures implemented by ALDi Foods Inc. The researchers analyzed how production data was generated, recorded, processed, monitored, and reported from the initial receipt of raw materials up to the completion of the production cycle. Particular attention was given to the recording of raw material inputs, extraction outputs, packed yield records, production computations, and report preparation. Understanding these procedures allowed the researchers to identify areas where digital technologies could improve operational efficiency and reduce dependence on manual recording practices.

In addition, the data gathering activities were intended to identify operational challenges that affected the efficiency of the company's current production monitoring process. These challenges included delayed report preparation, repetitive manual computations, inconsistent production records, difficulty retrieving historical information, data duplication, limited visibility of production performance, and human errors during recording and calculation. Identifying these issues enabled the researchers to establish the functional and non-functional requirements of the proposed digital production monitoring platform.

The information collected during this phase served as the foundation for designing the proposed system's database structure, user interface, system workflows, automated analytical computations, dashboard visualizations, and report generation features. Every major component of the proposed system was developed based on actual operational requirements gathered from the organization to ensure that the platform effectively supported daily production activities and management decision-making.

Furthermore, the data gathering process assisted the researchers in determining the information required for implementing the system's core functionalities, including digital production recording, automated yield percentage calculation, extraction efficiency monitoring, production stability interpretation, centralized historical data management, dashboard visualization, and periodic production reporting. By understanding how information was handled within the organization, the researchers developed a system that accurately reflected existing operational practices while improving efficiency through automation.

The use of multiple data gathering techniques also strengthened the credibility of the study by allowing information collected through one method to be verified using another. Observations of

actual production activities were compared with interview responses, while existing production documents were reviewed to validate operational procedures and reporting practices. This process helped ensure that the proposed system was based on factual operational information rather than assumptions.

Ultimately, the data gathering phase served as the foundation of the entire system development process. The findings obtained during this stage guided the succeeding phases of system analysis, design, development, testing, implementation, and evaluation. Through this comprehensive approach, the researchers aimed to develop a digital production monitoring platform that effectively addressed the operational needs of AIDi Foods Inc. while improving production efficiency, reporting accuracy, resource utilization, and managerial decision-making.

Interviews

The interview technique will be conducted with selected personnel of AIDi Foods Inc., including production supervisors, production personnel, and other authorized representatives involved in the production monitoring process. The purpose of the interviews is to obtain first-hand information regarding the organization's current production workflow, existing monitoring procedures, reporting practices, operational challenges, and expectations for the proposed system.

Semi-structured interviews will be used to allow respondents to describe their daily responsibilities, explain the current production recording procedures, identify difficulties encountered in manual monitoring, and provide suggestions for improving the overall production monitoring process. The researchers will also gather information regarding the production data currently being recorded, the frequency of report preparation, methods of calculating yield and

extraction efficiency, and the information needed by management for operational decision-making.

The responses collected through the interviews will provide valuable insights into user requirements and will serve as an important basis for identifying the necessary features and functionalities of the proposed production monitoring platform.

Observation

The researchers will conduct direct observation of the production monitoring activities within ALDi Foods Inc. to obtain a clear understanding of how production information flows throughout the manufacturing process. During this activity, the researchers will observe the procedures used in recording raw material inputs, monitoring extraction outputs, documenting packed yields, calculating production results, and preparing production reports.

Observation enables the researchers to examine actual operational practices, identify workflow inefficiencies, and understand how production personnel interact with existing recording procedures. This technique also allows the researchers to verify whether the information obtained during interviews accurately reflects actual production activities.

Through direct observation, the researchers can identify repetitive manual tasks, unnecessary documentation procedures, delays in information processing, and opportunities where automation can improve production monitoring efficiency. The observations will also assist in designing system workflows that closely match the organization's existing operational processes while reducing unnecessary manual activities.

Document Review

The researchers will conduct a comprehensive review of the existing documents, records, and production reports utilized by ALDi Foods Inc. as part of the data gathering process. Document review is an essential technique because it provides objective, factual, and historical information regarding the organization's production monitoring practices. Unlike interviews and observations, which primarily rely on current activities and stakeholder responses, document review enables the researchers to examine actual production records that have been used in daily operations. These documents serve as reliable references for understanding how production information is currently collected, organized, processed, and reported within the organization.

The primary objective of the document review is to analyze the existing production monitoring system and identify the information requirements needed for the development of the proposed digital production monitoring platform. By examining the company's operational records, the researchers can determine the types of production data that must be captured, stored, processed, and presented within the new system. This activity also allows the researchers to understand the format, structure, and frequency of existing reports, ensuring that the proposed system supports the organization's current reporting requirements while improving efficiency through automation.

The researchers will review various production-related documents currently maintained by ALDi Foods Inc., including but not limited to:

- Production Logs
- Daily Production Reports
- Raw Material Usage Records
- Extraction Output Records

- Packed Yield Reports
- Inventory Records
- Historical Production Data
- Existing Monitoring Forms

Each of these documents contained valuable information regarding different stages of the production process. Production logs and daily production reports provided detailed records of manufacturing activities conducted during each production cycle, while raw material usage records documented the quantity of materials consumed during processing. Extraction output records and packed yield reports contained information necessary for monitoring production efficiency, evaluating yield performance, and analyzing extraction results. Inventory records provided additional information regarding the availability and utilization of raw materials, while historical production records enabled the researchers to examine production performance over extended periods.

A major focus of the document review was the examination of the company's existing production monitoring forms. These forms served as the primary source of production information under the current manual monitoring system and illustrated how production personnel recorded operational data throughout the manufacturing process. By carefully reviewing these forms, the researchers gained a comprehensive understanding of how production information was organized, categorized, and documented. This analysis enabled the researchers to identify the sequence of production activities, determine the relationships among various production records, and understand how information flowed from raw material preparation through extraction, packaging, and final reporting.

The researchers also analyzed the layout, content, and structure of the existing documents to determine the specific data elements required for the proposed system. Particular attention was given to identifying the production variables that had to be recorded during each production cycle, including raw material quantities, extraction outputs, packed yields, production dates, batch information, operator details, and other operational data necessary for monitoring production performance. Identifying these essential data elements ensured that the proposed system captured complete and accurate information required for automated computations and report generation.

Furthermore, document review assisted the researchers in identifying the methods used to calculate production performance indicators, particularly yield percentages and extraction efficiency measurements. By examining existing reports and production summaries, the researchers understood how these values were computed, documented, and interpreted by production personnel and management. This information served as the basis for designing automated computation modules that accurately replicated and improved the organization's existing production monitoring procedures.

Historical production records were also analyzed to identify long-term production trends, recurring operational patterns, and variations in production performance over time. Reviewing historical data enabled the researchers to establish baseline production performance, identify seasonal or operational fluctuations, evaluate raw material utilization, and examine extraction efficiency across different production periods. These historical records provided valuable insights that supported the design of the system's historical data management, trend analysis, and dashboard visualization features.

In addition, document review enabled the researchers to identify weaknesses associated with the existing manual documentation process. The analysis revealed issues such as duplicate data entry, inconsistent record formats, incomplete production information, delayed report preparation, difficulty retrieving historical records, and the possibility of computational errors resulting from manual calculations. Identifying these limitations allowed the researchers to design system features that addressed these operational challenges through centralized data management, automated computations, standardized record formats, and digital report generation.

The findings obtained through the document review played a significant role in the development of the proposed system's database structure. The researchers used the identified production data elements to design database tables, establish relationships among production records, and organize information in a manner that supported efficient storage, retrieval, and analysis. A well-designed database ensured that production information remained accurate, consistent, secure, and readily accessible for operational monitoring and management decision-making.

Moreover, the information gathered through document review supported the development of the system's reporting and analytics modules. Existing production reports served as references for designing automated weekly, monthly, quarterly, and yearly production summaries. Similarly, the production data identified during document analysis was utilized to develop dashboard visualizations that displayed yield percentages, extraction efficiency, production stability status, raw material utilization, and historical production trends in an organized and meaningful manner.

The document review also served as a validation mechanism for information obtained through interviews and direct observation. By comparing documented production records with the

responses of production personnel and observations conducted within the facility, the researchers verified the consistency and accuracy of the collected information. This process strengthened the reliability of the study and ensured that the proposed production monitoring system accurately reflected the actual operational practices of ALDi Foods Inc.

Ultimately, the document review provided a solid foundation for the successful design and development of the proposed digital production monitoring platform. The information obtained through this technique guided the identification of system requirements, database design, workflow development, dashboard visualization, automated yield analytics, extraction efficiency monitoring, production stability interpretation, historical data management, and report generation. By thoroughly examining the organization's existing production documents, the researchers ensured that the proposed system was fully aligned with the operational needs of ALDi Foods Inc. and effectively supported the organization's transition from manual production monitoring to a centralized, data-driven digital information system.

Summary of Data Gathering Process

The data gathering process served as one of the most critical stages in the development of the proposed system entitled "Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc." This phase provided the researchers with a comprehensive understanding of the organization's existing production monitoring procedures, operational workflows, and information management practices. By collecting accurate and reliable information from multiple sources, the researchers established a strong foundation for designing a system that effectively addressed the operational needs of the company.

To obtain a complete and accurate understanding of the existing production environment, the researchers employed a combination of interviews, direct observation, and document review. These complementary data gathering techniques enabled the researchers to collect information from different perspectives, allowing the findings from one method to be verified through the others. This multi-method approach increased the credibility, reliability, and completeness of the collected information while providing a more detailed picture of the organization's production monitoring process.

The interviews provided firsthand information from production managers, supervisors, operators, and other personnel directly involved in manufacturing operations. Through these discussions, the researchers gained valuable insights into the production monitoring procedures, reporting practices, operational challenges, and user expectations for the proposed digital production monitoring system. The information obtained from these interviews helped identify user requirements and determine the features necessary to improve production efficiency, reporting accuracy, and decision-making.

Direct observation complemented the interview findings by allowing the researchers to examine actual production activities as they occurred within the facility. Observing the production environment enabled the researchers to understand how production information was recorded, processed, and utilized throughout each stage of manufacturing. This technique also assisted in identifying workflow inefficiencies, repetitive manual tasks, delays in report preparation, and opportunities where automation could improve operational performance without disrupting existing production processes.

The document review further strengthened the data gathering process by providing objective and historical information regarding the organization's production activities. Existing production logs, raw material usage records, extraction reports, packed yield reports, inventory records, historical production data, and monitoring forms were examined to determine the information being collected and the reporting formats utilized by the organization. These documents helped the researchers identify the specific production data elements required for the proposed system and ensured that the application's database structure and reporting modules accurately reflected the organization's operational requirements.

By integrating the information obtained through interviews, direct observation, and document review, the researchers developed a comprehensive understanding of the complete production monitoring workflow of AlDi Foods Inc. This process enabled the researchers to trace the movement of production information from the recording of raw material inputs through extraction, packed yield documentation, production analysis, and final report generation. Understanding this complete flow of information was essential in designing a digital system that supported existing operational practices while improving efficiency through automation and centralized information management.

The collected data also enabled the researchers to identify operational challenges associated with the organization's existing manual production monitoring process. These challenges included delayed report preparation, repetitive manual computations, inconsistent record-keeping, duplication of production data, difficulty retrieving historical records, limited real-time production visibility, and the possibility of human error during production recording and calculation. Identifying these operational concerns allowed the researchers to establish clear functional and non-functional requirements for the proposed system.

Furthermore, the findings gathered during this phase served as the basis for designing the system architecture, database structure, user interface, workflow processes, and analytical modules of the proposed application. The information obtained from the production environment guided the development of essential system functionalities, including digital production recording, automated yield percentage computation, extraction efficiency monitoring, production stability interpretation, centralized historical data management, interactive dashboard visualization, and automated generation of weekly, monthly, quarterly, and yearly production reports.

The data gathered also supported the development of the system's automated analytical capabilities. By understanding how production data was collected and analyzed, the researchers designed computation processes that accurately calculated yield percentages, evaluated extraction efficiency, monitored raw material utilization, and classified production performance as either Stable or Not Stable based on the organization's predefined operational standards. These automated processes were intended to minimize manual computations while providing management with accurate and timely production information for decision-making.

In addition, the information collected during this phase assisted in designing a centralized database capable of securely storing production records, historical information, user accounts, production batches, extraction results, packed yields, and generated reports. A well-structured database improved information consistency, eliminated duplicate records, facilitated rapid retrieval of historical production information, and supported long-term production analysis.

The comprehensive data gathering process also contributed to the successful implementation of real-time dashboard visualization. The researchers identified the production indicators that management regularly monitored, enabling the proposed system to display relevant production

metrics through charts, tables, graphical summaries, and performance indicators. These dashboards provided immediate visibility into production performance, allowing supervisors and managers to monitor yield percentages, extraction efficiency, production stability, and raw material utilization as production activities occurred.

Ultimately, the information collected through interviews, direct observation, and document review served as the primary foundation for the successful development of the proposed digital production monitoring platform. Every stage of the system development process—including requirements analysis, system design, software development, database implementation, testing, deployment, and evaluation—relied on the findings obtained during data gathering. By ensuring that the proposed system was designed based on actual operational requirements rather than assumptions, the researchers developed a reliable, efficient, and user-centered application that supported the digital transformation initiatives of AIDi Foods Inc.

Through this comprehensive and systematic data gathering process, the proposed system was capable of replacing manual production monitoring procedures with a centralized digital platform that improved data accuracy, enhanced real-time operational visibility, automated yield analytics and extraction efficiency monitoring, simplified report generation, and provided management with reliable information for timely and informed decision-making. The successful completion of this phase ensured that the proposed system was closely aligned with the operational objectives and production monitoring requirements of AIDi Foods Inc., thereby contributing to improved productivity, resource utilization, and overall manufacturing performance.

3.6 Testing Procedures

The proposed system underwent a series of structured testing procedures before its official deployment at ALDi Foods Inc. This rigorous evaluation phase was designed to ensure that all developed functionalities operated correctly, efficiently, and reliably under real-world conditions. By systematically walking through each feature in a controlled environment, the project team guaranteed that the platform was robust, stable, and completely safe to integrate into daily operations.

These thorough testing procedures were essential to validate the system's capability to handle complex, data-driven manufacturing workflows. Specifically, the technical team tested the platform's ability to support real-time yield analytics and extraction efficiency monitoring. Ensuring these features worked flawlessly meant that the data flowing from the production floor was captured accurately and processed without delay.

Furthermore, the testing process focused heavily on the system's advanced analytical capabilities, such as automated production stability interpretation and periodic raw material usage reporting. The software had to be able to evaluate production trends automatically and generate precise, dependable summaries of resource consumption. Testing these components ensured that management received highly accurate insights to help optimize resources and eliminate waste.

Ultimately, the primary aim of this comprehensive testing process was to identify hidden software errors, eliminate system bugs, and verify overall performance. By matching the system's actual behavior against its original design blueprints, the team confidently ensured that the developed platform met all functional and non-functional requirements of the study. This final validation guaranteed a high-quality digital.

Unit Testing

Unit testing was conducted to evaluate individual components or modules of the system independently. Each function was tested to ensure that it performed its intended task accurately before being integrated into the full system.

The following modules will be subjected to unit testing:

- User authentication and login module
- Raw material input module
- Extraction output and packed yield input module
- Yield percentage computation module
- Extraction efficiency calculation module
- Production stability interpretation module (Stable / Not Stable classification)
- Dashboard data retrieval module
- Report generation module (weekly, monthly, quarterly, yearly)

During unit testing, the researchers verified the correctness of calculations, particularly the accuracy of yield percentage computation and extraction efficiency results. The stability classification logic was also tested to ensure that production status was correctly determined based on predefined thresholds.

System Testing

System testing was performed to evaluate the complete and integrated system as a whole. This stage ensured that all modules worked together properly and that data flowed correctly across the Client, Application, and Database layers.

The system was tested under simulated production scenarios that reflected the actual workflow of ALDi Foods Inc., including:

- Input of raw material quantities during production
- Recording of extraction and packed yield outputs
- Real-time computation of yield percentage
- Continuous monitoring of extraction efficiency
- Automatic classification of production stability
- Generation of dashboard analytics and visual reports
- Creation of periodic raw material usage reports

System testing ensured that all components interacted seamlessly and that real-time updates were reflected accurately on the dashboard without data loss, delay, or inconsistency.

User Acceptance Testing (UAT)

User Acceptance Testing was conducted with selected personnel from ALDi Foods Inc., including production managers, supervisors, and production staff. This phase evaluated the system based on actual user experience and operational suitability.

The respondents assessed whether the system met the organization's requirements in terms of:

- Ease of use of the interface
- Accuracy of yield and efficiency calculations
- Clarity of dashboard visualizations
- Effectiveness of stability interpretation (Stable / Not Stable)
- Usefulness of periodic reports (weekly, monthly, quarterly, yearly)
- Overall system performance in real production monitoring scenarios

Feedback gathered during UAT was used to identify necessary improvements and refinements to ensure that the system effectively supported production monitoring, decision-making, and operational efficiency. Through these testing procedures, the system was validated as a reliable digital solution capable of replacing manual production monitoring methods and providing real-time analytics for improved production management at AIDi Foods Inc.

3.7 Evaluation Criteria

The proposed system underwent a formal evaluation to determine its overall effectiveness, performance, and suitability for the operational needs of AIDi Foods Inc. This assessment was designed to verify that the digital platform could reliably support the demands of a fast-paced manufacturing environment. By gathering targeted feedback from key stakeholders, the evaluation aimed to prove that the system was ready for full-scale implementation.

To ensure a rigorous and standardized assessment, the evaluation was conducted using a Likert scale questionnaire carefully adapted from the ISO 25010 Software Quality Model. Utilizing this internationally recognized standard ensured that the system was not just judged on opinion, but

systematically measured against proven software quality characteristics like usability, reliability, and functional suitability. This framework provided a benchmarked, professional foundation for the entire feedback process.

A primary focus of this evaluation was determining whether the new system successfully replaced the company's legacy manual production monitoring processes. Moving away from traditional paper-and-pen logging or siloed spreadsheets represented a major shift for the production floor. The survey closely examined how well the technology digitized day-to-day operations and whether it created a more streamlined, error-free workflow for the staff.

In addition to evaluating daily operations, the assessment evaluated user adoption and the overall usability of the digital interface. Gathering structured input from both frontline operators and supervisory personnel ensured that the software catered effectively to all user groups. This helped verify that the platform minimized operational friction and allowed staff to transition smoothly away from traditional tracking methods.

Furthermore, the evaluation measured the system's capacity to handle advanced, data-driven manufacturing tasks. Specifically, it assessed how effectively the platform supported real-time yield analytics, extraction efficiency monitoring, and automated stability interpretation. The results confirmed that these automated capabilities provided operational visibility that far surpassed previous manual methods.

Ultimately, the process validated the system's ability to generate accurate periodic reporting for raw material usage and key performance metrics. By delivering reliable, automated analytics, the evaluation confirmed that management gained the precise insights needed to optimize resources, minimize operational waste, and support long-term production goals at AIDi Foods Inc.

Functionality

Functionality referred to the degree to which the system performed its intended tasks accurately and completely. This included the system's ability to correctly process production inputs, compute yield percentages, and generate extraction efficiency results.

The system was evaluated based on its capability to:

- Accurately record raw material inputs and production outputs
- Compute real-time yield percentage without errors
- Calculate extraction efficiency correctly based on input-output relationships
- Automatically interpret production status as Stable or Not Stable based on defined thresholds
- Generate accurate production reports, including weekly, monthly, quarterly, and yearly summaries
- Provide real-time dashboard analytics for production monitoring

This criterion ensures that the system fulfills all required functionalities for digitized production monitoring.

Usability measured how easy and efficient the system was for users to operate. It evaluated the user interface design, navigation structure, and overall user experience of the system.

The system was assessed based on:

- Ease of navigating the dashboard and system modules
- Clarity of data presentation in charts, tables, and reports

- Simplicity of entering production data such as raw materials and output yields
- Accessibility of real-time analytics and reports
- User-friendliness for production staff and managers with minimal technical background

This ensured that the system could be effectively used by ALDi Foods Inc. personnel without extensive training.

Accuracy

Accuracy refers to the correctness and precision of the system's data processing and output generation. This was a critical criterion because the system was focused on production analytics and efficiency monitoring.

The system was evaluated based on:

- Correct computation of yield percentage values
- Accurate extraction efficiency calculations
- Precise classification of production status (Stable or Not Stable)
- Consistency of reported data across different time periods
- Accuracy of raw material usage reports and historical records

This ensured that all analytical outputs reflected the actual production performance of the organization.

Efficiency

Efficiency measures the system's ability to process data and deliver results with minimal delay and optimal resource usage. Since the system supported real-time monitoring, efficiency was a critical requirement.

The system was evaluated based on:

- Speed of data processing from input to dashboard display
- Real-time updating of production metrics and analytics
- Fast generation of reports (weekly, monthly, quarterly, yearly)
- Minimal system lag during simultaneous data input and monitoring
- Effective handling of multiple production records without performance degradation

This ensures that the system supports timely decision-making in production operations.

Reliability

Reliability refers to the system's ability to operate consistently and correctly over time without failure. It ensures that the system remains stable during continuous use in a production environment.

The system will be evaluated based on:

- Stability of system operations during continuous data input
- Consistency of real-time analytics and dashboard updates
- Ability to recover or handle errors without data loss
- Proper storage and retrieval of historical production records
- Dependability of system outputs under normal operational conditions

This criterion ensures that the system can be trusted for daily production monitoring and long-term operational use.

Summary of Evaluation Approach

To evaluate the overall impact of the new system, selected users from AIDi Foods Inc., including frontline production staff and upper management, completed a comprehensive Likert scale questionnaire. This structured survey captured standardized feedback regarding user experience, system functionality, and daily operational changes. By involving both the staff on the factory floor and the decision-makers in leadership, the evaluation ensured a well-rounded perspective on how the system performed across different levels of the organization.

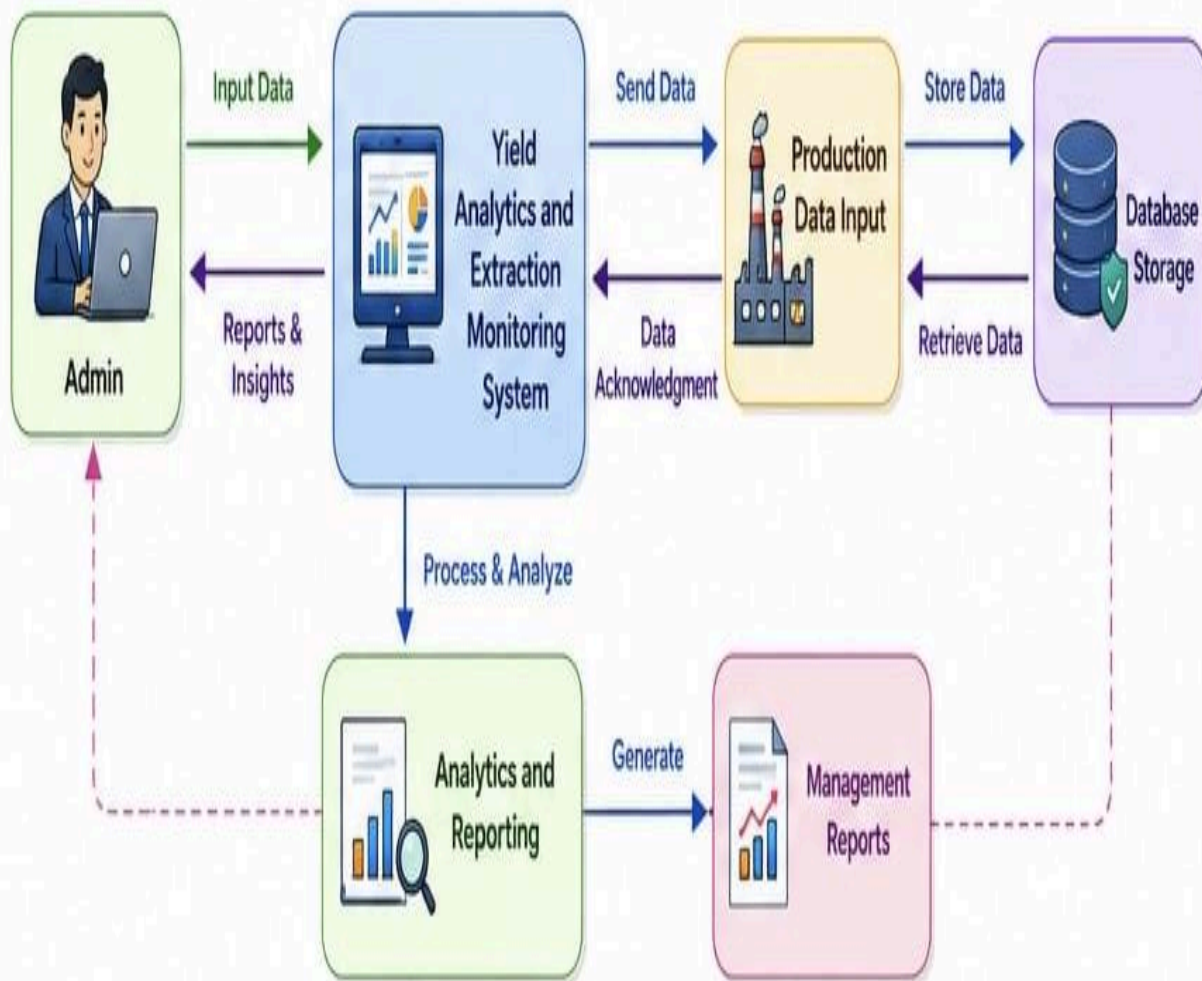
The data gathered from these evaluation results served as the primary metric to determine the overall effectiveness of the platform. Specifically, the feedback measured how successfully the digital platform replaced long-standing manual processes across the facility. Transitioning away from paper-based tracking represented a major operational shift, and this assessment pinpointed how smoothly the team had adapted to the digital transition.

Furthermore, the evaluation heavily focused on the system's ability to support real-time digital production monitoring throughout daily operations. Frontline data entry and automated tracking had to seamlessly translate into live updates that reflected current floor conditions. The outcome of this assessment confirmed whether the technology consistently delivered the accurate, up-to-the-minute data stream required for modern manufacturing.

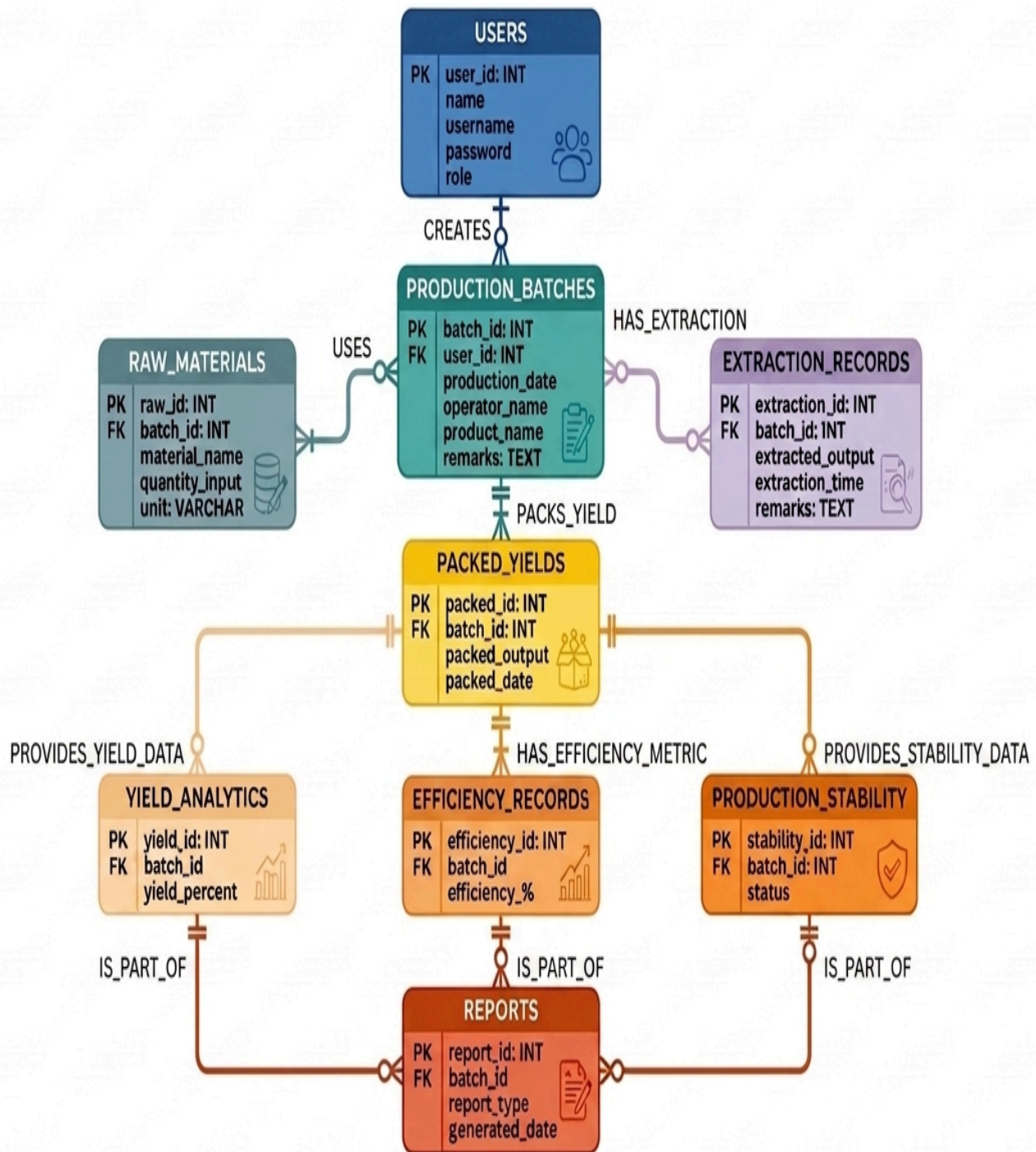
In addition to real-time tracking, the survey examined how well the system improved collaboration between different operational tiers. By capturing metrics on data accessibility, the evaluation demonstrated how communication between floor workers and management improved once manual bottlenecks were eliminated. This ensured that operational feedback was converted into actionable insights for continuous floor improvements.

Ultimately, the final results validated whether the system successfully achieved its core operational goals across all parameters. A positive evaluation proved that the platform improved production visibility, enhanced management decision-making, and drastically reduced manual data-entry errors. Most importantly, it confirmed that the system effectively supported the precise, efficient management of critical yield and extraction processes.

Overall, the comprehensive evaluation process confirmed that the technological transition yielded significant improvements in workflow efficiency, operational control, and data accuracy. By successfully replacing manual record-keeping with an automated solution, ALDi Foods Inc. established a solid foundation for sustained, data-driven production management.

DFD (Data Flow Diagram)**Figure 4.5 – Data Flow Diagram (DFD – Level 0)**

ERD (Entity Relationship Diagram)



Users

The Users entity represented production personnel who interacted with the system. Users were responsible for creating production batches, encoding raw material data, and recording packed yield outputs. Users played a central role in the system because they provided the primary production data used for real-time analytics and efficiency monitoring.

Relationships:

- A User could create multiple Production Batches.
- A User could record multiple Pack Yield entries under different batches.

Production Batches

The Production Batch entity represented a single production run or cycle within ALDi Foods Inc. Each batch contained detailed information about the production process, including raw materials used and outputs produced. This entity served as the main container of production data used for yield computation and extraction efficiency analysis.

Relationships:

- A Production Batch belonged to one User.
- A Production Batch contained multiple Batch Materials records. A
- Production Batch could generate multiple Pack Yield records.

Raw Materials

The Raw Materials entity stored information about raw materials used in each production batch. This included the quantity of raw inputs such as fruits or raw juice materials used during

the extraction process. This entity was essential for calculating production efficiency and yield percentage.

Relationships:

- Batch Materials belonged to a specific Production Batch.
- Each Production Batch could have multiple material entries depending on production requirements.

Extraction Records

The Extraction Records entity captures intermediate output data obtained during the processing or extraction phase of a production batch. It logs extracted quantities, timestamps, and operator remarks to track step-by-step progress before final packaging. This entity helps monitor inline operational yield and spot losses occurring mid-process.

Relationships:

- Multiple Extraction Records can be logged for one Production Batch.

Pack Yields

The Pack Yield entity represented the final output of the production process. It recorded the amount of finished products (e.g., bottled or packed juice) produced from a specific production batch. This data was crucial for real-time yield analytics and extraction efficiency computation. ***Relationships:***

- Pack Yield belonged to a Production Batch. A Production Batch could produce multiple
- Pack Yield records depending on product variations or packaging types.

Yield Analytics

The Analytics entity was a standalone system module responsible for processing production data into meaningful insights. It handled real-time computation and visualization of production performance. This entity was central to the system's objective of providing digital production intelligence.

Functions of Analytics:

- Computed real-time yield percentage
- Analyzed extraction efficiency performance
- Generated production trends and visual reports
- Classified production status (Stable / Not Stable)
- Generated periodic reports (weekly, monthly, quarterly, yearly)

Relationships:

- Analytics was independent but used data from Production Batch, Batch Materials, and Pack Yield.

Efficiency Records

The Efficiency Records entity tracks performance efficiency rates for individual production runs. It evaluates expected output standard rates against actual packed output to identify operational bottlenecks and measure worker/machine throughput.

Relationships:

- An Efficiency Record belongs to one Production Batch.
- Efficiency Records data can be compiled into multiple Reports.

Production Stability

The Production Stability entity logs batch consistency metrics and quality status updates throughout the production cycle. It ensures that outputs meet safety, stability, and quality standards before final distribution.

Relationships:

- A Production Stability record belongs to one Production Batch.
- Production Stability data can be compiled into multiple Reports.

Reports

The Reports entity aggregates operational, yield, efficiency, and stability metrics into formal documents. It tracks report types and generation dates to support management oversight, compliance audits, and long-term decision-making.

Relationships:

- A Report compiles analytical data originating from a Production Batch (including Yield Analytics, Efficiency Records, and Production Stability).

Summary of Relationships

Users

- **Creates** multiple Production Batches.
- **Records** multiple Pack Yield entries across various batches.

Production Batches

- **Belongs to** one User.
- **Contains** multiple Raw Materials (Batch Materials) entries.
- **Generates** multiple Extraction Records.
- **Produces** multiple Pack Yield records (for different variations or packaging types).
- **Has** one Efficiency Record.
- **Has** one Production Stability record.

Raw Materials (Batch Materials)

- **Belongs to** one specific Production Batch.

Extraction Records

- **Belongs to** one specific Production Batch (logs mid-process intermediate step data).

Pack Yields

- **Belongs to** one Production Batch.
- **Recorded by** a User.

Yield Analytics

- **Operates independently** while drawing source data directly from Production Batches, Batch Materials, and Pack Yields.

Efficiency Records

- **Belongs to** one Production Batch.

- **Compiles into** multiple Reports.

Production Stability

- **Belongs to** one Production Batch.
- **Compiles into** multiple Reports.

Reports

- **Aggregates** analytical data originating from Production Batches, including Yield Analytics, Efficiency Records, and Production Stability metrics.

System Role in Production Monitoring

This ERD structure supports the main goal of the system, which is to digitize production monitoring at AlDi Foods Inc. It enables:

- **Real-time tracking of raw material usage** by logging specific inputs, quantities, and units mapped directly to each active production batch.
- **Automated yield computation** through the continuous integration of raw material inputs and packed yield outputs to determine yield percentages.
- **Extraction efficiency monitoring** by capturing mid-process operational logs, timestamps, and output volumes during intermediate extraction stages.
- **Production stability classification** by logging batch consistency metrics to automatically categorize quality and operational status.
- **Centralized reporting and analytics** by consolidating yield metrics, efficiency records, and stability status into downloadable management reports.

By structuring the system in this way, the platform ensured accurate, efficient, and real-time monitoring of production activities, replacing manual record-keeping processes and improving decision-making capabilities.

Use Case Diagram

The Use Case Diagram illustrated the interactions between the system and its primary actors within the Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for ALDi Foods Inc. It clearly defined the functional requirements of the platform and demonstrated how each actor engaged with the system to facilitate production monitoring, data analytics, and operational reporting.

The system replaced manual production tracking by establishing a centralized digital platform. This transformation enabled the real-time computation of yield metrics, extraction efficiency analysis, production stability interpretations, and automated report generation across the entire manufacturing workflow.

Actors

The system consisted of two primary actors:

User

The User represented production staff or operators responsible for encoding production data into the system. Users were directly involved in recording real-time production information such as raw material usage, extraction output, and packed yield data. The User played a critical

role in ensuring that production data was accurate, complete, and updated in real time, which was essential for generating reliable analytics and efficiency reports.

Administrator

The Administrator was responsible for overseeing the entire system. This actor managed user accounts, monitored production performance, reviewed analytics dashboards, and generated reports for management decision-making. The Administrator had full access to system data and was responsible for evaluating production performance trends, monitoring efficiency levels, and ensuring data integrity across all modules.

Register / Manage Admin Account

This use case allowed the Administrator to create, update, and manage admin accounts within the system. It ensured that only authorized personnel had access to sensitive production analytics and system configuration features.

User Login

Both Users and Administrators had to authenticate through the login system to access their respective functionalities. This ensured system security and controlled access to production data and analytics.

Production Management

This was the core module of the system where Users input all production-related data. It supported real-time monitoring and served as the foundation for yield analytics and extraction efficiency computation.

Sub-functions included:

Input Raw Materials (Kilos)

Users encoded the quantity of raw materials used in production batches. This data was essential for computing yield and efficiency.

Input Extraction Kilos

Users recorded the amount of juice or extracted output obtained from the raw materials during production.

Input Packed Yield

Users entered the final packaged product output, which was used to evaluate production losses and efficiency.

Inventory Yield Tracking

This use case allowed the system to automatically track and compute production yield based on raw material input and output data. It provided real-time updates on:

- Yield percentage per batch
- Production output trends
- Material utilization efficiency
- Extraction loss monitoring

This feature supports continuous monitoring of production performance and helps identify inefficiencies early.

Analytics & Reporting (Administrator)

This use case enabled the Administrator to access and analyze system-generated production data. It included real-time dashboards and structured reports that supported decision-making.

Key functionalities included:

- Real-time yield analytics visualization
- Extraction efficiency performance analysis
- Production stability interpretation (Stable / Not Stable)
- Trend analysis of production performance
- Generation of periodic reports:
 - Weekly raw material usage report
 - Monthly production summary report
 - Quarterly performance analysis
 - Yearly production evaluation report

This module transforms raw production data into meaningful insights for management and operational improvement.

Manage History Logs

This use case allowed the Administrator to view and manage historical production records stored in the system database. It provided access to past production batches, yield records, and efficiency reports.

This supports:

- Audit trail of production activities
- Historical comparison of yield performance
- Detection of recurring inefficiencies
- Long-term production analysis and reporting

System Interaction Summary

The system operated through a structured and seamless interaction between Users and the Administrator, establishing a cohesive workflow across the entire operational environment. By defining clear roles and responsibilities, this framework ensured that data flowed efficiently from initial entry to high-level management review.

Frontline Users were primarily responsible for encoding real-time production data directly at the operational level. By capturing critical metrics such as raw material usage, extraction output, and final packaged yield right as they occurred, these operators provided the essential groundwork for accurate data collection.

Once these inputs were recorded, the system systematically processed the raw operational data into actionable intelligence. It automatically calculated real-time yield percentages, measured extraction efficiency, and categorized overall production stability, effectively replacing the need for manual calculations.

The Administrator then utilized these automated outputs to oversee plant-wide operations, monitor ongoing performance trends, and generate detailed analytical reports. Access to these clear, data-driven insights allowed management to identify operational bottlenecks quickly and make informed, strategic decisions to optimize output.

Through this structured interaction, the platform successfully achieved its primary objective of digitizing production processes at ALDi Foods Inc. By integrating real-time monitoring, automated analytics, and streamlined reporting, the system significantly improved operational efficiency and transformed daily manufacturing workflows.

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULT, AND DISCUSSION

4.2 Overview of the Developed System

The Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for ALDi Foods Inc. was a web-based application developed to modernize and improve the company's existing production monitoring process. The proposed system replaced the traditional manual and paper-based recording of production information with a centralized digital platform that enabled administrators to record, process, analyze, and retrieve production data efficiently. By utilizing digital technologies, the system addressed the limitations of manual production monitoring while providing accurate and timely information that supported operational efficiency and management decision-making.

The development of the proposed system was based on the findings presented in Chapters 1, 2, and 3 of the study. The researchers identified several operational challenges in the existing production monitoring process, including delayed recording of production information, inaccurate manual computations, duplication of records, difficulty in retrieving historical production data, and limited visibility of production performance. These problems affected the company's ability to monitor production yield, evaluate extraction efficiency, and make timely production decisions. Consequently, the researchers designed and developed a web-based production monitoring system that automated these processes while maintaining a centralized database for efficient data management.

The proposed application provided an integrated platform that allowed administrators to encode production batch information, automatically compute production yield and extraction efficiency, monitor production performance through graphical dashboards, and generate production reports. Unlike the existing manual process, the developed system performed automatic calculations once production information was entered, significantly reducing computational errors and minimizing the time required to prepare production summaries. This automation enabled production personnel and management to obtain accurate information immediately after data entry, thereby improving production monitoring and operational control.

The system was developed using PHP as the server-side programming language, HTML, CSS, and JavaScript for designing the user interface, Chart.js for graphical data visualization, and MySQL as the centralized database management system. These technologies were selected because they were reliable, cost-effective, widely supported, and suitable for developing web-based information systems for small and medium-sized enterprises (SMEs) such as ALDi Foods Inc. The application was deployed through the XAMPP development environment, allowing administrators to access the system through a standard web browser within the organization's local network.

One of the primary features of the proposed system was the Production Monitoring Module, which enabled administrators to encode production batch information such as production date, raw material quantity, operator information, and production output. These production records served as the foundation for all succeeding computations and analyses performed by the system. By maintaining complete and organized production records, the application provided a reliable source of information for evaluating production activities and monitoring operational performance.

Another important feature of the system was the Yield Analytics Module, which automatically computed the production yield percentage using the recorded production input and output values. The computed yield percentage allowed administrators to evaluate how efficiently raw materials were converted into finished products. Since the computation was performed automatically by the system, the possibility of manual computational errors was significantly reduced. The resulting values were immediately displayed on the dashboard through charts and summary tables, enabling administrators to evaluate production performance without performing manual calculations.

Complementing the yield analysis was the Extraction Efficiency Monitoring Module, which evaluated the effectiveness of the juice extraction process by computing the extraction efficiency percentage for every production batch. This module allowed production personnel and management to monitor extraction performance continuously and determine whether the extraction process was operating efficiently. The generated information assisted management in identifying production trends, recognizing variations in extraction performance, and implementing corrective actions whenever necessary to improve operational efficiency.

The proposed system also included a Reporting Module that automatically generated production reports based on the stored production information. These reports summarized production activities, production yield, extraction efficiency, and other relevant production data within a selected period. The reporting function minimized the time required to prepare production documents and enabled administrators to retrieve historical production reports whenever needed. Through centralized report management, management personnel were able to compare production performance across multiple production batches and evaluate long-term production trends.

To ensure data integrity and accessibility, all production information was stored in a centralized MySQL database. Centralized storage eliminated redundant records, improved data consistency, and allowed authorized administrators to retrieve historical production information quickly. Since all production records were maintained within a single database, the system provided better organization of production information while reducing the possibility of data loss associated with paper-based record keeping.

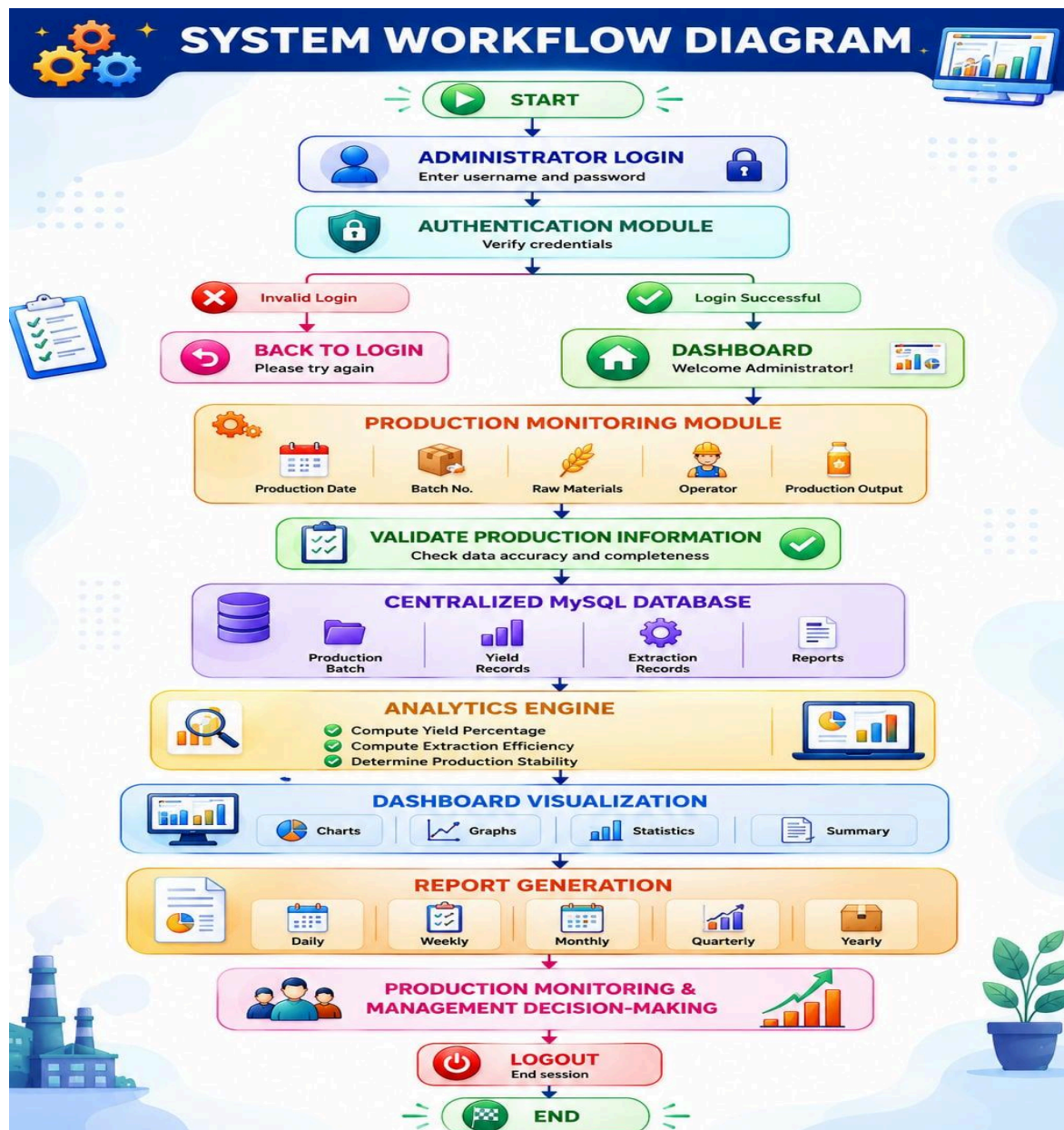
Security was also incorporated into the proposed application through the Administrator Module, which required authorized users to log in before accessing the system. User authentication restricted unauthorized access to production information and ensured that only designated administrators could encode production records, modify existing information, monitor production performance, and generate reports. This security mechanism protected the integrity and confidentiality of the company's production data.

The integration of these functional modules provided ALDi Foods Inc. with a comprehensive production monitoring solution capable of supporting daily production operations. By replacing manual recording procedures with an automated web-based application, the proposed system improved production monitoring, enhanced data accuracy, reduced processing time, minimized computational errors, and supported data-driven decision-making. Furthermore, the centralized storage of production information provided management with reliable historical records that could be used to evaluate operational performance, monitor production consistency, and identify opportunities for continuous process improvement.

Overall, the developed system demonstrated how digital technologies could be applied within the food manufacturing industry to improve production monitoring and operational efficiency.

Through automated computations, centralized database management, graphical visualization, and report generation, the proposed application provided a practical and cost-effective solution that addressed the production monitoring requirements of ALDi Foods Inc. while supporting the objectives of the study.

Figure 4.1 illustrates the **System Workflow Diagram** of the system.



The workflow began with the Administrator logging in and passing authentication to access the dashboard, where production details such as batch numbers, raw materials, and output were encoded and validated before being stored in the centralized MySQL database. Once stored, the system processed this data through an analytics engine to calculate yield and efficiency, presenting visual insights on the dashboard and generating periodic reports that empowered management decision-making prior to session logout.

4.3 System Development Methodology

The development of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for ALDi Foods Inc.* followed the Agile Software Development Methodology, an iterative and incremental approach to software development that emphasized flexibility, continuous collaboration, and regular evaluation throughout the project lifecycle. Agile was selected because it enabled the researchers to divide the project into manageable phases while allowing continuous refinement of the system based on user feedback and evaluation. Unlike traditional software development methodologies, Agile encouraged frequent testing, evaluation, and enhancement of each completed module before proceeding to the next development stage. This approach ensured that the proposed system remained aligned with the operational requirements of ALDi Foods Inc. and effectively addressed the identified problems in the existing production monitoring process.

The adoption of the Agile methodology also supported the developmental nature of this research. Since the objective of the study was to design, develop, implement, and evaluate a web-based production monitoring system, Agile provided an organized framework for developing the

application through successive iterations. Every completed feature was reviewed, tested, and improved before additional functionalities were implemented. This continuous cycle of planning, development, testing, and feedback enabled the researchers to identify and resolve system issues early in the development process, thereby improving software quality, minimizing development risks, and ensuring that the final application satisfied the functional and non-functional requirements identified during the requirement analysis phase.

The Agile development process adopted in this study consisted of seven major phases, namely Planning, Requirement Analysis, System Design, Development, Testing, Deployment, and Review and Feedback. Each phase contributed to the successful completion of the proposed system and ensured that every module functioned efficiently before the application was deployed.

Planning

The development process began with the Planning phase, where the researchers established the overall direction of the project. During this stage, the objectives, scope, limitations, expected outputs, project timeline, and development resources were identified and documented. The researchers carefully examined the existing production monitoring practices of ALDi Foods Inc. to understand how production information was being recorded, processed, and reported.

Through interviews with production personnel and direct observation of the company's production operations, the researchers identified several challenges associated with the existing manual recording process. These included delayed submission of production reports, inaccurate manual computations, inconsistent recording of production data, duplication of information,

difficulty in retrieving historical production records, and limited visibility of production performance. These findings became the foundation for defining the functional scope of the proposed web-based system.

During the planning stage, the researchers also selected the technologies that would be used to develop the application. The system was designed using PHP for server-side programming, HTML, CSS, and JavaScript for developing the user interface, Chart.js for graphical visualization of production data, MySQL as the centralized database management system, and XAMPP as the local web server environment. These technologies were selected because they were reliable, open-source, cost-effective, and appropriate for small and medium-sized manufacturing enterprises.

Requirement Analysis

After completing the planning activities, the researchers proceeded to the Requirement Analysis phase. The primary objective of this stage was to identify all functional and non-functional requirements needed to develop the proposed system successfully.

The researchers gathered information through interviews with production supervisors and personnel, direct observation of the juice extraction process, and examination of existing production records and manual log sheets. These activities enabled the researchers to understand the complete production workflow and determine the information required for production monitoring.

Based on the gathered data, the researchers identified several essential functional requirements of the system. These included secure administrator login, recording of production batch

information, automatic computation of production yield, automatic computation of extraction efficiency, dashboard visualization of production performance, centralized database management, historical data retrieval, and automated generation of production reports.

In addition to the functional requirements, several non-functional requirements were also identified. The system needed to provide a user-friendly interface, ensure reliable system performance, maintain data accuracy and integrity, support secure administrator access, and provide efficient retrieval of production information. These requirements served as the basis for designing the system architecture and implementing the application.

System Design

After identifying the system requirements, the researchers proceeded to the System Design phase. During this stage, the overall structure of the application was carefully planned before actual coding began. The researchers prepared the system architecture, database structure, user interface layouts, process flow diagrams, and database relationships to serve as blueprints during software development.

The System Workflow Diagram was designed to illustrate the Agile development process followed throughout the project. The System Architecture Diagram defined the interaction between the administrator and the major system components responsible for production monitoring, analytics, reporting, and database management. The Entity Relationship Diagram (ERD) described the relationships among the five primary database entities, namely Admin, Production Batch, Yield Record, Extraction Record, and Reports. The Data Flow Diagram (DFD Level 0) illustrated the overall movement of production data from the administrator to the database and ultimately to report generation.

The graphical user interface was also designed during this phase. The interface emphasized simplicity, consistency, and ease of navigation so that administrators could efficiently record production information, monitor production performance, generate reports, and access historical production records with minimal training.

Development

The Development phase involved translating the completed system design into a fully functional web-based application. The researchers implemented the application using PHP as the primary server-side programming language, while HTML, CSS, and JavaScript were utilized for developing responsive and interactive web pages. Chart.js was integrated into the application to provide graphical visualization of production yield and extraction efficiency, while MySQL was used to store and manage production records within a centralized database.

The development process followed Agile principles by implementing each module incrementally. The Administrator Module was developed first to provide secure user authentication and system access. This was followed by the Production Monitoring Module, which enabled administrators to encode production batch information such as production date, raw material quantity, operator information, and production output.

The Analytics Module was then developed to automatically compute production yield and extraction efficiency using the encoded production information. The module also generated graphical charts and summary tables that allowed administrators to monitor production performance more effectively. Finally, the Reporting Module was implemented to generate production reports and summaries that assisted management in evaluating production operations and making informed decisions.

Continuous integration and module verification were performed throughout development to ensure that newly implemented features operated correctly with previously completed modules.

Testing

Following the completion of system development, the application underwent a comprehensive Testing phase to verify that all modules operated according to the specified requirements. The primary objective of testing was to ensure that the application functioned correctly, produced accurate computations, maintained data integrity, and satisfied the operational requirements of AIDi Foods Inc.

The researchers initially performed Unit Testing, where each module was tested individually to verify its functionality. The login module, production monitoring module, analytics module, reporting module, and database operations were evaluated independently before integration.

After verifying individual modules, Integration Testing was conducted to ensure that all modules communicated correctly with one another. This phase confirmed that production information entered into the system was successfully stored in the database, automatically processed by the analytics module, and correctly displayed in the reports and dashboard.

The researchers also conducted System Testing to evaluate the performance of the complete application under normal operating conditions. Finally, User Acceptance Testing (UAT) was performed with selected personnel from AIDi Foods Inc. to determine whether the developed system met the company's operational requirements and expectations. Any identified software defects were corrected before the application was deployed.

Deployment

Upon successful completion of testing, the project entered the Deployment phase. During this stage, the developed application was installed and configured within the operational environment of ALDi Foods Inc. The researchers configured the Apache web server using XAMPP, initialized the MySQL database, created administrator accounts, and imported the required database tables.

The intended users were oriented on the proper use of the application, including procedures for logging into the system, encoding production information, monitoring production performance, generating reports, and retrieving historical production records. Sample production batches were also encoded to verify that the deployed application functioned correctly within the production environment.

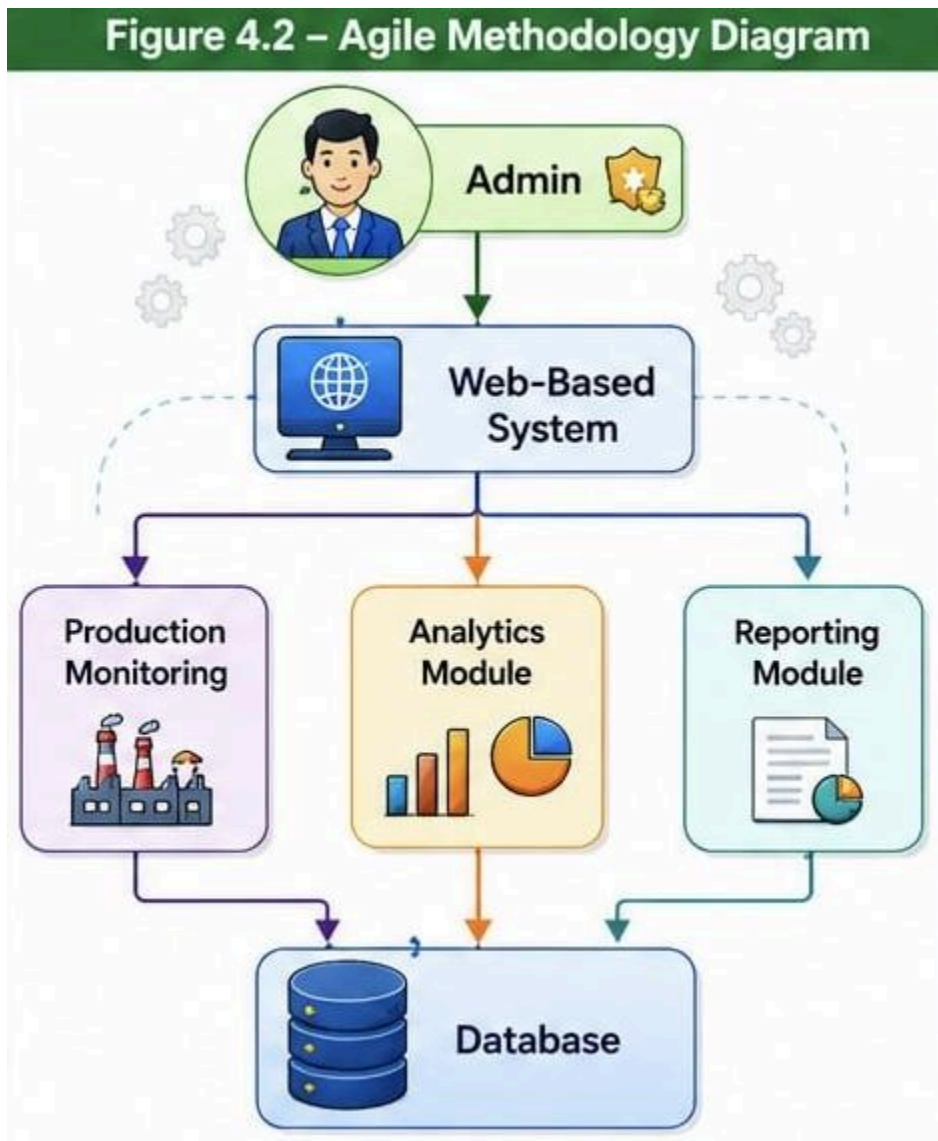
Deployment marked the transition of the proposed system from the development environment to actual operational use.

Review and Feedback

The final phase of the Agile methodology involved Review and Feedback. After deployment, the developed application was evaluated by its intended users to determine its overall effectiveness, usability, reliability, and functionality. Administrators and production personnel provided comments and recommendations regarding the operation of the application, user interface, report generation, dashboard visualization, and overall system performance. The researchers documented all observations and suggestions for future enhancement.

Consistent with Agile principles, the collected feedback served as the foundation for future development iterations. The continuous evaluation and improvement of the application ensured that the system remained responsive to the operational needs of ALDi Foods Inc. while providing reliable production monitoring, accurate yield analytics, efficient extraction efficiency monitoring, and effective production report generation.

Figure 4.2 presents the interaction between the Administrator and the Web-Based System, highlighting the major functional modules of the proposed application.



As illustrated in Figure 4.2, the Administrator serves as the primary user of the web-based application and is responsible for managing all production monitoring activities. After successfully logging into the system, the administrator interacts with the Web-Based System, which serves as the central platform for recording, processing, storing, and presenting production information.

The application is composed of three major functional modules: the Production Monitoring Module, the Analytics Module, and the Reporting Module. The Production Monitoring Module enables administrators to encode production batch information, including the production date, raw material quantity, operator information, and production output. These production records are stored securely in the centralized MySQL database, ensuring data consistency and efficient retrieval of historical production information.

The Analytics Module retrieves production data from the database and automatically computes the production yield percentage and extraction efficiency for each production batch. The computed results are displayed through graphical dashboards, charts, and summary tables, allowing administrators to monitor production performance, evaluate extraction efficiency, identify operational trends, and assess overall production stability.

The Reporting Module generates comprehensive production reports based on the processed information stored in the database. These reports provide summarized production information, yield analytics, extraction efficiency results, and other relevant operational data that support production monitoring, managerial evaluation, and data-driven decision-making.

The interaction illustrated in Figure 4.2 demonstrates how the proposed system integrates production monitoring, analytics, database management, and reporting into a single web-based

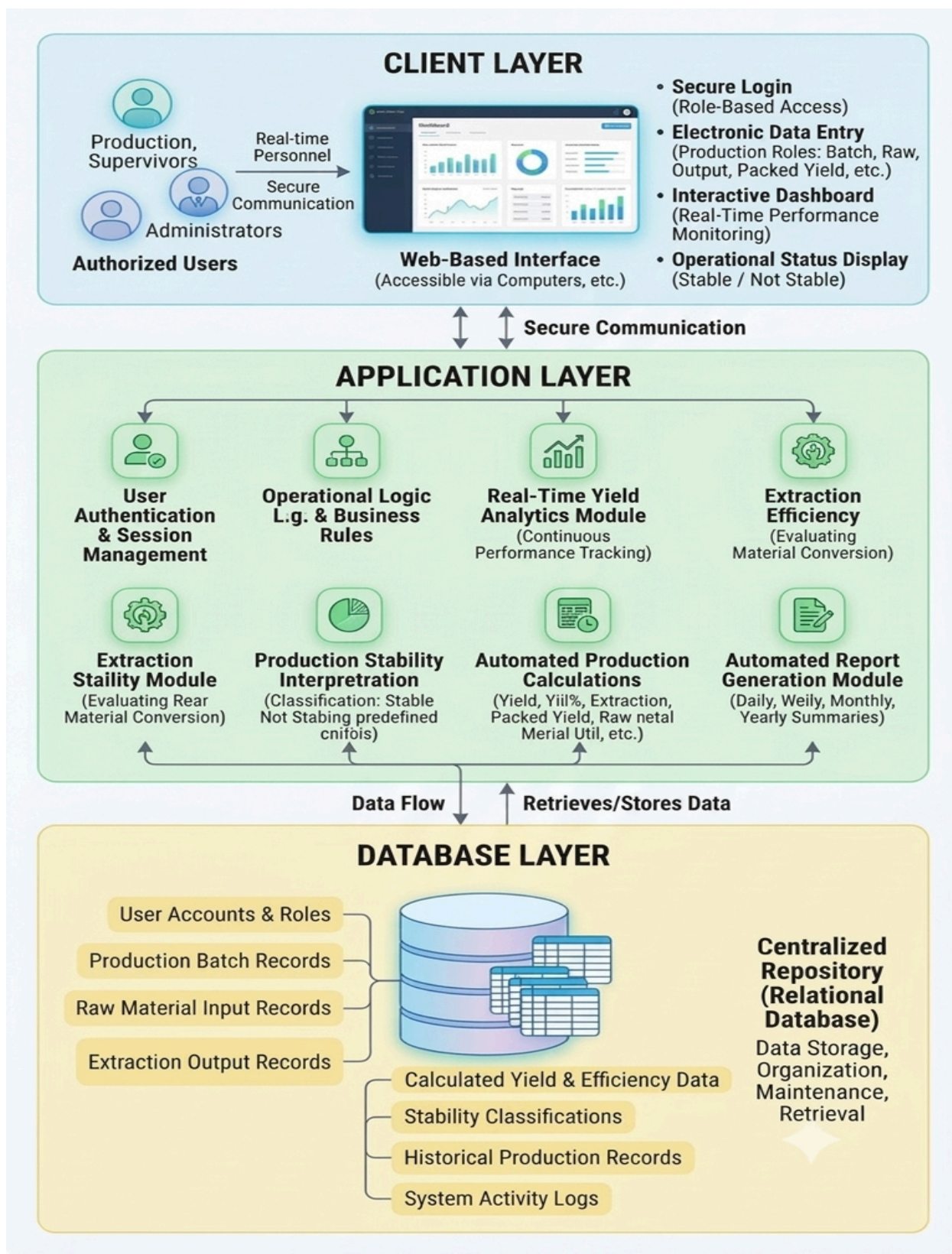
platform. Throughout the Agile development process, feedback obtained from administrators and intended users was continuously incorporated into succeeding iterations, ensuring that the developed application remained reliable, user-friendly, efficient, and aligned with the production monitoring requirements of ALDi Foods Inc.

4.4 System Architecture

The system architecture for ALDi Foods Inc. is separated into three core layers (a three-tier architecture):

- Client Layer (Presentation Layer): The web-based user interface where authorized users (production personnel, supervisors, and administrators) securely log in, enter electronic production data, and view real-time dashboards and reports.
- Application Layer (Logic Layer / Central Processing Engine): The core engine that processes requests, validates submitted data according to business rules, performs automated calculations (yield %, extraction efficiency), handles status classifications ("Stable" vs. "Not Stable"), and manages automated report generation.
- Database Layer (Data Management Layer / Centralized Repository): The centralized relational database responsible for securely storing, organizing, retrieving, and maintaining all system records, including batch data, calculations, user roles, stability

Figure 4.3 illustrated the overall System Architecture of the Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for ALDi Foods Inc.



4.5 Hardware and Software Requirements

The successful implementation of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for ALDi Foods Inc.* requires appropriate hardware and software resources to ensure the efficient operation of the proposed web-based application. Since the system is responsible for recording production batch information, automatically computing production yield and extraction efficiency, generating graphical dashboards, and producing management reports, reliable computing resources are necessary to maintain system performance, data accuracy, and operational efficiency.

The hardware and software requirements were carefully selected based on the functional requirements identified during the system analysis and design phases. These resources provide the necessary environment for the installation, development, testing, deployment, and daily operation of the proposed application. The selected technologies are widely available, cost-effective, and compatible with the existing information technology infrastructure of ALDi Foods Inc., making the proposed system practical for implementation without requiring expensive specialized equipment.

The hardware specifications ensure that the application can efficiently process production data, perform automated computations, retrieve historical records, generate graphical visualizations, and support report generation with minimal processing delays. Likewise, the software requirements provide the development framework, programming environment, database management capabilities, and web technologies needed to build and operate the application successfully.

Since the proposed system utilizes a centralized database, all production information is securely stored and managed within a single repository. This centralized approach improves data consistency, minimizes data redundancy, and enables administrators to retrieve production records quickly whenever monitoring, analysis, or report generation is required. The recommended hardware specifications also provide sufficient processing power and storage capacity to accommodate increasing production records as the company continues to expand its operations.

The software technologies selected for the development of the system are based on open-source and industry-standard applications that are widely used in web development. These technologies support the implementation of the system's major functionalities, including secure administrator authentication, production batch management, automated yield computation, extraction efficiency monitoring, dashboard visualization, centralized database management, and automated report generation. The use of open-source software also minimizes implementation costs while ensuring that the system remains maintainable, scalable, and easy to update in the future.

Furthermore, the combination of the recommended hardware and software provides a stable and reliable environment for the continuous operation of the proposed application. The selected components ensure that the system can perform real-time production monitoring, efficiently process production information, maintain accurate historical records, and provide administrators with timely analytical reports that support operational planning and decision-making.

The following sections present the **minimum and recommended hardware specifications** as well as the **software requirements** necessary for the successful development, deployment, and implementation of the proposed **Digitizing Production: Real-Time Yield Analytics and**

Extraction Efficiency Monitoring System for ALDi Foods Inc. These requirements serve as the technical foundation that enables the application to operate efficiently while achieving the objectives of improving production monitoring, minimizing manual computation, enhancing data accuracy, and supporting data-driven decision-making within the company.

4.5.1 Hardware Requirements

The hardware requirements specify the minimum and recommended computer specifications necessary for the successful implementation and operation of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for ALDi Foods Inc.* These hardware components provide the computing resources needed to support the installation, execution, and maintenance of the proposed web-based application. Selecting appropriate hardware ensures that the system can perform efficiently while processing production information, executing automated computations, storing production records, and generating analytical reports without experiencing significant delays or performance issues.

Since the proposed system operates as a web-based application, the hardware must be capable of supporting server-side processing, database management, and real-time interaction between the administrator and the application. The recommended specifications were selected based on the expected workload of the system, including encoding production batch information, computing yield percentage and extraction efficiency, generating graphical dashboards through Chart.js, retrieving historical production records, and producing management reports.

Adequate processing power is essential because the application performs multiple operations simultaneously, including validating user inputs, processing production data, executing automated computations, communicating with the MySQL database, and displaying analytical results through graphical visualizations. A computer with a higher-performance processor enables the application to execute these operations more efficiently, resulting in faster response times and improved user experience.

Memory (RAM) also plays an important role in ensuring smooth system performance. Sufficient memory allows the web browser, web server, database server, and other supporting applications to run simultaneously without affecting system responsiveness. As production records continue to increase over time, additional memory helps maintain stable system performance while retrieving historical production information and generating reports.

Storage capacity is another important hardware consideration because the application stores production batch information, yield records, extraction efficiency records, generated reports, and other related system files within the database. The use of **Solid-State Drive (SSD)** storage is recommended because it provides significantly faster data access, shorter application loading times, quicker database transactions, and improved overall system performance compared to traditional hard disk drives (HDD). Faster storage also enhances the retrieval of historical production records and improves the efficiency of report generation.

The monitor specification ensures that administrators can comfortably view the system's user interface, production forms, dashboards, charts, graphs, and generated reports. A higher display resolution provides clearer visualization of production analytics, allowing administrators to analyze production performance more effectively. Standard USB keyboard and mouse devices

provide the necessary input tools for encoding production information, navigating the system, and performing administrative tasks efficiently.

Although the proposed system can operate within a local network environment, a reliable internet or network connection is recommended to support system accessibility, software updates, and future deployment in a networked production environment. A stable connection also facilitates faster communication between the web server and client devices when the application is accessed through multiple workstations.

Table 4.1 presents the minimum and recommended hardware specifications required for the implementation of the proposed system.

Table 4.1 Hardware Requirements of the Proposed System

Hardware Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 or AMD Ryzen 3	Intel Core i5 or AMD Ryzen 5 and above
Memory (RAM)	4 GB	8 GB or higher
Storage	256 GB SSD	512 GB SSD or higher

Monitor	1366 × 768 Resolution	Full HD (1920 × 1080)
Keyboard and Mouse	Standard USB Keyboard and Mouse	Standard USB Keyboard and Mouse
Internet Connection	10 Mbps	50 Mbps or higher

The minimum hardware specifications are sufficient to install and operate the proposed web-based application under normal production monitoring conditions. These specifications allow administrators to encode production information, perform automated yield and extraction efficiency computations, access production records, and generate reports with acceptable system performance.

However, the recommended hardware specifications provide improved processing capability, faster database operations, greater storage capacity, and enhanced responsiveness, particularly when handling larger volumes of production data and generating graphical dashboards and management reports. The additional computing resources also improve the overall reliability of the application during continuous daily operation and reduce the likelihood of performance bottlenecks.

Overall, the recommended hardware configuration provides a stable and efficient computing environment for the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for ALDi Foods Inc.* By utilizing these hardware resources, the proposed

system can efficiently process production information, maintain accurate production records, generate real-time analytical results, and support effective production monitoring and decision-making. Furthermore, the selected hardware specifications provide sufficient capacity to accommodate future expansion of the system as additional production data and functional requirements are introduced.

4.5.2 Software Requirements

The proposed *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for ALDi Foods Inc.* was developed using widely available, reliable, and open-source software technologies to ensure cost-effectiveness, compatibility, ease of maintenance, and long-term scalability. The selected software components provide the necessary development environment, web server, database management system, programming languages, and development tools required to build, deploy, and operate the proposed web-based application successfully.

The software requirements were carefully selected based on the functional and technical requirements identified during the system analysis and design phases of the study. Each software component performs a specific role in supporting the overall operation of the application, from developing the user interface and implementing the system's business logic to storing production information and generating graphical dashboards and management reports.

The use of open-source technologies significantly reduces implementation and maintenance costs for ALDi Foods Inc. while ensuring that the application remains flexible, reliable, and easy to

upgrade. Since the proposed system is intended for small- and medium-sized manufacturing operations, the selected software provides a practical and sustainable solution without requiring expensive proprietary software licenses.

The operating environment of the application is based on Microsoft Windows 10 or Windows 11, which provides a stable platform for system development, deployment, and daily operation. The operating system supports all required development tools and server applications while providing compatibility with the selected programming languages and database management system.

For local development and deployment, the researchers utilized XAMPP, an integrated web server package that includes Apache, PHP, and MySQL. XAMPP simplifies the installation and configuration of the development environment by providing all essential server components within a single package. This environment enables the application to run locally during software development, testing, debugging, and demonstration before deployment in the production environment.

The Apache Web Server serves as the application's web server by processing client requests and delivering web pages to users through a web browser. Apache manages communication between the administrator and the application, allowing production information to be entered, processed, retrieved, and displayed efficiently. Its stability and widespread adoption make it suitable for supporting the daily operation of the proposed production monitoring system.

The core business logic of the application was developed using PHP (Hypertext Preprocessor). PHP is responsible for processing administrator requests, validating user inputs, communicating with the MySQL database, performing automated yield and extraction efficiency computations, and generating production reports. Through PHP, the application integrates all functional

modules into a unified web-based platform that supports production monitoring and management decision-making.

The system interface was developed using HTML5, CSS3, and JavaScript, which work together to create a responsive, organized, and user-friendly web application. HTML5 provides the structural foundation of the web pages, CSS3 enhances the visual appearance and layout of the interface, while JavaScript adds interactivity and dynamic functionality. These technologies allow administrators to navigate the application efficiently, encode production information, view analytical dashboards, and generate reports with ease.

To improve data interpretation and production monitoring, the proposed system utilizes Chart.js for data visualization. This JavaScript library enables the application to display production information using charts, graphs, and graphical dashboards. Through visual representations of production yield and extraction efficiency, administrators can quickly identify production trends, monitor operational performance, compare production batches, and evaluate the effectiveness of the juice extraction process. The graphical dashboards provide management with meaningful insights that support timely and data-driven decision-making.

The proposed system uses MySQL as its Database Management System (DBMS). MySQL stores all production-related information, including administrator accounts, production batches, yield records, extraction records, and generated reports. The centralized database ensures that production information is stored securely, organized efficiently, and easily retrievable whenever monitoring, analysis, or report generation is required. The relational database structure also maintains data integrity through the use of primary keys and foreign key relationships, minimizing data redundancy while ensuring consistency across all production records.

During system development, Visual Studio Code served as the primary source code editor. This lightweight yet powerful development environment enabled the researchers to write, organize, debug, and maintain the application's source code efficiently. Its support for multiple programming languages, extensions, and integrated development features significantly improved coding productivity and software quality throughout the implementation phase.

The application is accessed using modern web browsers such as Google Chrome and Microsoft Edge, both of which fully support HTML5, CSS3, JavaScript, and Chart.js technologies. These browsers ensure proper rendering of the system interface, responsive navigation, accurate graphical visualization, and reliable execution of all application functions. Their compatibility with current web standards contributes to a consistent user experience during daily system operation.

To support collaborative software development and maintain version history, the researchers utilized Git as the version control system. Git enabled the researchers to monitor code revisions, manage updates, restore previous versions when necessary, and maintain an organized development workflow throughout the implementation of the project. Version control also improved collaboration among the development team by ensuring that modifications to the source code were properly documented and managed.

Table 4.2 presents the software requirements used in the development and implementation of the proposed system.

Table 4.2 Software Requirements of the Proposed System

Software	Purpose
Windows 10/11	Operating System
XAMPP	Local Web Server Environment
Apache	Web Server
PHP	Server-side Programming Language
HTML5	Web Page Structure
CSS3	User Interface Design
JavaScript	Client-side Scripting
Chart.js	Data Visualization and Graph Generation
MySQL	Database Management System

Visual Studio Code

Source Code Editor

Google Chrome / Microsoft Edge Web Browser

Git

Version Control (Development)

Each of these software components contributes to the successful operation of the proposed *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for ALDi Foods Inc.* The integration of these technologies enables the application to automate production recording, compute yield percentage and extraction efficiency, manage production records within a centralized database, generate graphical dashboards, and produce management reports efficiently.

Furthermore, the selected software technologies provide a stable, secure, and scalable environment that supports the long-term implementation of the proposed system. Their compatibility with one another ensures seamless communication between the web interface, application logic, and database, resulting in reliable system performance and efficient data processing. As production requirements continue to evolve, these technologies also provide a flexible foundation for future system enhancements, including additional analytical features, expanded reporting capabilities, multiple user roles, and integration with other production management functions.

Overall, the software requirements presented in this section provide the technological foundation necessary for the successful development, deployment, and operation of the proposed web-based production monitoring system. By utilizing proven and widely adopted software technologies, the developed application delivers a reliable, user-friendly, and cost-effective solution that supports the production monitoring objectives of ALDi Foods Inc. while improving data accuracy, operational efficiency, and management decision-making.

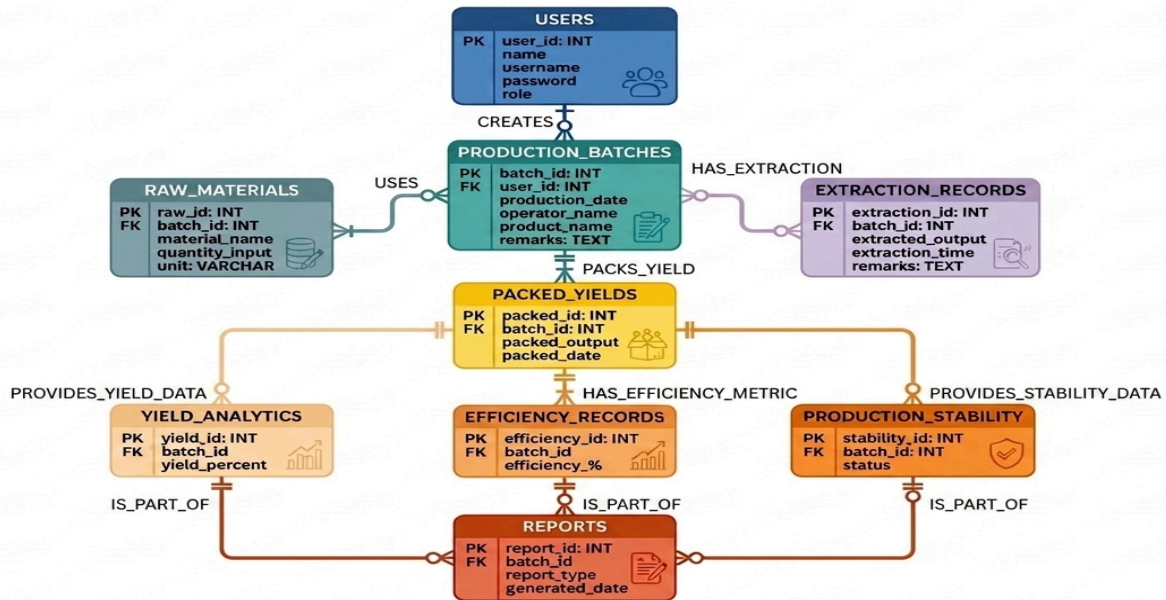
4.6 System Design

The relational database model serves as the foundational structure for the proposed system by organizing operational information into distinct, normalized tables designed to minimize redundancy and maintain data integrity. At the core of the database is the `USERS` table, which manages user credentials and access permissions, connected directly to `PRODUCTION_BATCHES` to track batch creation. Each production batch acts as a central hub linking various operational processes, including raw material inputs (`RAW_MATERIALS`), output metrics from extraction operations (`EXTRACTION_RECORDS`), and final packaged output logs (`PACKED_YIELDS`).

To support real-time efficiency monitoring, the database structures analytical outputs into specialized tables derived directly from packed production data. Calculated performance indicators are systematically organized across `YIELD_ANALYTICS`, `EFFICIENCY_RECORDS`, and `PRODUCTION_STABILITY` to evaluate output ratios and assign operational status classifications such as *Stable* or *Not Stable*. These computational records seamlessly feed into the `REPORTS` entity, enabling the automated generation of

comprehensive daily, weekly, and monthly summaries without compromising system efficiency or data consistency.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.



The primary entities of the database include:

- Users
- Raw Materials
- Production Batches
- Extraction Records
- Packed Yields
- Yield Analytics
- Efficiency Records
- Production Stability

- Reports

Users Table

This table stores account credentials and access permissions for system users.

Field	Description
user_id	Primary Key
name	Complete Name
username	Login Username
password	Account Password
role	User Access Level

Production Batches Table

This table acts as the central hub tracking production batch information.

Field	Description
batch_id	Primary Key
user_id	Foreign Key (Created by User)

production_date	Date of Production
operator_name	Name of Assigned Operator
product_name	Name of Product
remarks	Additional Operational Notes

Raw Materials Table

This table records the raw materials utilized in each production batch.

Field	Description
raw_id	Primary Key
batch_id	Foreign Key
material_name	Name of Raw Material
quantity_input	Amount of Material Used
unit	Measurement Unit (e.g., kg, L)

Extraction Records Table

This table tracks the volume and details of extracted outputs from production.

Field	Description
extraction_id	Primary Key
batch_id	Foreign Key
extracted_output	Volume of Extracted Output
extraction_time	Time of Extraction Process
remarks	Extraction Notes

Packed Yields Table

This table stores the final packaged yield measurements derived from production batches.

Field	Description
packed_id	Primary Key
batch_id	Foreign Key

packed_output	Total Packaged Volume/Units
packed_date	Date Packaging Completed

Yield Analytics Table

This table stores calculated yield percentages to measure overall output performance.

Field	Description
yield_id	Primary Key
batch_id	Foreign Key
yield_percent	Calculated Yield Percentage

Efficiency Records Table

This table records resource-to-output conversion efficiency metrics.

Field	Description
efficiency_id	Primary Key
batch_id	Foreign Key

efficiency_%	Extraction Efficiency Metric (%)
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Production Stability Table

This table evaluates batch performance stability against organizational criteria.

Field	Description
stability_id	Primary Key
batch_id	Foreign Key
status	Operational Status (e.g., Stable / Not Stable)

Reports Table

This table logs generated production summaries derived from analytical and stability metrics.

Field	Description
report_id	Primary Key
batch_id	Foreign Key
report_type	Type of Summary (e.g., Daily, Monthly)

generated_date	Date Report Created
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The relationships among these tables allowed the application to track raw material consumption, measure extraction performance, calculate packaged yields, evaluate production stability, and generate comprehensive management reports. Utilizing proper indexing and foreign key constraints helped maintain referential integrity while optimizing query performance across all operational and analytical workflows.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presented the conclusion and recommendations of the study entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc."* It summarized the overall findings of the research, determined whether the objectives established in the study had been successfully achieved, and discussed the significance of the developed system in addressing the operational challenges experienced by ALDi Foods Inc. Furthermore, this chapter provided recommendations for future system enhancements, implementation strategies, and possible directions for future researchers who intended to continue or improve the proposed system.

The study was conducted in response to the company's existing manual production monitoring process, which relied heavily on handwritten records and manual computation of production data. Although the existing process enabled the company to document production activities, it presented several limitations, including delays in recording production information, difficulty in monitoring production performance in real time, increased possibility of human errors during computation, and challenges in retrieving historical production records for analysis and reporting. These issues affected the organization's ability to monitor production efficiency promptly and make timely operational decisions.

To address these concerns, the researchers designed and developed a web-based production monitoring system capable of digitizing production records and providing real-time yield

analytics and extraction efficiency monitoring. The system centralized production information into a single database where authorized users could record raw material usage, extraction output, and packed yield data. It also automated the computation of yield percentage and extraction efficiency while providing dashboard visualizations and periodic reports that supported production monitoring and management decision-making.

Throughout the development process, the researchers followed the Agile Software Development Methodology to ensure that the system was continuously refined based on identified requirements and feedback gathered during each phase of development. The proposed system integrated several functional modules, including user authentication, production management, raw material recording, extraction monitoring, packed yield monitoring, production stability interpretation, dashboard visualization, historical record management, and automated report generation. These modules worked together to provide an efficient and centralized digital platform for production monitoring.

The findings presented in this chapter were based on the successful completion of the system development process described in the previous chapters. The discussion highlighted how the developed system addressed the problems identified in Chapter 1, incorporated concepts and best practices gathered from the related literature and studies discussed in Chapter 2, and implemented the methodologies presented in Chapter 3. By comparing the objectives of the study with the actual capabilities of the developed system presented in Chapter 4, this chapter demonstrated the extent to which the proposed solution satisfied the operational requirements of ALDi Foods Inc.

Finally, this chapter emphasized the importance of adopting digital technologies in manufacturing environments, particularly for organizations that continued to rely on manual production monitoring procedures. The implementation of the proposed system demonstrated that integrating web-based information systems into daily production operations improved data accuracy, enhanced production visibility, reduced manual workload, and supported more informed and timely decision-making. The conclusions and recommendations presented in this chapter provided valuable insights not only for ALDi Foods Inc. but also for future researchers and organizations seeking to modernize their production monitoring processes through digital transformation.

5.2 Summary of Findings

The study successfully achieved its primary objective of designing and developing a web-based production monitoring system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc."* The proposed system was developed to replace the company's existing manual production recording process with a centralized digital platform capable of providing real-time monitoring, automated computations, and production analytics.

Based on the requirements gathered through interviews, direct observations, and analysis of the company's current production workflow, the researchers identified several operational challenges affecting production monitoring. These challenges included the use of manual recording methods, delayed availability of production information, difficulty in monitoring production performance in real time, manual computation of yield percentages, limited visibility of

extraction efficiency, and the absence of a centralized repository for historical production records. These identified problems served as the basis for designing the proposed system and guided the development of its core functionalities.

The completed system successfully digitized the recording of production information by allowing authorized users to encode production batches, raw material quantities, extraction output, and packed yield through an organized web-based interface. Instead of relying on handwritten logs and manual computation, all production records were securely stored in a centralized MySQL database, making information more accessible, organized, and easier to retrieve whenever needed.

One of the major accomplishments of the proposed system was the automation of yield percentage computation. After users recorded the raw material input and production output, the system automatically calculated the corresponding yield percentage using predefined computational formulas. This automation significantly reduced the possibility of human error associated with manual calculations while improving the accuracy and consistency of production monitoring. The automated computation also enabled production personnel and managers to obtain immediate production performance results without performing repetitive mathematical calculations.

The system also successfully performed extraction efficiency monitoring by comparing raw material input with extracted production output. This functionality allowed production managers to evaluate the effectiveness of the extraction process for every production batch. Through continuous monitoring of extraction efficiency, management could identify production losses,

monitor resource utilization, and evaluate whether production processes were operating within acceptable performance standards.

Another significant finding of the study was the successful implementation of production stability interpretation. Based on predefined production thresholds established during system development, the application automatically classified production performance as either Stable or Not Stable. This feature enabled production supervisors and managers to immediately recognize production inconsistencies and investigate potential causes before they affected succeeding production batches. The automated stability interpretation supported faster decision-making and contributed to improved operational control.

The dashboard module developed for the system provided real-time visualization of production performance through charts, graphs, summary cards, and production statistics. The dashboard consolidated production information into an easy-to-understand interface, allowing administrators and production managers to monitor key production indicators without manually reviewing individual production records. The availability of real-time graphical reports improved production visibility and enabled management to evaluate operational performance more efficiently.

The developed system also supported historical production record management by storing all encoded production information within a centralized relational database. This enabled authorized users to retrieve previous production records for analysis, comparison, auditing, and reporting purposes. Historical production information could be utilized to evaluate long-term production trends, identify recurring operational issues, and support future production planning.

Furthermore, the report generation module automatically produced production reports based on weekly, monthly, quarterly, and yearly production records. These reports summarized raw material usage, extraction output, yield percentage, extraction efficiency, and production stability results, providing management with comprehensive documentation of production performance. Automated report generation reduced the time required to prepare production summaries while improving the consistency and accuracy of organizational records.

The researchers also found that the integration of the different system modules—including user authentication, production management, raw material recording, extraction monitoring, packed yield monitoring, dashboard visualization, historical record management, and report generation—resulted in a unified production monitoring platform capable of supporting the day-to-day operational requirements of AlDi Foods Inc. The centralized nature of the system improved communication among users and ensured that production information was consistently updated and available for authorized personnel.

Overall, the findings demonstrated that the proposed system successfully fulfilled the objectives established in Chapter 1. The developed application effectively digitized production monitoring, automated production computations, provided real-time analytics, centralized production information, and supported data-driven decision-making within AlDi Foods Inc. By replacing traditional manual recording methods with an integrated digital platform, the study contributed to improving production efficiency, reducing human error, and supporting the organization's ongoing digital transformation initiatives.

5.3 Conclusion

The primary objective of this study was to design and develop a web-based production monitoring system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc."* The study was conducted in response to the operational challenges experienced by ALDi Foods Inc., particularly the company's reliance on manual production recording, delayed access to production information, manual computation of yield values, and the absence of a centralized platform for monitoring production performance. These limitations affected the efficiency of production monitoring and reduced the organization's ability to make timely and informed operational decisions.

Based on the findings presented in the previous chapters, the researchers concluded that the proposed system successfully achieved the objectives established at the beginning of the study. The developed application provided a centralized digital platform that replaced manual production recording with a more organized, accurate, and efficient method of managing production information. Through the implementation of computerized data recording and automated processing, the system improved the accessibility, consistency, and reliability of production records while minimizing errors commonly associated with handwritten documentation and manual computation.

One of the major accomplishments of the proposed system was the successful digitization of the production monitoring process. Instead of relying on traditional paper-based recording methods, authorized users were able to record production information through an organized web-based interface. Production batches, raw material quantities, extraction outputs, and packed yield records were securely stored in a centralized database where information could be retrieved

efficiently whenever needed. This centralized approach not only improved record management but also reduced the possibility of data loss and duplication, which were common issues in manual record-keeping systems.

The developed system also successfully automated the computation of yield percentage and extraction efficiency. These calculations were performed automatically based on the production data entered by authorized users, eliminating the need for repetitive manual computations. Automation improved the accuracy of production calculations, reduced the likelihood of computational errors, and enabled production personnel to focus more on operational activities rather than mathematical processing. The availability of immediate computation results also allowed managers to evaluate production performance without unnecessary delays.

Another significant contribution of the study was the implementation of real-time dashboard analytics. The dashboard provided production managers and administrators with visual representations of production performance through charts, graphs, summary statistics, and production indicators. These visualizations simplified the interpretation of production information by presenting complex production data in an organized and understandable format. Real-time monitoring enabled management to quickly identify operational issues, monitor production trends, and make informed decisions based on current production conditions rather than relying solely on historical records.

The system's production stability interpretation feature further enhanced production monitoring by automatically classifying production performance as either *Stable* or *Not Stable* according to predefined production thresholds. This functionality provided immediate feedback regarding the consistency of production operations and assisted supervisors in identifying potential production

inefficiencies before they became significant operational problems. By providing timely production status information, the system contributed to better production control and continuous process improvement.

The automated report generation module also played an important role in improving production management. Weekly, monthly, quarterly, and yearly reports could be generated directly from the centralized database without requiring extensive manual preparation. These reports provided comprehensive summaries of raw material consumption, production output, yield percentage, extraction efficiency, and production stability, enabling management to evaluate production performance over different reporting periods. The availability of organized reports also supported documentation requirements, performance evaluation, and strategic planning within the organization.

The successful integration of user authentication, production management, raw material recording, extraction monitoring, packed yield monitoring, dashboard visualization, historical data management, production stability interpretation, and report generation demonstrated that the proposed system functioned as a complete production monitoring solution tailored to the operational needs of AlDi Foods Inc. The integration of these modules created a unified platform that streamlined production monitoring activities while ensuring that production information remained accurate, secure, and easily accessible.

Furthermore, the findings of this study supported the importance of digital transformation within the food manufacturing industry. As discussed in the related literature and studies presented in Chapter 2, many manufacturing organizations continued to adopt digital technologies to improve operational efficiency, reduce production waste, enhance data accuracy, and strengthen

decision-making processes. The developed system demonstrated that even without implementing advanced industrial technologies such as IoT or machine learning, a well-designed web-based information system could significantly improve production monitoring through centralized data management, automated computations, and real-time visualization.

The use of the Agile Software Development Methodology also contributed significantly to the successful completion of the project. The iterative development process allowed the researchers to continuously evaluate system functionalities, address identified issues during development, and ensure that the completed application met the operational requirements gathered from ALDi Foods Inc. The structured development process resulted in a system that aligned with the organization's existing workflow while remaining flexible enough to accommodate future improvements and expansion.

Overall, the researchers concluded that the proposed *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc.* system was a practical, reliable, and effective digital solution for modernizing production monitoring activities. The system successfully addressed the operational problems identified in the study by replacing manual production recording with an automated, centralized, and data-driven platform capable of supporting daily production operations. The implementation of the proposed system improved production visibility, enhanced data accuracy, simplified production analysis, and supported timely managerial decision-making.

Finally, the researchers concluded that the developed system contributed not only to the operational improvement of ALDi Foods Inc. but also to the broader objective of promoting digital transformation within the manufacturing sector. By demonstrating the successful

application of information technology in production monitoring, the study provided evidence that digital solutions could effectively improve organizational efficiency while serving as a foundation for future technological advancements. The proposed system therefore represented a meaningful contribution to both the field of Information Technology and the continuing modernization of manufacturing processes in the Philippine food industry.

5.4 Recommendations

Based on the findings, conclusions, and overall evaluation of the proposed system, the researchers recommended several improvements and future directions to further enhance the effectiveness, usability, scalability, and long-term sustainability of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* These recommendations were intended for AlDi Foods Inc., future researchers, and software developers who might continue the development and enhancement of the proposed web-based production monitoring system.

The researchers recommended that AlDi Foods Inc. fully implement the developed system in its daily production operations to maximize the benefits of digital production monitoring. The adoption of the system was expected to improve the accuracy of production recording, reduce manual computation, enhance production monitoring, and support faster and more informed managerial decision-making. The company was also encouraged to provide adequate training for administrators and production personnel to ensure the proper utilization and maintenance of the application.

The researchers also recommended the implementation of regular database backup and maintenance procedures to protect production records from accidental data loss and system failures. Routine software updates and preventive maintenance were likewise recommended to maintain the reliability, security, and efficiency of the application throughout its operational use.

For future system enhancement, the researchers recommended expanding the functionalities of the application by incorporating additional production management features such as inventory management, raw material tracking, supplier management, production scheduling, quality assurance monitoring, and employee performance monitoring. The integration of these modules would further improve the overall production management capabilities of the system.

The researchers further recommended the implementation of role-based access control by allowing multiple authorized users with different levels of system privileges. Future versions of the application could include separate user accounts for production supervisors, quality assurance personnel, warehouse staff, and management to improve collaboration while maintaining data security and accountability.

To improve accessibility and operational flexibility, the researchers recommended developing a mobile-responsive version of the application or a dedicated mobile application. This enhancement would enable production personnel and management to monitor production activities, review reports, and access production information using smartphones or tablet devices.

The researchers also recommended the integration of advanced data analytics and predictive reporting features in future versions of the system. Additional analytical tools, such as production forecasting, trend analysis, and performance prediction, could assist management in

making more strategic operational decisions and identifying potential production issues before they occurred.

Although the current study intentionally excluded Industrial Internet of Things (IIoT) technologies and advanced automation to maintain affordability for small and medium-sized enterprises, future researchers were encouraged to investigate the possible integration of sensor-based data collection, automated production monitoring, or machine learning techniques if organizational resources and operational requirements permitted such enhancements.

Future researchers were also encouraged to conduct further evaluation of the system using larger groups of users and longer implementation periods. Additional studies could assess the long-term effects of the system on production efficiency, operational performance, user satisfaction, and organizational productivity using different evaluation methodologies and performance indicators.

Lastly, the researchers recommended that future studies continue exploring innovative information technology solutions that support digital transformation within the manufacturing industry. Similar production monitoring systems could be developed for other food processing companies and manufacturing organizations to improve production management, enhance operational efficiency, reduce manual workloads, and promote data-driven decision-making through modern web-based technologies.

Overall, these recommendations were intended to support the continuous improvement, sustainability, and future expansion of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for ALDi Foods Inc.*, ensuring that the developed application remained relevant, adaptable, and capable of meeting the evolving operational requirements of the organization.

System Improvements

Although the developed system successfully achieved its primary objectives of digitizing production monitoring, automating yield computation, monitoring extraction efficiency, and generating production reports, there were still opportunities to improve its functionality and overall performance.

The researchers recommended incorporating an automated notification and alert feature into future versions of the application. This feature could notify administrators whenever production results fell below the expected yield percentage or extraction efficiency, or when production stability was classified as Not Stable. Such notifications would allow production personnel to immediately review production records and take appropriate corrective actions.

Future versions of the system could also include a more comprehensive dashboard with additional visualization features. While the current system provided charts, graphs, and production summaries, future enhancements could include interactive filters, comparison of production batches by date, graphical trend analysis, and customizable dashboard layouts. These features would provide management with clearer insights into production performance and support more effective decision-making.

Another recommended enhancement was the integration of barcode or QR code technology for production batch identification. This feature would simplify production batch recording, reduce manual encoding errors, improve the traceability of production records, and make information retrieval more efficient.

To strengthen system security, future versions should implement additional authentication and security features such as stronger password encryption, password recovery options, user activity logs, automatic session timeout, and role-based access control if additional user roles were introduced. These improvements would help protect production information from unauthorized access while improving accountability among system users.

The researchers also recommended enhancing the report generation module by allowing reports to be exported in different formats such as PDF, Microsoft Excel, and CSV. Providing multiple export options would make it easier for management to archive production reports, prepare documentation, and perform further data analysis whenever necessary.

To ensure the safety of production records, future versions of the application should include automatic database backup and recovery functions. Regular backup procedures would help protect valuable production information from accidental deletion, hardware failure, or unexpected system interruptions.

Finally, the researchers recommended improving the usability of the system by continuously refining the user interface based on user feedback. Enhancing navigation, simplifying data entry forms, and improving dashboard organization would make the system easier to learn and use for production personnel and administrators.

Future Research Directions

This study focused on developing a web-based production monitoring system for AlDi Foods Inc. that digitized production recording, computed yield percentage and extraction efficiency, stored historical production records, and generated production reports. Although the

proposed system successfully addressed the identified operational problems, the researchers recommended that future researchers could further improve the system by expanding its functionality while remaining aligned with the needs of manufacturing organizations.

The researchers recommended that future researchers could enhance the dashboard by incorporating more advanced statistical summaries, production comparisons, and additional graphical reports that would provide deeper insights into production performance. These enhancements could help management evaluate long-term production trends and identify areas for operational improvement.

The researchers also recommended that future researchers could develop a mobile-responsive or dedicated mobile application version of the system to enable administrators and production supervisors to access production information using smartphones or tablets. This enhancement would improve accessibility and would allow authorized users to monitor production records even when they were away from the production office.

Another possible enhancement was the expansion of the system to include additional management modules such as inventory management, raw material inventory tracking, supplier information management, quality assurance monitoring, production scheduling, and employee management. Integrating these modules would create a more comprehensive manufacturing information system while maintaining the centralized database established in the current study.

The researchers also recommended that future studies could focus on improving report generation by incorporating more detailed production summaries, customized report templates, and comparative analyses of production performance over different reporting periods. Such

improvements would provide management with more comprehensive information for production planning and operational evaluation.

The researchers likewise encouraged future researchers to conduct comparative studies involving other food manufacturing companies to determine whether the proposed production monitoring framework could be adapted and implemented in different production environments. These studies could help evaluate the flexibility and applicability of the developed system across various manufacturing processes.

Finally, the researchers recommended that future researchers could evaluate the long-term impact of implementing digital production monitoring systems on operational efficiency, production consistency, data accuracy, reporting efficiency, and overall organizational productivity. Such studies would provide additional evidence regarding the effectiveness of digital production monitoring systems in supporting continuous operational improvement.

Here is the past tense version of your Implementation Strategies section. Since this is under Recommendations, the recommendations are expressed as "recommended", while future actions remain as should/would where appropriate.

Implementation Strategies

For the successful implementation of the proposed system, the researchers recommended that ALDi Foods Inc. should establish a structured deployment strategy that included proper preparation, employee training, continuous monitoring, and regular system evaluation. Proper implementation would help maximize the benefits of the developed application while ensuring consistent and accurate production monitoring.

Before full implementation, the researchers recommended that production personnel and administrators should undergo comprehensive user training covering system navigation, production data encoding, dashboard interpretation, report generation, and basic troubleshooting procedures. Adequate training would improve user confidence, minimize encoding errors, and promote the proper utilization of the system.

The researchers also recommended that management should establish standardized operating procedures (SOPs) that clearly defined responsibilities for recording production information, validating encoded data, reviewing production reports, and maintaining database accuracy. Standardized procedures would ensure consistency in production monitoring activities and would improve the reliability of stored production information.

To maintain data integrity, the researchers recommended that regular database backups should be performed according to an established schedule. Backup files should be stored securely to minimize the risk of data loss caused by hardware failure, software malfunction, or other unforeseen circumstances.

The researchers further recommended that periodic system maintenance should be conducted to ensure continuous system reliability and performance. This included updating software components, correcting identified system errors, optimizing database performance, reviewing security settings, and monitoring overall system functionality.

The researchers also recommended that management should regularly analyze the production reports and dashboard information generated by the system during production meetings and operational planning. Continuous evaluation of yield percentage, extraction efficiency,

production stability, and production output would support informed decision-making and encourage continuous process improvement.

Finally, the researchers recommended that user feedback should be collected regularly after system implementation. Suggestions and observations from administrators and production personnel should be documented and considered in future system updates. Continuous improvement based on actual user experience would help ensure that the system remained effective, user-friendly, and responsive to the operational requirements of AIDi Foods Inc.

Here is the past tense version of Section 5.5 Impact of the Study. Since this section discusses the completed study, all present-tense descriptions have been converted to past tense while maintaining academic style.

5.5 Impact of the Study

The development of the Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AIDi Foods Inc. contributed significantly to the advancement of digital transformation in food manufacturing, particularly among small and medium-sized enterprises (SMEs). By replacing traditional manual production monitoring with a centralized digital platform, the study demonstrated how information technology could improve operational efficiency, production visibility, and management decision-making.

For AIDi Foods Inc., the developed system served as an effective production monitoring tool that digitized the recording of raw material usage, extraction outputs, packed yield, and production performance. Through real-time yield analytics and extraction efficiency monitoring,

management could immediately identify production trends, monitor operational performance, and detect potential production losses before they became significant. This enabled supervisors and production managers to make timely, evidence-based decisions that contributed to improved productivity and reduced waste.

The study also minimized the limitations associated with manual record-keeping. Traditional paper-based production logs were susceptible to human error, misplaced records, delayed reporting, and inconsistencies in data recording. By implementing a centralized database with automated calculations, the proposed system improved data accuracy, consistency, and accessibility. Historical production records were securely stored and could be retrieved whenever needed for operational reviews, audits, and production planning.

Another significant contribution of the study was the enhancement of production transparency. Because production information was updated in real time, authorized users could monitor production activities as they occurred rather than waiting until the end of the production cycle. This immediate availability of information allowed management to respond quickly to operational issues, helping maintain production stability and optimize resource utilization.

The system also promoted a culture of data-driven decision-making within the organization. Rather than relying solely on experience or manually compiled reports, production managers could use real-time dashboards, graphical reports, and automated analytics to evaluate production performance objectively. Information such as yield percentage, extraction efficiency, production stability classification, and raw material utilization provided valuable insights that supported strategic planning and continuous process improvement.

From an academic perspective, this study contributed to the field of Information Technology, particularly in the areas of systems analysis and design, database management, web application development, and business analytics. The project demonstrated how software engineering principles could be applied to solve practical operational problems in the manufacturing industry through the development of an integrated production monitoring system. It also illustrated the use of modern web technologies to create digital solutions that improved organizational efficiency.

The study likewise benefited future BSIT students and researchers by providing a practical reference for similar projects involving production monitoring, manufacturing information systems, dashboard development, and real-time analytics. The methodologies, system architecture, database design, Agile development approach, and evaluation procedures documented in this research could serve as useful guides for future academic and industry-based system development projects.

Furthermore, the study contributed to the growing body of knowledge supporting the digital transformation of Philippine manufacturing SMEs. Many local businesses continued to rely on manual production monitoring because of financial limitations and limited access to advanced manufacturing technologies. The proposed system demonstrated that digital transformation could be achieved through practical, affordable, and scalable information systems without requiring expensive industrial automation or complex infrastructure. This made the solution more accessible to organizations with limited technological resources.

Despite its contributions, the researchers recognized several limitations of the study. The system was developed as a prototype specifically tailored to the production workflow of AlDi Foods

Inc., limiting its immediate applicability to other manufacturing environments without modification. In addition, the study focused only on production monitoring, yield analytics, extraction efficiency monitoring, and report generation. Other operational areas such as inventory management, purchasing, sales, accounting, supplier management, and predictive analytics were intentionally excluded from the project scope. Likewise, the prototype did not incorporate IoT devices or automated sensor technologies, as these were beyond the defined scope of the study.

Nevertheless, these limitations did not diminish the overall value of the developed system. Instead, they provided opportunities for future enhancements and research. The study successfully demonstrated that a carefully designed digital production monitoring platform could significantly improve production management, enhance operational visibility, reduce manual recording errors, and support informed managerial decision-making. Ultimately, the project served as a practical example of how information technology could drive digital transformation and operational excellence within food manufacturing enterprises such as AlDi Foods Inc.

5.6 Summary

This chapter presented the overall conclusions and recommendations of the study entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* It summarized the major findings obtained during the design, development, implementation, testing, and evaluation of the proposed system. The chapter also discussed how the study achieved its research objectives and addressed the production monitoring challenges identified in Chapters 1, 2, 3, and 4.

The study successfully developed a web-based production monitoring system that digitized the recording of production data, automated the computation of yield percentage and extraction efficiency, and provided real-time dashboard analytics for production management. By replacing manual recording practices with a centralized digital platform, the system improved data accuracy, enhanced production visibility, minimized human errors, and supported timely decision-making. The results of the system testing and user evaluation further demonstrated that the developed system was functional, reliable, accurate, efficient, and user-friendly for its intended users at ALDi Foods Inc.

The recommendations presented in this chapter emphasized the importance of continuous system enhancement through additional security features, expanded reporting capabilities, mobile accessibility, predictive analytics, and integration with other production management functions. The study also recommended future research involving emerging technologies such as Artificial Intelligence (AI), Machine Learning, and the Internet of Things (IoT) to further improve production monitoring and operational efficiency.

Overall, the study demonstrated that the application of information technology could significantly improve production management in food manufacturing by providing accurate, real-time production information that supported evidence-based decision-making. The developed system offered a practical and cost-effective solution for ALDi Foods Inc. and served as a valuable reference for future researchers, software developers, and organizations seeking to implement digital production monitoring systems.

In conclusion, the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for ALDi Foods Inc.* project successfully achieved its goal of designing and

developing a digital production monitoring platform that aligned with the operational needs of the company. The study contributed to the growing adoption of digital transformation among manufacturing organizations by demonstrating how real-time analytics and centralized production monitoring could improve efficiency, reduce production losses, and support continuous operational improvement. It also provided a strong foundation for future developments aimed at creating more intelligent, scalable, and integrated manufacturing information systems.

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